

AUTOMATIC LICENSE NUMBER PLATE DETECTION AND RECOGNITION



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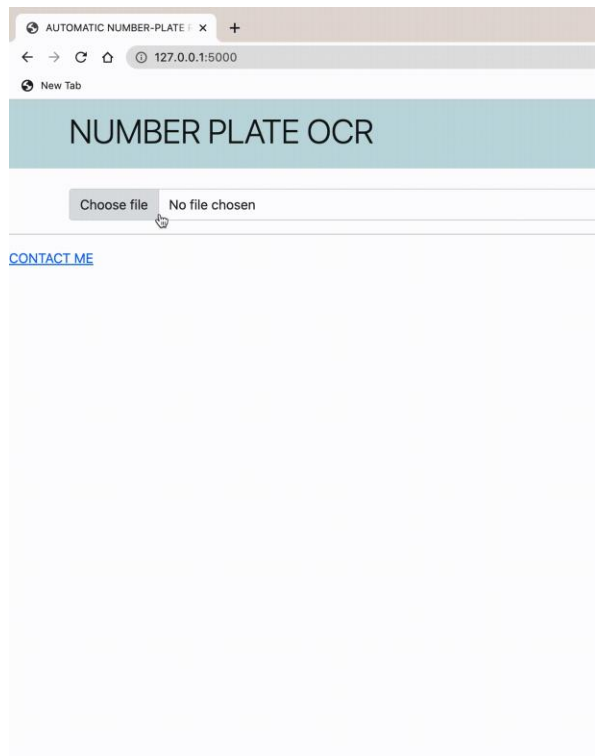
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1. INTRODUCTION



In Today's Day and Age Security has become one of the biggest concerns for any organization, and automation of such security is essential. However, many of the current solutions are still not robust in real-world situations, commonly depending on many constraints. In the following project, we will understand how to recognize License number plates using the Python programming language. We will utilize OpenCV for this project in order to identify the license number plates and the python pytesseract for the characters and digits extraction from the plate. As well this project will presents a robust and efficient ALPR system based on the state-of-the-art YOLO object detector. We will Web App with a Python program that automatically recognizes the License Number Plate by the end of this journi. The results have shown that the trained neural network is able to perform with high accuracy of nearly 90-95 percent in recognizing license plates in low resolution images using this system.

1.1 USE CASES

License plate detection is identifying the part of the car that is predicted to be the number plate. Recognition is identifying the values that make up the license plate. License plate detection and recognition is the technology that uses computer vision to detect and recognize a license plate from an input image of a car. This technology applies in many areas. On roads, it is used to identify the cars that are breaking the traffic rules. In security, it is used to capture the license plates of the vehicles getting into and out of certain premises. In parking lots, it is used to capture the license plates of the cars being parked. The list of its applications goes on and on.



1.2 PROJECT ARCHITECTURE

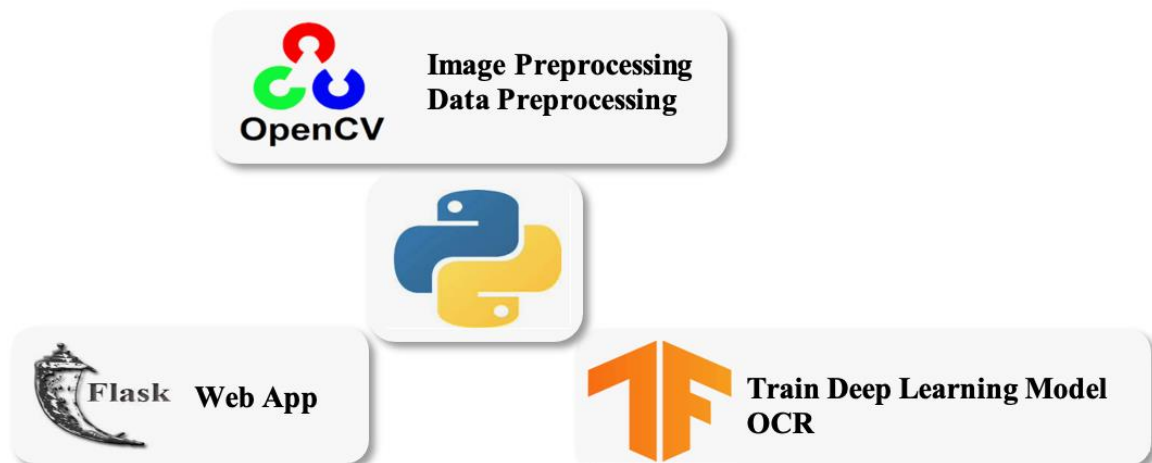


Figure 1 Framework

In the below architecture *Figure 2*, there are six modules. Labeling, Training, Save Model, OCR and Pipeline, and RESTful API. The process is as follows. First, we will collect the image. Then we have to label images for object detection of License Plate or Number Plate using Image Annotation Tool [Github](#) which is open-source software developed in python GUI. Then after labeling the image we will work on data preprocessing, build and train a deep learning object detection model (Inception Resnet V2) in TensorFlow 2. Once we have done with the Object Detection model training process, then using this model we will crop the image which contains the license plate which is also called the region of interest (ROI), and pass the ROI to Optical Character Recognition API Tesseract in Python (PyTesseract). As well extract text from images. Now, we will put it all together and build a Pipeline Deep Learning model. In the final module, we will learn to create a web app project using FLASK Python. With that, we are finally ready with our App.

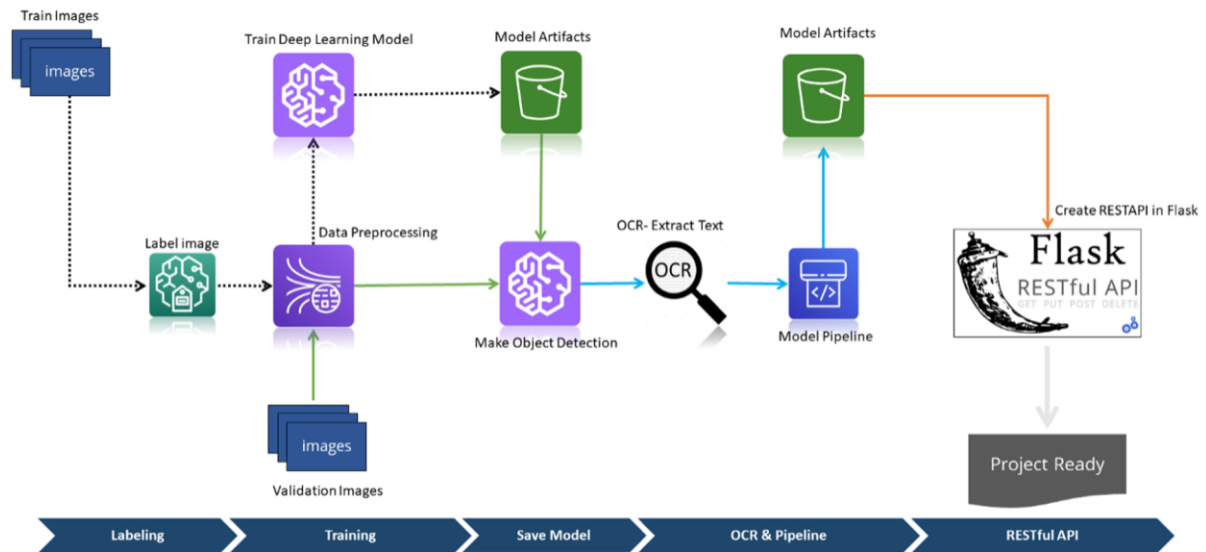


Figure 2 Number Plate Recognition Project Architecture

2. LABELING

2.1 UNDERSTAND & COLLECT REQUIRED DATA

For building the license plate recognition we need data. For that, we need to collect the vehicle images where the number plate appears on it as much as we can. [Here](#) is the sample data that I used for building this project. Below you can see few examples *Figure 3,4*.



Figure 3 Sample 1



Figure 4 Sample 2

2.2 LABEL IMAGES

For label images, I used the Labelling image annotation tool. Download the labelling from [GitHub](#) and follow the instruction to install the package. After that open, the GUI as give the instruction and click on CreateRectBox and draw the rectangle box as shown below and save the output in XML.

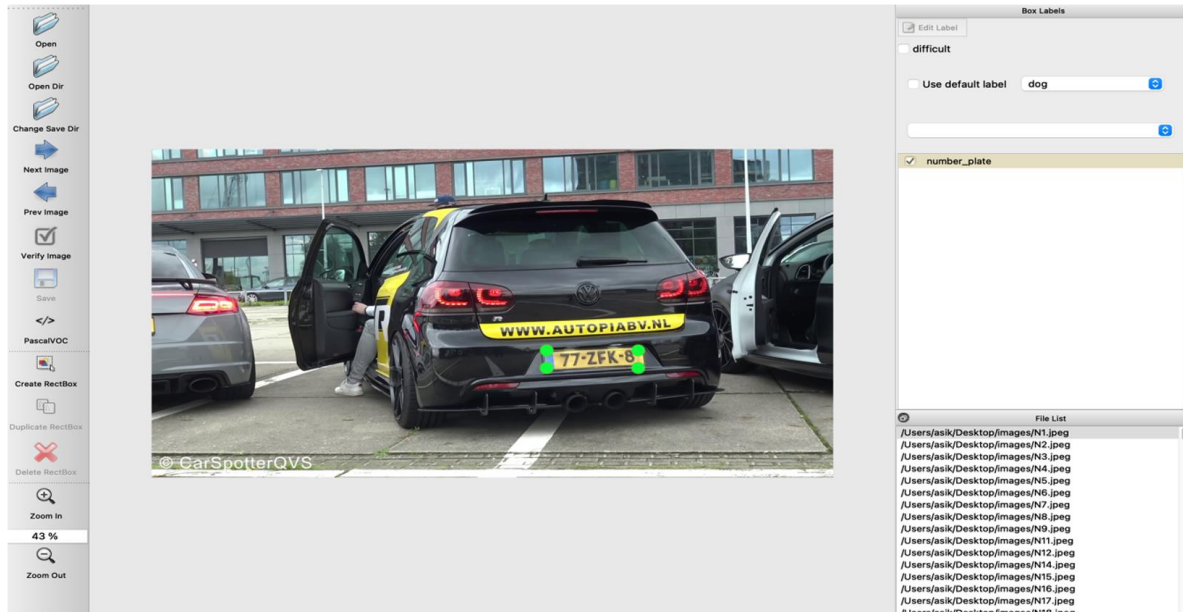


Figure 5 Labeling

This is a manual process and you need to do it for all the images. Be careful while doing labeling because the labeling process has a direct impact on the accuracy of the model. [Click here for a video tutorial](#). On *Figure 5* you can find main window.

2.3 PARSING INFORMATION FROM XML

Example of our .xml files will look as below.

```
<annotation>

  <folder>images</folder>

  <filename>N1.jpeg</filename>

  <path>/Users/asik/Desktop/ANPR/imagesN1.jpeg</path>

  <source>

    <database>Unknown</database>

  </source>

  <size>

    <width>1920</width>

    <height>1080</height>

    <depth>3</depth>

  </size>

  <segmented>0</segmented>
```

```

<object>

  <name>number_plate</name>

  <pose>Unspecified</pose>

  <truncated>0</truncated>

  <difficult>0</difficult>

  <bndbox>

    <xmin>1093</xmin>

    <ymin>645</ymin>

    <xmax>1396</xmax>

    <ymax>727</ymax>

  </bndbox>

</object>

</annotation>

```

Once you are done with the labeling process now we need to do some data preprocessing. Since the output of the label is XML in order to use this for the training process we need data in array format. For that, we will take the useful information from the label which are the diagonal points of the rectangle box or bounding box which are *xmin*, *ymin*, *xmax*, *ymax* respectively. This is available in XML. So we need to extract the information and save it in any convenient format, here I will convert bounding information into CSV and later on, I will convert that into an array using Pandas. Now let's see how to parse the information using Python.

2.4 PARSING DATA FROM XML AND CONVERTING IT INTO CSV

Firstly let me load all libraries which I will use in this project at one time. As well I will use **xml.etree** python library to parse the data from XML and also import pandas and glob. Using glob let first get all the XML files that are produced during labeling.

In [1]:

```

import os
import cv2
import numpy as np
import pandas as pd
import tensorflow as tf
import pytesseract as pt
import plotly.express as px
import matplotlib.pyplot as plt
import xml.etree.ElementTree as xet

from glob import glob

```

```

from skimage import io
from shutil import copy
from tensorflow.keras.models import Model
from tensorflow.keras.callbacks import TensorBoard
from sklearn.model_selection import train_test_split
from tensorflow.keras.applications import InceptionResNetV2
from tensorflow.keras.layers import Dense, Dropout, Flatten, Input
from tensorflow.keras.preprocessing.image import load_img, img_to_array

```

In [2]:

```

path = glob('../input/number-plate-detection/images/*.xml')
labels_dict = dict(filepath=[], xmin=[], xmax=[], ymin=[], ymax=[])
for filename in path:

    info = xet.parse(filename)
    root = info.getroot()
    member_object = root.find('object')
    labels_info = member_object.find('bndbox')
    xmin = int(labels_info.find('xmin').text)
    xmax = int(labels_info.find('xmax').text)
    ymin = int(labels_info.find('ymin').text)
    ymax = int(labels_info.find('ymax').text)

    labels_dict['filepath'].append(filename)
    labels_dict['xmin'].append(xmin)
    labels_dict['xmax'].append(xmax)
    labels_dict['ymin'].append(ymin)
    labels_dict['ymax'].append(ymax)

```

In the above code, we individually take each file and parse into xml.etree and find the object -> bndbox. Then we extract xmin,xmax,ymin,ymax and saved those values in the dictionary. After we convert it into a pandas data frame and save that into CSV file and save it in project folder as shown below.

In [3]:

```

df = pd.DataFrame(labels_dict)
df.to_csv('labels.csv', index=False)
df.head()

```

Out[3]:

	filepath	xmin	xmax	ymin	ymax
0	../input/number-plate-detection/images/N148.xml	244	369	240	293
1	../input/number-plate-detection/images/N177.xml	331	538	263	317
2	../input/number-plate-detection/images/N173.xml	80	335	150	243

	filepath	xmin	xmax	ymin	ymax
3	../input/number-plate-detection/images/N213.xml	131	209	129	153
4	../input/number-plate-detection/images/N119.xml	180	559	216	314

With the above code, we successfully extract the diagonal position of each image and convert the data from an unstructured to a structured format. You can have a look at the data above. Now also extract the respective image filename of the XML.

In [4]:

```
filename = df['filepath'][0]
def getFilename(filename):
    filename_image = xet.parse(filename).getroot().find('filename').text
    filepath_image = os.path.join('../input/number-plate-detection/images',
    filename_image)
    return filepath_image
getFilename(filename)
```

Out[4]:

```
'../input/number-plate-detection/images/N148.jpeg'
```

In [5]:

```
image_path = list(df['filepath'].apply(getFilename))
image_path[:10]#random check
```

Out[5]:

```
['../input/number-plate-detection/images/N148.jpeg',
 '../input/number-plate-detection/images/N177.jpeg',
 '../input/number-plate-detection/images/N173.jpeg',
 '../input/number-plate-detection/images/N213.jpeg',
 '../input/number-plate-detection/images/N119.jpeg',
 '../input/number-plate-detection/images/N103.jpeg',
 '../input/number-plate-detection/images/N166.jpeg',
 '../input/number-plate-detection/images/N127.jpeg',
 '../input/number-plate-detection/images/N162.jpeg',
 '../input/number-plate-detection/images/N198.jpeg']
```

2.5 VERIFY THE DATA

As till now we did the manual process it is important to verify the information we got is valid or not. For that just verify the bounding box is appearing properly for a given image. Here I consider the image N2.jpeg and the corresponding diagonal position can be found in df. Result you can see on *Figure 8*

In [6]:

```
file_path = image_path[87] #path of our image N2.jpeg
img = cv2.imread(file_path) #read the image
# xmin-1804/ymin-1734/xmax-2493/ymax-1882
```

```
img = io.imread(file_path) #Read the image
fig = px.imshow(img)
fig.update_layout(width=600, height=500, margin=dict(l=10, r=10, b=10, t=10),
xaxis_title='Figure 8 - N2.jpeg with bounding box')
fig.add_shape(type='rect',x0=1804, x1=2493, y0=1734, y1=1882, xref='x', yref='y',line_color='cyan')
0500100015002000250030002000150010005000
Figure 8 - N2.jpeg with bounding box
```

3. DATA PROCESSING

3.1 READ DATA

This is a very important step, in this process we will take each and every image and convert it into an array using OpenCV and resize the image into 224 x 224 which is the standard compatible size of the pre-trained transfer learning model.

In [7]:

```
#Targeting all our values in array selecting all columns
labels = df.iloc[:,1:].values
data = []
output = []
for ind in range(len(image_path)):
    image = image_path[ind]
    img_arr = cv2.imread(image)
    h,w,d = img_arr.shape
    # Preprocessing
    load_image = load_img(image,target_size=(224,224))
    load_image_arr = img_to_array(load_image)
    norm_load_image_arr = load_image_arr/255.0 # Normalization
    # Normalization to labels
    xmin,xmax,ymin,ymax = labels[ind]
    nxmin,nxmax = xmin/w,xmax/w
    nymin,nymax = ymin/h,ymax/h
    label_norm = (nxmin,nxmax,nymin,nymax) # Normalized output
    # Append
    data.append(norm_load_image_arr)
    output.append(label_norm)
```

After that, we will normalize the image just by dividing with maximum number as we know that the maximum number for an 8-bit image is $2^8 - 1 = 255$. That the reason we will divide our image 255.0. The way of dividing an array with the maximum value is called Normalization (Min-Max Scaler). We also need to normalize our labels too. Because for the deep learning model the output range should be between 0 to 1. For normalizing labels, we need to divide the diagonal points with the width and height of the image. And finally values in a python list.

3.2 SPLIT TRAIN AND TEST SET

In the next step, we will convert the list into an array using **Numpy**.

In [8]:

```
# Convert data to array
X = np.array(data, dtype=np.float32)
y = np.array(output, dtype=np.float32)

Now split the data into training and testing set using sklearn.
```

In [9]:

```
# Split the data into training and testing set using sklearn.
x_train, x_test, y_train, y_test = train_test_split(X, y, train_size=0.8, random_
state=0)
x_train.shape, x_test.shape, y_train.shape, y_test.shape
```

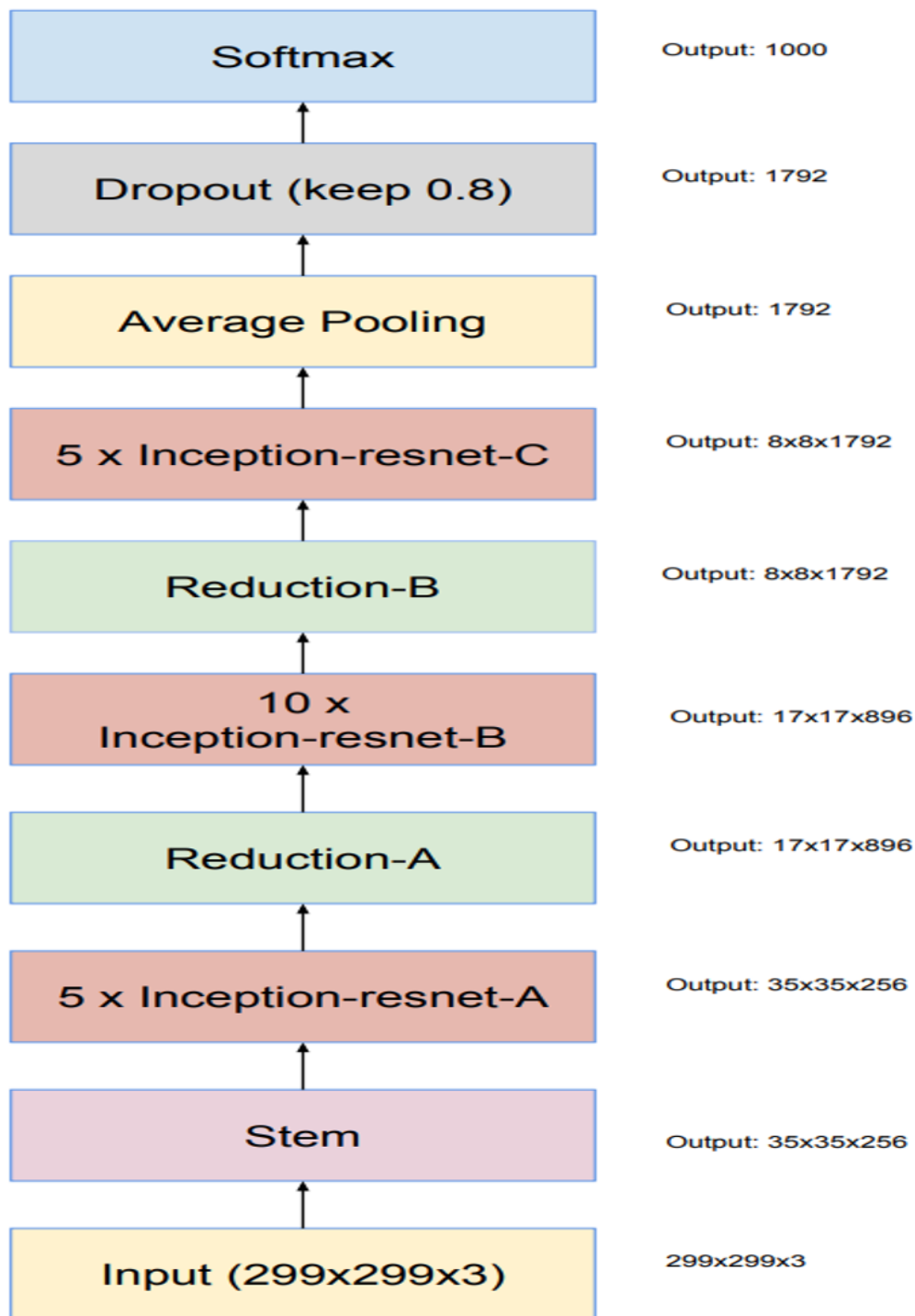
Out[9]:

```
((180, 224, 224, 3), (45, 224, 224, 3), (180, 4), (45, 4))
```

4. DEEP LEARNING FOR OBJECT DETECTION

4.1 INCEPTION-RESNET-V2 MODEL BUILDING

Inception-ResNet-v2 is a convolutional neural network that is trained on more than a million images from the ImageNet database. The network is 164 layers deep and can classify images into 1000 object categories, such as keyboard, mouse, pencil, and many animals. As a result, the network has learned rich feature representations for a wide range of images. The Inception-ResNet-v2 was used for the classification task. The architecture of the network is shown in Figure 9 . Inception-Resnet-v2 is formulated based on a combination of the Inception structure and the Residual connection. In the Inception-Resnet block multiple sized convolutional filters are combined by residual connections. The usage of residual connections not only avoids the degradation problem caused by deep structures but also reduces the training time.



- *Figure 9* -

We are ready to train a deep learning model for object detection. Here we will use the Inception-ResNet-v2 model with pre-trained weights and train this to our data. We are already import necessary libraries from TensorFlow previously, lets continue.

```

In [10]:
inception_resnet = InceptionResNetV2(weights="imagenet", include_top=False,
input_tensor=Input(shape=(224, 224, 3)))
# -----
headmodel = inception_resnet.output

```

```
headmodel = Flatten()(headmodel)
headmodel = Dense(500,activation="relu")(headmodel)
headmodel = Dense(250,activation="relu")(headmodel)
headmodel = Dense(4,activation='sigmoid')(headmodel)
```

```
# ----- model
```

```
model = Model(inputs=inception_resnet.input,outputs=headmodel)
```

```
2022-06-17 11:06:59.087750: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:06:59.092729: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:06:59.093468: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:06:59.094728: I tensorflow/core/platform/cpu_feature_guard.cc:142] This TensorFlow binary is optimized with oneAPI Deep Neural Network Library (oneDNN) to use the following CPU instructions in performance-critical operations: AVX2 AVX512F FMA
```

To enable them in other operations, rebuild TensorFlow with the appropriate compiler flags.

```
2022-06-17 11:06:59.095013: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:06:59.095684: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:06:59.096340: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:07:01.372375: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:07:01.373241: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:07:01.373896: I tensorflow/stream_executor/cuda/cuda_gpu_executor.cc:937] successful NUMA node read from SysFS had negative value (-1), but there must be at least one NUMA node, so returning NUMA node zero
```

```
2022-06-17 11:07:01.374503: I tensorflow/core/common_runtime/gpu/gpu_device.cc:1510] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 15385 MB memory: -> device: 0, name: Tesla P100-PCIE-16GB, pci bus id: 0000:00:04.0, compute capability: 6.0
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/inception_resnet_v2/inception_resnet_v2_weights_tf_dim_ordering_tf_kernels_notop.h5
```

```
219062272/219055592 [=====] - 1s 0us/step
```

```
219070464/219055592 [=====] - 1s 0us/step
```

Now compile the model and have a look at our summary. Don't be surprised, the summary will be a bit massive. The summary is textual and includes information about: The layers and their order in the model. The output shape of each layer. The number of parameters (weights) in each layer.

In [11]:

```
linkcode
```

```
# Compile model
```

```
model.compile(loss='mse', optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4))
```

```
model.summary()
```

```
Model: "model"
```

Layer (type)	Output Shape	Param #	Connected to
=====			
input_1 (InputLayer)	[(None, 224, 224, 3)]	0	

conv2d (Conv2D)	(None, 111, 111, 32)	864	input_1[0][0]

batch_normalization (Batch Normalization)	(None, 111, 111, 32)	96	conv2d[0][0]

activation (Activation)	(None, 111, 111, 32)	0	batch_normalization[0][0]

conv2d_1 (Conv2D)	(None, 109, 109, 32)	9216	activation[0][0]

batch_normalization_1 (Batch Normalization)	(None, 109, 109, 32)	96	conv2d_1[0][0]

activation_1 (Activation)	(None, 109, 109, 32)	0	batch_
normalization_1[0][0]			

conv2d_2 (Conv2D)	(None, 109, 109, 64)	18432	activa
tion_1[0][0]			

batch_normalization_2 (BatchNor	(None, 109, 109, 64)	192	conv2d
_2[0][0]			

activation_2 (Activation)	(None, 109, 109, 64)	0	batch_
normalization_2[0][0]			

max_pooling2d (MaxPooling2D)	(None, 54, 54, 64)	0	activa
tion_2[0][0]			

conv2d_3 (Conv2D)	(None, 54, 54, 80)	5120	max_po
oling2d[0][0]			

batch_normalization_3 (BatchNor	(None, 54, 54, 80)	240	conv2d
_3[0][0]			

activation_3 (Activation)	(None, 54, 54, 80)	0	batch_
normalization_3[0][0]			

conv2d_4 (Conv2D)	(None, 52, 52, 192)	138240	activa
tion_3[0][0]			

batch_normalization_4 (BatchNor	(None, 52, 52, 192)	576	conv2d
_4[0][0]			

activation_4 (Activation)	(None, 52, 52, 192)	0	batch_
normalization_4[0][0]			

max_pooling2d_1 (MaxPooling2D)	(None, 25, 25, 192)	0	activa
tion_4[0][0]			

conv2d_8 (Conv2D)	(None, 25, 25, 64)	12288	max_po
oling2d_1[0][0]			

batch_normalization_8 (BatchNormalizer)	(None, 25, 25, 64)	192	conv2d_8[0][0]
activation_8 (Activation)	(None, 25, 25, 64)	0	batch_normalization_8[0][0]
conv2d_6 (Conv2D)	(None, 25, 25, 48)	9216	max_pooling2d_1[0][0]
conv2d_9 (Conv2D)	(None, 25, 25, 96)	55296	activation_8[0][0]
batch_normalization_6 (BatchNormalizer)	(None, 25, 25, 48)	144	conv2d_6[0][0]
batch_normalization_9 (BatchNormalizer)	(None, 25, 25, 96)	288	conv2d_9[0][0]
activation_6 (Activation)	(None, 25, 25, 48)	0	batch_normalization_6[0][0]
activation_9 (Activation)	(None, 25, 25, 96)	0	batch_normalization_9[0][0]
average_pooling2d (AveragePooling2D)	(None, 25, 25, 192)	0	max_pooling2d_1[0][0]
conv2d_5 (Conv2D)	(None, 25, 25, 96)	18432	max_pooling2d_1[0][0]
conv2d_7 (Conv2D)	(None, 25, 25, 64)	76800	activation_6[0][0]
conv2d_10 (Conv2D)	(None, 25, 25, 96)	82944	activation_9[0][0]

conv2d_11 (Conv2D)	(None, 25, 25, 64)	12288	average

batch_normalization_5 (BatchNor	(None, 25, 25, 96)	288	conv2d
_5[0][0]			

batch_normalization_7 (BatchNor	(None, 25, 25, 64)	192	conv2d
_7[0][0]			

batch_normalization_10 (BatchNo	(None, 25, 25, 96)	288	conv2d
_10[0][0]			

batch_normalization_11 (BatchNo	(None, 25, 25, 64)	192	conv2d
_11[0][0]			

activation_5 (Activation)	(None, 25, 25, 96)	0	batch_
normalization_5[0][0]			

activation_7 (Activation)	(None, 25, 25, 64)	0	batch_
normalization_7[0][0]			

activation_10 (Activation)	(None, 25, 25, 96)	0	batch_
normalization_10[0][0]			

activation_11 (Activation)	(None, 25, 25, 64)	0	batch_
normalization_11[0][0]			

mixed_5b (Concatenate)	(None, 25, 25, 320)	0	activa
			activa
			activa
			activa
tion_5[0][0]			
tion_7[0][0]			
tion_10[0][0]			
tion_11[0][0]			

conv2d_15 (Conv2D)	(None, 25, 25, 32)	10240	mixed_
5b[0][0]			

batch_normalization_15 (BatchNo	(None, 25, 25, 32)	96	conv2d
_15[0][0]			

activation_15 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_15[0][0]			

conv2d_13 (Conv2D)	(None, 25, 25, 32)	10240	mixed_
5b[0][0]			

conv2d_16 (Conv2D)	(None, 25, 25, 48)	13824	activa
tion_15[0][0]			

batch_normalization_13 (BatchNo	(None, 25, 25, 32)	96	conv2d
_13[0][0]			

batch_normalization_16 (BatchNo	(None, 25, 25, 48)	144	conv2d
_16[0][0]			

activation_13 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_13[0][0]			

activation_16 (Activation)	(None, 25, 25, 48)	0	batch_
normalization_16[0][0]			

conv2d_12 (Conv2D)	(None, 25, 25, 32)	10240	mixed_
5b[0][0]			

conv2d_14 (Conv2D)	(None, 25, 25, 32)	9216	activa
tion_13[0][0]			

conv2d_17 (Conv2D)	(None, 25, 25, 64)	27648	activa
tion_16[0][0]			

batch_normalization_12 (BatchNo	(None, 25, 25, 32)	96	conv2d
_12[0][0]			

batch_normalization_14 (BatchNo	(None, 25, 25, 32)	96	conv2d
_14[0][0]			

batch_normalization_17 (BatchNo	(None, 25, 25, 64)	192	conv2d
_17[0][0]			
activation_12 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_12[0][0]			
activation_14 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_14[0][0]			
activation_17 (Activation)	(None, 25, 25, 64)	0	batch_
normalization_17[0][0]			
block35_1_mixed (Concatenate)	(None, 25, 25, 128)	0	activa
tion_12[0][0]			
tion_14[0][0]			activa
tion_17[0][0]			activa
block35_1_conv (Conv2D)	(None, 25, 25, 320)	41280	block3
5_1_mixed[0][0]			
block35_1 (Lambda)	(None, 25, 25, 320)	0	mixed_
5b[0][0]			
5_1_conv[0][0]			block3
block35_1_ac (Activation)	(None, 25, 25, 320)	0	block3
5_1[0][0]			
conv2d_21 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_1_ac[0][0]			
batch_normalization_21 (BatchNo	(None, 25, 25, 32)	96	conv2d
_21[0][0]			
activation_21 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_21[0][0]			

conv2d_19 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_1_ac[0][0]			
conv2d_22 (Conv2D)	(None, 25, 25, 48)	13824	activation_21[0][0]
batch_normalization_19 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_19[0][0]
batch_normalization_22 (Batch Normalization)	(None, 25, 25, 48)	144	conv2d_22[0][0]
activation_19 (Activation)	(None, 25, 25, 32)	0	batch_normalization_19[0][0]
activation_22 (Activation)	(None, 25, 25, 48)	0	batch_normalization_22[0][0]
conv2d_18 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_1_ac[0][0]			
conv2d_20 (Conv2D)	(None, 25, 25, 32)	9216	activation_19[0][0]
conv2d_23 (Conv2D)	(None, 25, 25, 64)	27648	activation_22[0][0]
batch_normalization_18 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_18[0][0]
batch_normalization_20 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_20[0][0]
batch_normalization_23 (Batch Normalization)	(None, 25, 25, 64)	192	conv2d_23[0][0]

activation_18 (Activation) normalization_18[0][0]	(None, 25, 25, 32)	0	batch_

activation_20 (Activation) normalization_20[0][0]	(None, 25, 25, 32)	0	batch_

activation_23 (Activation) normalization_23[0][0]	(None, 25, 25, 64)	0	batch_

block35_2_mixed (Concatenate) activation_18[0][0]	(None, 25, 25, 128)	0	activa

tion_20[0][0]			activa

tion_23[0][0]			activa

block35_2_conv (Conv2D) 5_2_mixed[0][0]	(None, 25, 25, 320)	41280	block3

block35_2 (Lambda) 5_1_ac[0][0]	(None, 25, 25, 320)	0	block3

5_2_conv[0][0]			block3

block35_2_ac (Activation) 5_2[0][0]	(None, 25, 25, 320)	0	block3

conv2d_27 (Conv2D) 5_2_ac[0][0]	(None, 25, 25, 32)	10240	block3

batch_normalization_27 (BatchNo _27[0][0]	(None, 25, 25, 32)	96	conv2d

activation_27 (Activation) normalization_27[0][0]	(None, 25, 25, 32)	0	batch_

conv2d_25 (Conv2D) 5_2_ac[0][0]	(None, 25, 25, 32)	10240	block3

conv2d_28 (Conv2D)	(None, 25, 25, 48)	13824	activation_27[0][0]

batch_normalization_25 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_25[0][0]

batch_normalization_28 (Batch Normalization)	(None, 25, 25, 48)	144	conv2d_28[0][0]

activation_25 (Activation)	(None, 25, 25, 32)	0	batch_normalization_25[0][0]

activation_28 (Activation)	(None, 25, 25, 48)	0	batch_normalization_28[0][0]

conv2d_24 (Conv2D)	(None, 25, 25, 32)	10240	block3_5_2_ac[0][0]

conv2d_26 (Conv2D)	(None, 25, 25, 32)	9216	activation_25[0][0]

conv2d_29 (Conv2D)	(None, 25, 25, 64)	27648	activation_28[0][0]

batch_normalization_24 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_24[0][0]

batch_normalization_26 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_26[0][0]

batch_normalization_29 (Batch Normalization)	(None, 25, 25, 64)	192	conv2d_29[0][0]

activation_24 (Activation)	(None, 25, 25, 32)	0	batch_normalization_24[0][0]

activation_26 (Activation)	(None, 25, 25, 32)	0	batch_normalization_26[0][0]

activation_29 (Activation) normalization_29[0][0]	(None, 25, 25, 64)	0	batch_
block35_3_mixed (Concatenate) tion_24[0][0]	(None, 25, 25, 128)	0	activa
tion_26[0][0]			activa
tion_29[0][0]			activa
block35_3_conv (Conv2D) 5_3_mixed[0][0]	(None, 25, 25, 320)	41280	block3
block35_3 (Lambda) 5_2_ac[0][0]	(None, 25, 25, 320)	0	block3
5_3_conv[0][0]			block3
block35_3_ac (Activation) 5_3[0][0]	(None, 25, 25, 320)	0	block3
conv2d_33 (Conv2D) 5_3_ac[0][0]	(None, 25, 25, 32)	10240	block3
batch_normalization_33 (BatchNo _33[0][0])	(None, 25, 25, 32)	96	conv2d
activation_33 (Activation) normalization_33[0][0]	(None, 25, 25, 32)	0	batch_
conv2d_31 (Conv2D) 5_3_ac[0][0]	(None, 25, 25, 32)	10240	block3
conv2d_34 (Conv2D) tion_33[0][0]	(None, 25, 25, 48)	13824	activa
batch_normalization_31 (BatchNo _31[0][0])	(None, 25, 25, 32)	96	conv2d

batch_normalization_34 (BatchNo	(None, 25, 25, 48)	144	conv2d_34[0][0]
activation_31 (Activation)	(None, 25, 25, 32)	0	batch_normalization_31[0][0]
activation_34 (Activation)	(None, 25, 25, 48)	0	batch_normalization_34[0][0]
conv2d_30 (Conv2D)	(None, 25, 25, 32)	10240	block3_5_3_ac[0][0]
conv2d_32 (Conv2D)	(None, 25, 25, 32)	9216	activation_31[0][0]
conv2d_35 (Conv2D)	(None, 25, 25, 64)	27648	activation_34[0][0]
batch_normalization_30 (BatchNo	(None, 25, 25, 32)	96	conv2d_30[0][0]
batch_normalization_32 (BatchNo	(None, 25, 25, 32)	96	conv2d_32[0][0]
batch_normalization_35 (BatchNo	(None, 25, 25, 64)	192	conv2d_35[0][0]
activation_30 (Activation)	(None, 25, 25, 32)	0	batch_normalization_30[0][0]
activation_32 (Activation)	(None, 25, 25, 32)	0	batch_normalization_32[0][0]
activation_35 (Activation)	(None, 25, 25, 64)	0	batch_normalization_35[0][0]

block35_4_mixed (Concatenate)	(None, 25, 25, 128)	0	activation_30[0][0]
			activation_32[0][0]
			activation_35[0][0]

block35_4_conv (Conv2D)	(None, 25, 25, 320)	41280	block35_4_mixed[0][0]

block35_4 (Lambda)	(None, 25, 25, 320)	0	block35_3_ac[0][0]
			block35_4_conv[0][0]

block35_4_ac (Activation)	(None, 25, 25, 320)	0	block35_4[0][0]

conv2d_39 (Conv2D)	(None, 25, 25, 32)	10240	block35_4_ac[0][0]

batch_normalization_39 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_39[0][0]

activation_39 (Activation)	(None, 25, 25, 32)	0	batch_normalization_39[0][0]

conv2d_37 (Conv2D)	(None, 25, 25, 32)	10240	block35_4_ac[0][0]

conv2d_40 (Conv2D)	(None, 25, 25, 48)	13824	activation_39[0][0]

batch_normalization_37 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_37[0][0]

batch_normalization_40 (Batch Normalization)	(None, 25, 25, 48)	144	conv2d_40[0][0]

activation_37 (Activation) normalization_37[0][0]	(None, 25, 25, 32)	0	batch_

activation_40 (Activation) normalization_40[0][0]	(None, 25, 25, 48)	0	batch_

conv2d_36 (Conv2D) 5_4_ac[0][0]	(None, 25, 25, 32)	10240	block3

conv2d_38 (Conv2D) tion_37[0][0]	(None, 25, 25, 32)	9216	activa

conv2d_41 (Conv2D) tion_40[0][0]	(None, 25, 25, 64)	27648	activa

batch_normalization_36 (BatchNo _36[0][0]	(None, 25, 25, 32)	96	conv2d

batch_normalization_38 (BatchNo _38[0][0]	(None, 25, 25, 32)	96	conv2d

batch_normalization_41 (BatchNo _41[0][0]	(None, 25, 25, 64)	192	conv2d

activation_36 (Activation) normalization_36[0][0]	(None, 25, 25, 32)	0	batch_

activation_38 (Activation) normalization_38[0][0]	(None, 25, 25, 32)	0	batch_

activation_41 (Activation) normalization_41[0][0]	(None, 25, 25, 64)	0	batch_

block35_5_mixed (Concatenate) tion_36[0][0]	(None, 25, 25, 128)	0	activa

tion_38[0][0]			activa

tion_41[0][0]			activa

block35_5_conv (Conv2D) 5_5_mixed[0][0]	(None, 25, 25, 320)	41280	block3
block35_5 (Lambda) 5_4_ac[0][0]	(None, 25, 25, 320)	0	block3
5_5_conv[0][0]			block3
block35_5_ac (Activation) 5_5[0][0]	(None, 25, 25, 320)	0	block3
conv2d_45 (Conv2D) 5_5_ac[0][0]	(None, 25, 25, 32)	10240	block3
batch_normalization_45 (BatchNo _45[0][0])	(None, 25, 25, 32)	96	conv2d
activation_45 (Activation) normalization_45[0][0]	(None, 25, 25, 32)	0	batch_
conv2d_43 (Conv2D) 5_5_ac[0][0]	(None, 25, 25, 32)	10240	block3
conv2d_46 (Conv2D) tion_45[0][0]	(None, 25, 25, 48)	13824	activa
batch_normalization_43 (BatchNo _43[0][0])	(None, 25, 25, 32)	96	conv2d
batch_normalization_46 (BatchNo _46[0][0])	(None, 25, 25, 48)	144	conv2d
activation_43 (Activation) normalization_43[0][0]	(None, 25, 25, 32)	0	batch_
activation_46 (Activation) normalization_46[0][0]	(None, 25, 25, 48)	0	batch_

conv2d_42 (Conv2D)	(None, 25, 25, 32)	10240	block35_5_ac[0][0]
conv2d_44 (Conv2D)	(None, 25, 25, 32)	9216	activation_43[0][0]
conv2d_47 (Conv2D)	(None, 25, 25, 64)	27648	activation_46[0][0]
batch_normalization_42 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_42[0][0]
batch_normalization_44 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_44[0][0]
batch_normalization_47 (Batch Normalization)	(None, 25, 25, 64)	192	conv2d_47[0][0]
activation_42 (Activation)	(None, 25, 25, 32)	0	batch_normalization_42[0][0]
activation_44 (Activation)	(None, 25, 25, 32)	0	batch_normalization_44[0][0]
activation_47 (Activation)	(None, 25, 25, 64)	0	batch_normalization_47[0][0]
block35_6_mixed (Concatenate)	(None, 25, 25, 128)	0	activation_42[0][0]
			activation_44[0][0]
			activation_47[0][0]
block35_6_conv (Conv2D)	(None, 25, 25, 320)	41280	block35_6_mixed[0][0]

block35_6 (Lambda)	(None, 25, 25, 320)	0	block3
5_5_ac[0][0]			
5_6_conv[0][0]			block3

block35_6_ac (Activation)	(None, 25, 25, 320)	0	block3
5_6[0][0]			

conv2d_51 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_6_ac[0][0]			

batch_normalization_51 (BatchNo	(None, 25, 25, 32)	96	conv2d
_51[0][0]			

activation_51 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_51[0][0]			

conv2d_49 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_6_ac[0][0]			

conv2d_52 (Conv2D)	(None, 25, 25, 48)	13824	activa
tion_51[0][0]			

batch_normalization_49 (BatchNo	(None, 25, 25, 32)	96	conv2d
_49[0][0]			

batch_normalization_52 (BatchNo	(None, 25, 25, 48)	144	conv2d
_52[0][0]			

activation_49 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_49[0][0]			

activation_52 (Activation)	(None, 25, 25, 48)	0	batch_
normalization_52[0][0]			

conv2d_48 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_6_ac[0][0]			

conv2d_50 (Conv2D)	(None, 25, 25, 32)	9216	activation_49[0][0]
conv2d_53 (Conv2D)	(None, 25, 25, 64)	27648	activation_52[0][0]
batch_normalization_48 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_48[0][0]
batch_normalization_50 (Batch Normalization)	(None, 25, 25, 32)	96	conv2d_50[0][0]
batch_normalization_53 (Batch Normalization)	(None, 25, 25, 64)	192	conv2d_53[0][0]
activation_48 (Activation)	(None, 25, 25, 32)	0	batch_normalization_48[0][0]
activation_50 (Activation)	(None, 25, 25, 32)	0	batch_normalization_50[0][0]
activation_53 (Activation)	(None, 25, 25, 64)	0	batch_normalization_53[0][0]
block35_7_mixed (Concatenate)	(None, 25, 25, 128)	0	activation_48[0][0]
			activation_50[0][0]
			activation_53[0][0]
block35_7_conv (Conv2D)	(None, 25, 25, 320)	41280	block35_7_mixed[0][0]
block35_7 (Lambda)	(None, 25, 25, 320)	0	block35_7_conv[0][0]
			block35_6_ac[0][0]
			block35_7_conv[0][0]

block35_7_ac (Activation)	(None, 25, 25, 320)	0	block3
5_7[0][0]			

conv2d_57 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_7_ac[0][0]			

batch_normalization_57 (BatchNo	(None, 25, 25, 32)	96	conv2d
_57[0][0]			

activation_57 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_57[0][0]			

conv2d_55 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_7_ac[0][0]			

conv2d_58 (Conv2D)	(None, 25, 25, 48)	13824	activa
tion_57[0][0]			

batch_normalization_55 (BatchNo	(None, 25, 25, 32)	96	conv2d
_55[0][0]			

batch_normalization_58 (BatchNo	(None, 25, 25, 48)	144	conv2d
_58[0][0]			

activation_55 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_55[0][0]			

activation_58 (Activation)	(None, 25, 25, 48)	0	batch_
normalization_58[0][0]			

conv2d_54 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_7_ac[0][0]			

conv2d_56 (Conv2D)	(None, 25, 25, 32)	9216	activa
tion_55[0][0]			

conv2d_59 (Conv2D)	(None, 25, 25, 64)	27648	activa
tion_58[0][0]			

batch_normalization_54 (BatchNo	(None, 25, 25, 32)	96	conv2d
_54[0][0]			
batch_normalization_56 (BatchNo	(None, 25, 25, 32)	96	conv2d
_56[0][0]			
batch_normalization_59 (BatchNo	(None, 25, 25, 64)	192	conv2d
_59[0][0]			
activation_54 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_54[0][0]			
activation_56 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_56[0][0]			
activation_59 (Activation)	(None, 25, 25, 64)	0	batch_
normalization_59[0][0]			
block35_8_mixed (Concatenate)	(None, 25, 25, 128)	0	activa
tion_54[0][0]			
			activa
tion_56[0][0]			
			activa
tion_59[0][0]			
block35_8_conv (Conv2D)	(None, 25, 25, 320)	41280	block3
5_8_mixed[0][0]			
block35_8 (Lambda)	(None, 25, 25, 320)	0	block3
5_7_ac[0][0]			
			block3
5_8_conv[0][0]			
block35_8_ac (Activation)	(None, 25, 25, 320)	0	block3
5_8[0][0]			
conv2d_63 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_8_ac[0][0]			

batch_normalization_63 (BatchNo	(None, 25, 25, 32)	96	conv2d_63[0][0]
activation_63 (Activation)	(None, 25, 25, 32)	0	batch_normalization_63[0][0]
conv2d_61 (Conv2D)	(None, 25, 25, 32)	10240	block3_5_8_ac[0][0]
conv2d_64 (Conv2D)	(None, 25, 25, 48)	13824	activation_63[0][0]
batch_normalization_61 (BatchNo	(None, 25, 25, 32)	96	conv2d_61[0][0]
batch_normalization_64 (BatchNo	(None, 25, 25, 48)	144	conv2d_64[0][0]
activation_61 (Activation)	(None, 25, 25, 32)	0	batch_normalization_61[0][0]
activation_64 (Activation)	(None, 25, 25, 48)	0	batch_normalization_64[0][0]
conv2d_60 (Conv2D)	(None, 25, 25, 32)	10240	block3_5_8_ac[0][0]
conv2d_62 (Conv2D)	(None, 25, 25, 32)	9216	activation_61[0][0]
conv2d_65 (Conv2D)	(None, 25, 25, 64)	27648	activation_64[0][0]
batch_normalization_60 (BatchNo	(None, 25, 25, 32)	96	conv2d_60[0][0]

batch_normalization_62 (BatchNo	(None, 25, 25, 32)	96	conv2d
_62[0][0]			

batch_normalization_65 (BatchNo	(None, 25, 25, 64)	192	conv2d
_65[0][0]			

activation_60 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_60[0][0]			

activation_62 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_62[0][0]			

activation_65 (Activation)	(None, 25, 25, 64)	0	batch_
normalization_65[0][0]			

block35_9_mixed (Concatenate)	(None, 25, 25, 128)	0	activa
tion_60[0][0]			
			activa
tion_62[0][0]			
			activa
tion_65[0][0]			

block35_9_conv (Conv2D)	(None, 25, 25, 320)	41280	block3
5_9_mixed[0][0]			

block35_9 (Lambda)	(None, 25, 25, 320)	0	block3
5_8_ac[0][0]			
			block3
5_9_conv[0][0]			

block35_9_ac (Activation)	(None, 25, 25, 320)	0	block3
5_9[0][0]			

conv2d_69 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_9_ac[0][0]			

batch_normalization_69 (BatchNo	(None, 25, 25, 32)	96	conv2d
_69[0][0]			

activation_69 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_69[0][0]			

conv2d_67 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_9_ac[0][0]			

conv2d_70 (Conv2D)	(None, 25, 25, 48)	13824	activa
tion_69[0][0]			

batch_normalization_67 (BatchNo	(None, 25, 25, 32)	96	conv2d
_67[0][0]			

batch_normalization_70 (BatchNo	(None, 25, 25, 48)	144	conv2d
_70[0][0]			

activation_67 (Activation)	(None, 25, 25, 32)	0	batch_
normalization_67[0][0]			

activation_70 (Activation)	(None, 25, 25, 48)	0	batch_
normalization_70[0][0]			

conv2d_66 (Conv2D)	(None, 25, 25, 32)	10240	block3
5_9_ac[0][0]			

conv2d_68 (Conv2D)	(None, 25, 25, 32)	9216	activa
tion_67[0][0]			

conv2d_71 (Conv2D)	(None, 25, 25, 64)	27648	activa
tion_70[0][0]			

batch_normalization_66 (BatchNo	(None, 25, 25, 32)	96	conv2d
_66[0][0]			

batch_normalization_68 (BatchNo	(None, 25, 25, 32)	96	conv2d
_68[0][0]			

batch_normalization_71 (BatchNo	(None, 25, 25, 64)	192	conv2d
_71[0][0]			

activation_66 (Activation) normalization_66[0][0]	(None, 25, 25, 32)	0	batch_
activation_68 (Activation) normalization_68[0][0]	(None, 25, 25, 32)	0	batch_
activation_71 (Activation) normalization_71[0][0]	(None, 25, 25, 64)	0	batch_
block35_10_mixed (Concatenate) activation_66[0][0] activation_68[0][0] activation_71[0][0]	(None, 25, 25, 128)	0	activa activa activa
block35_10_conv (Conv2D) 5_10_mixed[0][0]	(None, 25, 25, 320)	41280	block3
block35_10 (Lambda) 5_9_ac[0][0] 5_10_conv[0][0]	(None, 25, 25, 320)	0	block3 block3
block35_10_ac (Activation) 5_10[0][0]	(None, 25, 25, 320)	0	block3
conv2d_73 (Conv2D) 5_10_ac[0][0]	(None, 25, 25, 256)	81920	block3
batch_normalization_73 (BatchNo _73[0][0])	(None, 25, 25, 256)	768	conv2d
activation_73 (Activation) normalization_73[0][0]	(None, 25, 25, 256)	0	batch_
conv2d_74 (Conv2D) activation_73[0][0]	(None, 25, 25, 256)	589824	activa

batch_normalization_74 (BatchNo	(None, 25, 25, 256)	768	conv2d_74[0][0]
activation_74 (Activation)	(None, 25, 25, 256)	0	batch_normalization_74[0][0]
conv2d_72 (Conv2D)	(None, 12, 12, 384)	1105920	block3_5_10_ac[0][0]
conv2d_75 (Conv2D)	(None, 12, 12, 384)	884736	activation_74[0][0]
batch_normalization_72 (BatchNo	(None, 12, 12, 384)	1152	conv2d_72[0][0]
batch_normalization_75 (BatchNo	(None, 12, 12, 384)	1152	conv2d_75[0][0]
activation_72 (Activation)	(None, 12, 12, 384)	0	batch_normalization_72[0][0]
activation_75 (Activation)	(None, 12, 12, 384)	0	batch_normalization_75[0][0]
max_pooling2d_2 (MaxPooling2D)	(None, 12, 12, 320)	0	block3_5_10_ac[0][0]
mixed_6a (Concatenate)	(None, 12, 12, 1088)	0	activation_72[0][0]
			activation_75[0][0]
			max_pooling2d_2[0][0]
conv2d_77 (Conv2D)	(None, 12, 12, 128)	139264	mixed_6a[0][0]

batch_normalization_77 (BatchNo	(None, 12, 12, 128)	384	conv2d
_77[0][0]			

activation_77 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_77[0][0]			

conv2d_78 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_77[0][0]			

batch_normalization_78 (BatchNo	(None, 12, 12, 160)	480	conv2d
_78[0][0]			

activation_78 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_78[0][0]			

conv2d_76 (Conv2D)	(None, 12, 12, 192)	208896	mixed_
6a[0][0]			

conv2d_79 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_78[0][0]			

batch_normalization_76 (BatchNo	(None, 12, 12, 192)	576	conv2d
_76[0][0]			

batch_normalization_79 (BatchNo	(None, 12, 12, 192)	576	conv2d
_79[0][0]			

activation_76 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_76[0][0]			

activation_79 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_79[0][0]			

block17_1_mixed (Concatenate)	(None, 12, 12, 384)	0	activa
tion_76[0][0]			

tion_79[0][0]			activa

block17_1_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_1_mixed[0][0]			

block17_1 (Lambda)	(None, 12, 12, 1088)	0	mixed_
6a[0][0]			block1
7_1_conv[0][0]			

block17_1_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_1[0][0]			

conv2d_81 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_1_ac[0][0]			

batch_normalization_81 (BatchNo	(None, 12, 12, 128)	384	conv2d
_81[0][0]			

activation_81 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_81[0][0]			

conv2d_82 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_81[0][0]			

batch_normalization_82 (BatchNo	(None, 12, 12, 160)	480	conv2d
_82[0][0]			

activation_82 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_82[0][0]			

conv2d_80 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_1_ac[0][0]			

conv2d_83 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_82[0][0]			

batch_normalization_80 (BatchNo	(None, 12, 12, 192)	576	conv2d
_80[0][0]			

batch_normalization_83 (BatchNo	(None, 12, 12, 192)	576	conv2d
_83[0][0]			

activation_80 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_80[0][0]			

activation_83 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_83[0][0]			

block17_2_mixed (Concatenate)	(None, 12, 12, 384)	0	activa
tion_80[0][0]			

			activa
tion_83[0][0]			

block17_2_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_2_mixed[0][0]			

block17_2 (Lambda)	(None, 12, 12, 1088)	0	block1
7_1_ac[0][0]			

			block1
7_2_conv[0][0]			

block17_2_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_2[0][0]			

conv2d_85 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_2_ac[0][0]			

batch_normalization_85 (BatchNo	(None, 12, 12, 128)	384	conv2d
_85[0][0]			

activation_85 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_85[0][0]			

conv2d_86 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_85[0][0]			

batch_normalization_86 (BatchNo	(None, 12, 12, 160)	480	conv2d
_86[0][0]			

activation_86 (Activation) normalization_86[0][0]	(None, 12, 12, 160)	0	batch_
conv2d_84 (Conv2D) 7_2_ac[0][0]	(None, 12, 12, 192)	208896	block1
conv2d_87 (Conv2D) tion_86[0][0]	(None, 12, 12, 192)	215040	activa
batch_normalization_84 (BatchNo _84[0][0]	(None, 12, 12, 192)	576	conv2d
batch_normalization_87 (BatchNo _87[0][0]	(None, 12, 12, 192)	576	conv2d
activation_84 (Activation) normalization_84[0][0]	(None, 12, 12, 192)	0	batch_
activation_87 (Activation) normalization_87[0][0]	(None, 12, 12, 192)	0	batch_
block17_3_mixed (Concatenate) tion_84[0][0]	(None, 12, 12, 384)	0	activa
			activa
block17_3_conv (Conv2D) 7_3_mixed[0][0]	(None, 12, 12, 1088)	418880	block1
block17_3 (Lambda) 7_2_ac[0][0]	(None, 12, 12, 1088)	0	block1
7_3_conv[0][0]			block1
block17_3_ac (Activation) 7_3[0][0]	(None, 12, 12, 1088)	0	block1

conv2d_89 (Conv2D)	(None, 12, 12, 128)	139264	block17_3_ac[0][0]

batch_normalization_89 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_89[0][0]

activation_89 (Activation)	(None, 12, 12, 128)	0	batch_normalization_89[0][0]

conv2d_90 (Conv2D)	(None, 12, 12, 160)	143360	activation_89[0][0]

batch_normalization_90 (Batch Normalization)	(None, 12, 12, 160)	480	conv2d_90[0][0]

activation_90 (Activation)	(None, 12, 12, 160)	0	batch_normalization_90[0][0]

conv2d_88 (Conv2D)	(None, 12, 12, 192)	208896	block17_3_ac[0][0]

conv2d_91 (Conv2D)	(None, 12, 12, 192)	215040	activation_90[0][0]

batch_normalization_88 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_88[0][0]

batch_normalization_91 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_91[0][0]

activation_88 (Activation)	(None, 12, 12, 192)	0	batch_normalization_88[0][0]

activation_91 (Activation)	(None, 12, 12, 192)	0	batch_normalization_91[0][0]

block17_4_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_88[0][0]

tion_91[0][0]			activa

block17_4_conv (Conv2D) 7_4_mixed[0][0]	(None, 12, 12, 1088)	418880	block1

block17_4 (Lambda) 7_3_ac[0][0]	(None, 12, 12, 1088)	0	block1
7_4_conv[0][0]			block1

block17_4_ac (Activation) 7_4[0][0]	(None, 12, 12, 1088)	0	block1

conv2d_93 (Conv2D) 7_4_ac[0][0]	(None, 12, 12, 128)	139264	block1

batch_normalization_93 (BatchNo _93[0][0])	(None, 12, 12, 128)	384	conv2d

activation_93 (Activation) normalization_93[0][0]	(None, 12, 12, 128)	0	batch_

conv2d_94 (Conv2D) tion_93[0][0]	(None, 12, 12, 160)	143360	activa

batch_normalization_94 (BatchNo _94[0][0])	(None, 12, 12, 160)	480	conv2d

activation_94 (Activation) normalization_94[0][0]	(None, 12, 12, 160)	0	batch_

conv2d_92 (Conv2D) 7_4_ac[0][0]	(None, 12, 12, 192)	208896	block1

conv2d_95 (Conv2D) tion_94[0][0]	(None, 12, 12, 192)	215040	activa

batch_normalization_92 (BatchNo	(None, 12, 12, 192)	576	conv2d
_92[0][0]			

batch_normalization_95 (BatchNo	(None, 12, 12, 192)	576	conv2d
_95[0][0]			

activation_92 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_92[0][0]			

activation_95 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_95[0][0]			

block17_5_mixed (Concatenate)	(None, 12, 12, 384)	0	activa
tion_92[0][0]			
			activa
tion_95[0][0]			

block17_5_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_5_mixed[0][0]			

block17_5 (Lambda)	(None, 12, 12, 1088)	0	block1
7_4_ac[0][0]			
			block1
7_5_conv[0][0]			

block17_5_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_5[0][0]			

conv2d_97 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_5_ac[0][0]			

batch_normalization_97 (BatchNo	(None, 12, 12, 128)	384	conv2d
_97[0][0]			

activation_97 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_97[0][0]			

conv2d_98 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_97[0][0]			

batch_normalization_98 (BatchNo	(None, 12, 12, 160)	480	conv2d
_98[0][0]			
activation_98 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_98[0][0]			
conv2d_96 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_5_ac[0][0]			
conv2d_99 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_98[0][0]			
batch_normalization_96 (BatchNo	(None, 12, 12, 192)	576	conv2d
_96[0][0]			
batch_normalization_99 (BatchNo	(None, 12, 12, 192)	576	conv2d
_99[0][0]			
activation_96 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_96[0][0]			
activation_99 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_99[0][0]			
block17_6_mixed (Concatenate)	(None, 12, 12, 384)	0	activa
tion_96[0][0]			
			activa
tion_99[0][0]			
block17_6_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_6_mixed[0][0]			
block17_6 (Lambda)	(None, 12, 12, 1088)	0	block1
7_5_ac[0][0]			
			block1
7_6_conv[0][0]			

block17_6_ac (Activation)	(None, 12, 12, 1088)	0	block17_6[0][0]

conv2d_101 (Conv2D)	(None, 12, 12, 128)	139264	block17_6_ac[0][0]

batch_normalization_101 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_101[0][0]

activation_101 (Activation)	(None, 12, 12, 128)	0	batch_normalization_101[0][0]

conv2d_102 (Conv2D)	(None, 12, 12, 160)	143360	activation_101[0][0]

batch_normalization_102 (Batch Normalization)	(None, 12, 12, 160)	480	conv2d_102[0][0]

activation_102 (Activation)	(None, 12, 12, 160)	0	batch_normalization_102[0][0]

conv2d_100 (Conv2D)	(None, 12, 12, 192)	208896	block17_6_ac[0][0]

conv2d_103 (Conv2D)	(None, 12, 12, 192)	215040	activation_102[0][0]

batch_normalization_100 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_100[0][0]

batch_normalization_103 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_103[0][0]

activation_100 (Activation)	(None, 12, 12, 192)	0	batch_normalization_100[0][0]

activation_103 (Activation)	(None, 12, 12, 192)	0	batch_normalization_103[0][0]

block17_7_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_100[0][0]
			activation_103[0][0]
block17_7_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_7_mixed[0][0]
block17_7 (Lambda)	(None, 12, 12, 1088)	0	block17_6_ac[0][0]
			block17_7_conv[0][0]
block17_7_ac (Activation)	(None, 12, 12, 1088)	0	block17_7[0][0]
conv2d_105 (Conv2D)	(None, 12, 12, 128)	139264	block17_7_ac[0][0]
batch_normalization_105 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_105[0][0]
activation_105 (Activation)	(None, 12, 12, 128)	0	batch_normalization_105[0][0]
conv2d_106 (Conv2D)	(None, 12, 12, 160)	143360	activation_105[0][0]
batch_normalization_106 (Batch Normalization)	(None, 12, 12, 160)	480	conv2d_106[0][0]
activation_106 (Activation)	(None, 12, 12, 160)	0	batch_normalization_106[0][0]
conv2d_104 (Conv2D)	(None, 12, 12, 192)	208896	block17_7_ac[0][0]

conv2d_107 (Conv2D)	(None, 12, 12, 192)	215040	activation_106[0][0]

batch_normalization_104 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_104[0][0]

batch_normalization_107 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_107[0][0]

activation_104 (Activation)	(None, 12, 12, 192)	0	batch_normalization_104[0][0]

activation_107 (Activation)	(None, 12, 12, 192)	0	batch_normalization_107[0][0]

block17_8_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_104[0][0]
			activation_107[0][0]

block17_8_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_8_mixed[0][0]

block17_8 (Lambda)	(None, 12, 12, 1088)	0	block17_8_conv[0][0]
			block17_7_ac[0][0]

block17_8_ac (Activation)	(None, 12, 12, 1088)	0	block17_8_conv[0][0]

conv2d_109 (Conv2D)	(None, 12, 12, 128)	139264	block17_8_ac[0][0]

batch_normalization_109 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_109[0][0]

activation_109 (Activation)	(None, 12, 12, 128)	0	batch_normalization_109[0][0]

conv2d_110 (Conv2D)	(None, 12, 12, 160)	143360	activation_109[0][0]
batch_normalization_110 (Batch Normalization)	(None, 12, 12, 160)	480	conv2d_110[0][0]
activation_110 (Activation)	(None, 12, 12, 160)	0	batch_normalization_110[0][0]
conv2d_108 (Conv2D)	(None, 12, 12, 192)	208896	block17_8_ac[0][0]
conv2d_111 (Conv2D)	(None, 12, 12, 192)	215040	activation_110[0][0]
batch_normalization_108 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_108[0][0]
batch_normalization_111 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_111[0][0]
activation_108 (Activation)	(None, 12, 12, 192)	0	batch_normalization_108[0][0]
activation_111 (Activation)	(None, 12, 12, 192)	0	batch_normalization_111[0][0]
block17_9_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_108[0][0]
			activation_111[0][0]
block17_9_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_9_mixed[0][0]
block17_9 (Lambda)	(None, 12, 12, 1088)	0	block17_9_conv[0][0]

			block1
7_9_conv[0][0]			

block17_9_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_9[0][0]			

conv2d_113 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_9_ac[0][0]			

batch_normalization_113 (BatchN	(None, 12, 12, 128)	384	conv2d
_113[0][0]			

activation_113 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_113[0][0]			

conv2d_114 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_113[0][0]			

batch_normalization_114 (BatchN	(None, 12, 12, 160)	480	conv2d
_114[0][0]			

activation_114 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_114[0][0]			

conv2d_112 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_9_ac[0][0]			

conv2d_115 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_114[0][0]			

batch_normalization_112 (BatchN	(None, 12, 12, 192)	576	conv2d
_112[0][0]			

batch_normalization_115 (BatchN	(None, 12, 12, 192)	576	conv2d
_115[0][0]			

activation_112 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_112[0][0]			

activation_115 (Activation) normalization_115[0][0]	(None, 12, 12, 192)	0	batch_
block17_10_mixed (Concatenate) tion_112[0][0]	(None, 12, 12, 384)	0	activa
			activa
block17_10_conv (Conv2D) 7_10_mixed[0][0]	(None, 12, 12, 1088)	418880	block1
block17_10 (Lambda) 7_9_ac[0][0]	(None, 12, 12, 1088)	0	block1
7_10_conv[0][0]			block1
block17_10_ac (Activation) 7_10[0][0]	(None, 12, 12, 1088)	0	block1
conv2d_117 (Conv2D) 7_10_ac[0][0]	(None, 12, 12, 128)	139264	block1
batch_normalization_117 (BatchN _117[0][0]	(None, 12, 12, 128)	384	conv2d
activation_117 (Activation) normalization_117[0][0]	(None, 12, 12, 128)	0	batch_
conv2d_118 (Conv2D) tion_117[0][0]	(None, 12, 12, 160)	143360	activa
batch_normalization_118 (BatchN _118[0][0]	(None, 12, 12, 160)	480	conv2d
activation_118 (Activation) normalization_118[0][0]	(None, 12, 12, 160)	0	batch_

conv2d_116 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_10_ac[0][0]			

conv2d_119 (Conv2D)	(None, 12, 12, 192)	215040	activation_118[0][0]

batch_normalization_116 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_116[0][0]

batch_normalization_119 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_119[0][0]

activation_116 (Activation)	(None, 12, 12, 192)	0	batch_normalization_116[0][0]

activation_119 (Activation)	(None, 12, 12, 192)	0	batch_normalization_119[0][0]

block17_11_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_116[0][0]

block17_11_conv (Conv2D)	(None, 12, 12, 1088)	418880	activation_119[0][0]

block17_11 (Lambda)	(None, 12, 12, 1088)	0	block17_11_conv[0][0]

block17_11_ac (Activation)	(None, 12, 12, 1088)	0	block17_11[0][0]

conv2d_121 (Conv2D)	(None, 12, 12, 128)	139264	block17_11_ac[0][0]

batch_normalization_121 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_121[0][0]

activation_121 (Activation) normalization_121[0][0]	(None, 12, 12, 128)	0	batch_
conv2d_122 (Conv2D) tion_121[0][0]	(None, 12, 12, 160)	143360	activa
batch_normalization_122 (BatchN _122[0][0]	(None, 12, 12, 160)	480	conv2d
activation_122 (Activation) normalization_122[0][0]	(None, 12, 12, 160)	0	batch_
conv2d_120 (Conv2D) 7_11_ac[0][0]	(None, 12, 12, 192)	208896	block1
conv2d_123 (Conv2D) tion_122[0][0]	(None, 12, 12, 192)	215040	activa
batch_normalization_120 (BatchN _120[0][0]	(None, 12, 12, 192)	576	conv2d
batch_normalization_123 (BatchN _123[0][0]	(None, 12, 12, 192)	576	conv2d
activation_120 (Activation) normalization_120[0][0]	(None, 12, 12, 192)	0	batch_
activation_123 (Activation) normalization_123[0][0]	(None, 12, 12, 192)	0	batch_
block17_12_mixed (Concatenate) tion_120[0][0]	(None, 12, 12, 384)	0	activa
tion_123[0][0]			activa
block17_12_conv (Conv2D) 7_12_mixed[0][0]	(None, 12, 12, 1088)	418880	block1

block17_12 (Lambda)	(None, 12, 12, 1088)	0	block1
7_11_ac[0][0]			block1
7_12_conv[0][0]			
block17_12_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_12[0][0]			
conv2d_125 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_12_ac[0][0]			
batch_normalization_125 (BatchN	(None, 12, 12, 128)	384	conv2d
_125[0][0]			
activation_125 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_125[0][0]			
conv2d_126 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_125[0][0]			
batch_normalization_126 (BatchN	(None, 12, 12, 160)	480	conv2d
_126[0][0]			
activation_126 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_126[0][0]			
conv2d_124 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_12_ac[0][0]			
conv2d_127 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_126[0][0]			
batch_normalization_124 (BatchN	(None, 12, 12, 192)	576	conv2d
_124[0][0]			
batch_normalization_127 (BatchN	(None, 12, 12, 192)	576	conv2d
_127[0][0]			

activation_124 (Activation) normalization_124[0][0]	(None, 12, 12, 192)	0	batch_
activation_127 (Activation) normalization_127[0][0]	(None, 12, 12, 192)	0	batch_
block17_13_mixed (Concatenate) tion_124[0][0]	(None, 12, 12, 384)	0	activa activa
block17_13_conv (Conv2D) 7_13_mixed[0][0]	(None, 12, 12, 1088)	418880	block1
block17_13 (Lambda) 7_12_ac[0][0]	(None, 12, 12, 1088)	0	block1 block1
block17_13_ac (Activation) 7_13[0][0]	(None, 12, 12, 1088)	0	block1
conv2d_129 (Conv2D) 7_13_ac[0][0]	(None, 12, 12, 128)	139264	block1
batch_normalization_129 (BatchN _129[0][0]	(None, 12, 12, 128)	384	conv2d
activation_129 (Activation) normalization_129[0][0]	(None, 12, 12, 128)	0	batch_
conv2d_130 (Conv2D) tion_129[0][0]	(None, 12, 12, 160)	143360	activa
batch_normalization_130 (BatchN _130[0][0]	(None, 12, 12, 160)	480	conv2d

activation_130 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_130[0][0]			

conv2d_128 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_13_ac[0][0]			

conv2d_131 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_130[0][0]			

batch_normalization_128 (BatchN	(None, 12, 12, 192)	576	conv2d
_128[0][0]			

batch_normalization_131 (BatchN	(None, 12, 12, 192)	576	conv2d
_131[0][0]			

activation_128 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_128[0][0]			

activation_131 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_131[0][0]			

block17_14_mixed (Concatenate)	(None, 12, 12, 384)	0	activa
tion_128[0][0]			
			activa
tion_131[0][0]			

block17_14_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_14_mixed[0][0]			

block17_14 (Lambda)	(None, 12, 12, 1088)	0	block1
7_13_ac[0][0]			
			block1
7_14_conv[0][0]			

block17_14_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_14[0][0]			

conv2d_133 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_14_ac[0][0]			

batch_normalization_133 (BatchN	(None, 12, 12, 128)	384	conv2d_133[0][0]
activation_133 (Activation)	(None, 12, 12, 128)	0	batch_normalization_133[0][0]
conv2d_134 (Conv2D)	(None, 12, 12, 160)	143360	activation_133[0][0]
batch_normalization_134 (BatchN	(None, 12, 12, 160)	480	conv2d_134[0][0]
activation_134 (Activation)	(None, 12, 12, 160)	0	batch_normalization_134[0][0]
conv2d_132 (Conv2D)	(None, 12, 12, 192)	208896	block17_14_ac[0][0]
conv2d_135 (Conv2D)	(None, 12, 12, 192)	215040	activation_134[0][0]
batch_normalization_132 (BatchN	(None, 12, 12, 192)	576	conv2d_132[0][0]
batch_normalization_135 (BatchN	(None, 12, 12, 192)	576	conv2d_135[0][0]
activation_132 (Activation)	(None, 12, 12, 192)	0	batch_normalization_132[0][0]
activation_135 (Activation)	(None, 12, 12, 192)	0	batch_normalization_135[0][0]
block17_15_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_132[0][0]
			activation_135[0][0]

block17_15_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_15_mixed[0][0]			
block17_15 (Lambda)	(None, 12, 12, 1088)	0	block1
7_14_ac[0][0]			block1
7_15_conv[0][0]			
block17_15_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_15[0][0]			
conv2d_137 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_15_ac[0][0]			
batch_normalization_137 (BatchN	(None, 12, 12, 128)	384	conv2d
_137[0][0]			
activation_137 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_137[0][0]			
conv2d_138 (Conv2D)	(None, 12, 12, 160)	143360	activa
tion_137[0][0]			
batch_normalization_138 (BatchN	(None, 12, 12, 160)	480	conv2d
_138[0][0]			
activation_138 (Activation)	(None, 12, 12, 160)	0	batch_
normalization_138[0][0]			
conv2d_136 (Conv2D)	(None, 12, 12, 192)	208896	block1
7_15_ac[0][0]			
conv2d_139 (Conv2D)	(None, 12, 12, 192)	215040	activa
tion_138[0][0]			
batch_normalization_136 (BatchN	(None, 12, 12, 192)	576	conv2d
_136[0][0]			

batch_normalization_139 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_139[0][0]
activation_136 (Activation)	(None, 12, 12, 192)	0	batch_normalization_136[0][0]
activation_139 (Activation)	(None, 12, 12, 192)	0	batch_normalization_139[0][0]
block17_16_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_136[0][0] activation_139[0][0]
block17_16_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_16_mixed[0][0]
block17_16 (Lambda)	(None, 12, 12, 1088)	0	block17_16_conv[0][0]
block17_15_ac (Activation)	(None, 12, 12, 1088)	0	block17_16[0][0]
conv2d_141 (Conv2D)	(None, 12, 12, 128)	139264	block17_15_ac[0][0]
batch_normalization_141 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_141[0][0]
activation_141 (Activation)	(None, 12, 12, 128)	0	batch_normalization_141[0][0]
conv2d_142 (Conv2D)	(None, 12, 12, 160)	143360	activation_141[0][0]

batch_normalization_142 (BatchN	(None, 12, 12, 160)	480	conv2d_142[0][0]

activation_142 (Activation)	(None, 12, 12, 160)	0	batch_normalization_142[0][0]

conv2d_140 (Conv2D)	(None, 12, 12, 192)	208896	block17_16_ac[0][0]

conv2d_143 (Conv2D)	(None, 12, 12, 192)	215040	activation_142[0][0]

batch_normalization_140 (BatchN	(None, 12, 12, 192)	576	conv2d_140[0][0]

batch_normalization_143 (BatchN	(None, 12, 12, 192)	576	conv2d_143[0][0]

activation_140 (Activation)	(None, 12, 12, 192)	0	batch_normalization_140[0][0]

activation_143 (Activation)	(None, 12, 12, 192)	0	batch_normalization_143[0][0]

block17_17_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_140[0][0]

			activation_143[0][0]

block17_17_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_17_mixed[0][0]

block17_17 (Lambda)	(None, 12, 12, 1088)	0	block17_16_ac[0][0]

			block17_17_conv[0][0]

block17_17_ac (Activation)	(None, 12, 12, 1088)	0	block17_17[0][0]

conv2d_145 (Conv2D)	(None, 12, 12, 128)	139264	block1 7_17_ac[0][0]
batch_normalization_145 (BatchN	(None, 12, 12, 128)	384	conv2d _145[0][0]
activation_145 (Activation)	(None, 12, 12, 128)	0	batch_ normalization_145[0][0]
conv2d_146 (Conv2D)	(None, 12, 12, 160)	143360	activa tion_145[0][0]
batch_normalization_146 (BatchN	(None, 12, 12, 160)	480	conv2d _146[0][0]
activation_146 (Activation)	(None, 12, 12, 160)	0	batch_ normalization_146[0][0]
conv2d_144 (Conv2D)	(None, 12, 12, 192)	208896	block1 7_17_ac[0][0]
conv2d_147 (Conv2D)	(None, 12, 12, 192)	215040	activa tion_146[0][0]
batch_normalization_144 (BatchN	(None, 12, 12, 192)	576	conv2d _144[0][0]
batch_normalization_147 (BatchN	(None, 12, 12, 192)	576	conv2d _147[0][0]
activation_144 (Activation)	(None, 12, 12, 192)	0	batch_ normalization_144[0][0]
activation_147 (Activation)	(None, 12, 12, 192)	0	batch_ normalization_147[0][0]

block17_18_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_144[0][0]
			activation_147[0][0]

block17_18_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_18_mixed[0][0]

block17_18 (Lambda)	(None, 12, 12, 1088)	0	block17_17_ac[0][0]
			block17_18_conv[0][0]

block17_18_ac (Activation)	(None, 12, 12, 1088)	0	block17_18[0][0]

conv2d_149 (Conv2D)	(None, 12, 12, 128)	139264	block17_18_ac[0][0]

batch_normalization_149 (Batch Normalization)	(None, 12, 12, 128)	384	conv2d_149[0][0]

activation_149 (Activation)	(None, 12, 12, 128)	0	batch_normalization_149[0][0]

conv2d_150 (Conv2D)	(None, 12, 12, 160)	143360	activation_149[0][0]

batch_normalization_150 (Batch Normalization)	(None, 12, 12, 160)	480	conv2d_150[0][0]

activation_150 (Activation)	(None, 12, 12, 160)	0	batch_normalization_150[0][0]

conv2d_148 (Conv2D)	(None, 12, 12, 192)	208896	block17_18_ac[0][0]

conv2d_151 (Conv2D)	(None, 12, 12, 192)	215040	activation_150[0][0]

batch_normalization_148 (BatchN	(None, 12, 12, 192)	576	conv2d
_148[0][0]			
batch_normalization_151 (BatchN	(None, 12, 12, 192)	576	conv2d
_151[0][0]			
activation_148 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_148[0][0]			
activation_151 (Activation)	(None, 12, 12, 192)	0	batch_
normalization_151[0][0]			
block17_19_mixed (Concatenate)	(None, 12, 12, 384)	0	activa
tion_148[0][0]			
			activa
tion_151[0][0]			
block17_19_conv (Conv2D)	(None, 12, 12, 1088)	418880	block1
7_19_mixed[0][0]			
block17_19 (Lambda)	(None, 12, 12, 1088)	0	block1
7_18_ac[0][0]			
			block1
7_19_conv[0][0]			
block17_19_ac (Activation)	(None, 12, 12, 1088)	0	block1
7_19[0][0]			
conv2d_153 (Conv2D)	(None, 12, 12, 128)	139264	block1
7_19_ac[0][0]			
batch_normalization_153 (BatchN	(None, 12, 12, 128)	384	conv2d
_153[0][0]			
activation_153 (Activation)	(None, 12, 12, 128)	0	batch_
normalization_153[0][0]			

conv2d_154 (Conv2D)	(None, 12, 12, 160)	143360	activation_153[0][0]

batch_normalization_154 (Batch Normalization)	(None, 12, 12, 160)	480	conv2d_154[0][0]

activation_154 (Activation)	(None, 12, 12, 160)	0	batch_normalization_154[0][0]

conv2d_152 (Conv2D)	(None, 12, 12, 192)	208896	block17_19_ac[0][0]

conv2d_155 (Conv2D)	(None, 12, 12, 192)	215040	activation_154[0][0]

batch_normalization_152 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_152[0][0]

batch_normalization_155 (Batch Normalization)	(None, 12, 12, 192)	576	conv2d_155[0][0]

activation_152 (Activation)	(None, 12, 12, 192)	0	batch_normalization_152[0][0]

activation_155 (Activation)	(None, 12, 12, 192)	0	batch_normalization_155[0][0]

block17_20_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_152[0][0]

block17_20_mixed (Concatenate)	(None, 12, 12, 384)	0	activation_155[0][0]

block17_20_conv (Conv2D)	(None, 12, 12, 1088)	418880	block17_20_mixed[0][0]

block17_20 (Lambda)	(None, 12, 12, 1088)	0	block17_19_ac[0][0]

block17_20_conv (Conv2D)	(None, 12, 12, 1088)	0	block17_20_conv[0][0]

block17_20_ac (Activation)	(None, 12, 12, 1088)	0	block17_20[0][0]
conv2d_160 (Conv2D)	(None, 12, 12, 256)	278528	block17_20_ac[0][0]
batch_normalization_160 (Batch Normalization)	(None, 12, 12, 256)	768	conv2d_160[0][0]
activation_160 (Activation)	(None, 12, 12, 256)	0	batch_normalization_160[0][0]
conv2d_156 (Conv2D)	(None, 12, 12, 256)	278528	block17_20_ac[0][0]
conv2d_158 (Conv2D)	(None, 12, 12, 256)	278528	block17_20_ac[0][0]
conv2d_161 (Conv2D)	(None, 12, 12, 288)	663552	activation_160[0][0]
batch_normalization_156 (Batch Normalization)	(None, 12, 12, 256)	768	conv2d_156[0][0]
batch_normalization_158 (Batch Normalization)	(None, 12, 12, 256)	768	conv2d_158[0][0]
batch_normalization_161 (Batch Normalization)	(None, 12, 12, 288)	864	conv2d_161[0][0]
activation_156 (Activation)	(None, 12, 12, 256)	0	batch_normalization_156[0][0]
activation_158 (Activation)	(None, 12, 12, 256)	0	batch_normalization_158[0][0]

activation_161 (Activation)	(None, 12, 12, 288)	0	batch_
normalization_161[0][0]			

conv2d_157 (Conv2D)	(None, 5, 5, 384)	884736	activa
tion_156[0][0]			

conv2d_159 (Conv2D)	(None, 5, 5, 288)	663552	activa
tion_158[0][0]			

conv2d_162 (Conv2D)	(None, 5, 5, 320)	829440	activa
tion_161[0][0]			

batch_normalization_157 (BatchN	(None, 5, 5, 384)	1152	conv2d
_157[0][0]			

batch_normalization_159 (BatchN	(None, 5, 5, 288)	864	conv2d
_159[0][0]			

batch_normalization_162 (BatchN	(None, 5, 5, 320)	960	conv2d
_162[0][0]			

activation_157 (Activation)	(None, 5, 5, 384)	0	batch_
normalization_157[0][0]			

activation_159 (Activation)	(None, 5, 5, 288)	0	batch_
normalization_159[0][0]			

activation_162 (Activation)	(None, 5, 5, 320)	0	batch_
normalization_162[0][0]			

max_pooling2d_3 (MaxPooling2D)	(None, 5, 5, 1088)	0	block1
7_20_ac[0][0]			

mixed_7a (Concatenate)	(None, 5, 5, 2080)	0	activa
tion_157[0][0]			

			activa
tion_159[0][0]			

			activa
tion_162[0][0]			

			max_po
oling2d_3[0][0]			

conv2d_164 (Conv2D)	(None, 5, 5, 192)	399360	mixed_7a[0][0]

batch_normalization_164 (BatchN	(None, 5, 5, 192)	576	conv2d_164[0][0]

activation_164 (Activation)	(None, 5, 5, 192)	0	batch_normalization_164[0][0]

conv2d_165 (Conv2D)	(None, 5, 5, 224)	129024	activation_164[0][0]

batch_normalization_165 (BatchN	(None, 5, 5, 224)	672	conv2d_165[0][0]

activation_165 (Activation)	(None, 5, 5, 224)	0	batch_normalization_165[0][0]

conv2d_163 (Conv2D)	(None, 5, 5, 192)	399360	mixed_7a[0][0]

conv2d_166 (Conv2D)	(None, 5, 5, 256)	172032	activation_165[0][0]

batch_normalization_163 (BatchN	(None, 5, 5, 192)	576	conv2d_163[0][0]

batch_normalization_166 (BatchN	(None, 5, 5, 256)	768	conv2d_166[0][0]

activation_163 (Activation)	(None, 5, 5, 192)	0	batch_normalization_163[0][0]

activation_166 (Activation)	(None, 5, 5, 256)	0	batch_normalization_166[0][0]

block8_1_mixed (Concatenate)	(None, 5, 5, 448)	0	activation_163[0][0]
			activation_166[0][0]
block8_1_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8_1_mixed[0][0]
block8_1 (Lambda)	(None, 5, 5, 2080)	0	mixed_7a[0][0]
			block8_1_conv[0][0]
block8_1_ac (Activation)	(None, 5, 5, 2080)	0	block8_1[0][0]
conv2d_168 (Conv2D)	(None, 5, 5, 192)	399360	block8_1_ac[0][0]
batch_normalization_168 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_168[0][0]
activation_168 (Activation)	(None, 5, 5, 192)	0	batch_normalization_168[0][0]
conv2d_169 (Conv2D)	(None, 5, 5, 224)	129024	activation_168[0][0]
batch_normalization_169 (Batch Normalization)	(None, 5, 5, 224)	672	conv2d_169[0][0]
activation_169 (Activation)	(None, 5, 5, 224)	0	batch_normalization_169[0][0]
conv2d_167 (Conv2D)	(None, 5, 5, 192)	399360	conv2d_167[0][0]

conv2d_170 (Conv2D)	(None, 5, 5, 256)	172032	activation_169[0][0]

batch_normalization_167 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_167[0][0]

batch_normalization_170 (Batch Normalization)	(None, 5, 5, 256)	768	conv2d_170[0][0]

activation_167 (Activation)	(None, 5, 5, 192)	0	batch_normalization_167[0][0]

activation_170 (Activation)	(None, 5, 5, 256)	0	batch_normalization_170[0][0]

block8_2_mixed (Concatenate)	(None, 5, 5, 448)	0	activation_167[0][0]
			activation_170[0][0]

block8_2_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8_2_mixed[0][0]

block8_2 (Lambda)	(None, 5, 5, 2080)	0	block8_1_ac[0][0]
			block8_2_conv[0][0]

block8_2_ac (Activation)	(None, 5, 5, 2080)	0	block8_2[0][0]

conv2d_172 (Conv2D)	(None, 5, 5, 192)	399360	block8_2_ac[0][0]

batch_normalization_172 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_172[0][0]

activation_172 (Activation)	(None, 5, 5, 192)	0	batch_normalization_172[0][0]

conv2d_173 (Conv2D)	(None, 5, 5, 224)	129024	activation_172[0][0]
batch_normalization_173 (Batch Normalization)	(None, 5, 5, 224)	672	conv2d_173[0][0]
activation_173 (Activation)	(None, 5, 5, 224)	0	batch_normalization_173[0][0]
conv2d_171 (Conv2D)	(None, 5, 5, 192)	399360	block8_2_ac[0][0]
conv2d_174 (Conv2D)	(None, 5, 5, 256)	172032	activation_173[0][0]
batch_normalization_171 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_171[0][0]
batch_normalization_174 (Batch Normalization)	(None, 5, 5, 256)	768	conv2d_174[0][0]
activation_171 (Activation)	(None, 5, 5, 192)	0	batch_normalization_171[0][0]
activation_174 (Activation)	(None, 5, 5, 256)	0	batch_normalization_174[0][0]
block8_3_mixed (Concatenate)	(None, 5, 5, 448)	0	activation_171[0][0]
			activation_174[0][0]
block8_3_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8_3_mixed[0][0]
block8_3 (Lambda)	(None, 5, 5, 2080)	0	block8_2_ac[0][0]

			block8
_3_conv[0][0]			

block8_3_ac (Activation)	(None, 5, 5, 2080)	0	block8
_3[0][0]			

conv2d_176 (Conv2D)	(None, 5, 5, 192)	399360	block8
_3_ac[0][0]			

batch_normalization_176 (BatchN	(None, 5, 5, 192)	576	conv2d
_176[0][0]			

activation_176 (Activation)	(None, 5, 5, 192)	0	batch_
normalization_176[0][0]			

conv2d_177 (Conv2D)	(None, 5, 5, 224)	129024	activa
tion_176[0][0]			

batch_normalization_177 (BatchN	(None, 5, 5, 224)	672	conv2d
_177[0][0]			

activation_177 (Activation)	(None, 5, 5, 224)	0	batch_
normalization_177[0][0]			

conv2d_175 (Conv2D)	(None, 5, 5, 192)	399360	block8
_3_ac[0][0]			

conv2d_178 (Conv2D)	(None, 5, 5, 256)	172032	activa
tion_177[0][0]			

batch_normalization_175 (BatchN	(None, 5, 5, 192)	576	conv2d
_175[0][0]			

batch_normalization_178 (BatchN	(None, 5, 5, 256)	768	conv2d
_178[0][0]			

activation_175 (Activation)	(None, 5, 5, 192)	0	batch_
normalization_175[0][0]			

activation_178 (Activation) normalization_178[0][0]	(None, 5, 5, 256)	0	batch_
block8_4_mixed (Concatenate) tion_175[0][0]	(None, 5, 5, 448)	0	activa
			activa
block8_4_conv (Conv2D) _4_mixed[0][0]	(None, 5, 5, 2080)	933920	block8
block8_4 (Lambda) _3_ac[0][0]	(None, 5, 5, 2080)	0	block8
_4_conv[0][0]			block8
block8_4_ac (Activation) _4[0][0]	(None, 5, 5, 2080)	0	block8
conv2d_180 (Conv2D) _4_ac[0][0]	(None, 5, 5, 192)	399360	block8
batch_normalization_180 (BatchN _180[0][0]	(None, 5, 5, 192)	576	conv2d
activation_180 (Activation) normalization_180[0][0]	(None, 5, 5, 192)	0	batch_
conv2d_181 (Conv2D) tion_180[0][0]	(None, 5, 5, 224)	129024	activa
batch_normalization_181 (BatchN _181[0][0]	(None, 5, 5, 224)	672	conv2d
activation_181 (Activation) normalization_181[0][0]	(None, 5, 5, 224)	0	batch_

conv2d_179 (Conv2D)	(None, 5, 5, 192)	399360	block8
_4_ac[0][0]			

conv2d_182 (Conv2D)	(None, 5, 5, 256)	172032	activation_181[0][0]

batch_normalization_179 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_179[0][0]

batch_normalization_182 (Batch Normalization)	(None, 5, 5, 256)	768	conv2d_182[0][0]

activation_179 (Activation)	(None, 5, 5, 192)	0	batch_normalization_179[0][0]

activation_182 (Activation)	(None, 5, 5, 256)	0	batch_normalization_182[0][0]

block8_5_mixed (Concatenate)	(None, 5, 5, 448)	0	activation_179[0][0]

activation_182[0][0]			activation_182[0][0]

block8_5_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8_5_mixed[0][0]

block8_5 (Lambda)	(None, 5, 5, 2080)	0	block8_5_conv[0][0]

_4_ac[0][0]			block8_5_conv[0][0]

block8_5_ac (Activation)	(None, 5, 5, 2080)	0	block8_5[0][0]

conv2d_184 (Conv2D)	(None, 5, 5, 192)	399360	block8_5_ac[0][0]

batch_normalization_184 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_184[0][0]

activation_184 (Activation) normalization_184[0][0]	(None, 5, 5, 192)	0	batch_
conv2d_185 (Conv2D) tion_184[0][0]	(None, 5, 5, 224)	129024	activa
batch_normalization_185 (BatchN _185[0][0]	(None, 5, 5, 224)	672	conv2d
activation_185 (Activation) normalization_185[0][0]	(None, 5, 5, 224)	0	batch_
conv2d_183 (Conv2D) _5_ac[0][0]	(None, 5, 5, 192)	399360	block8
conv2d_186 (Conv2D) tion_185[0][0]	(None, 5, 5, 256)	172032	activa
batch_normalization_183 (BatchN _183[0][0]	(None, 5, 5, 192)	576	conv2d
batch_normalization_186 (BatchN _186[0][0]	(None, 5, 5, 256)	768	conv2d
activation_183 (Activation) normalization_183[0][0]	(None, 5, 5, 192)	0	batch_
activation_186 (Activation) normalization_186[0][0]	(None, 5, 5, 256)	0	batch_
block8_6_mixed (Concatenate) tion_183[0][0]	(None, 5, 5, 448)	0	activa
tion_186[0][0]			activa
block8_6_conv (Conv2D) _6_mixed[0][0]	(None, 5, 5, 2080)	933920	block8

block8_6 (Lambda)	(None, 5, 5, 2080)	0	block8
_5_ac[0][0]			block8
_6_conv[0][0]			
block8_6_ac (Activation)	(None, 5, 5, 2080)	0	block8
_6[0][0]			
conv2d_188 (Conv2D)	(None, 5, 5, 192)	399360	block8
_6_ac[0][0]			
batch_normalization_188 (BatchN	(None, 5, 5, 192)	576	conv2d
_188[0][0]			
activation_188 (Activation)	(None, 5, 5, 192)	0	batch_
normalization_188[0][0]			
conv2d_189 (Conv2D)	(None, 5, 5, 224)	129024	activa
tion_188[0][0]			
batch_normalization_189 (BatchN	(None, 5, 5, 224)	672	conv2d
_189[0][0]			
activation_189 (Activation)	(None, 5, 5, 224)	0	batch_
normalization_189[0][0]			
conv2d_187 (Conv2D)	(None, 5, 5, 192)	399360	block8
_6_ac[0][0]			
conv2d_190 (Conv2D)	(None, 5, 5, 256)	172032	activa
tion_189[0][0]			
batch_normalization_187 (BatchN	(None, 5, 5, 192)	576	conv2d
_187[0][0]			
batch_normalization_190 (BatchN	(None, 5, 5, 256)	768	conv2d
_190[0][0]			

activation_187 (Activation) normalization_187[0][0]	(None, 5, 5, 192)	0	batch_
activation_190 (Activation) normalization_190[0][0]	(None, 5, 5, 256)	0	batch_
block8_7_mixed (Concatenate) tion_187[0][0]	(None, 5, 5, 448)	0	activa
tion_190[0][0]			activa
block8_7_conv (Conv2D) _7_mixed[0][0]	(None, 5, 5, 2080)	933920	block8
block8_7 (Lambda) _6_ac[0][0]	(None, 5, 5, 2080)	0	block8
_7_conv[0][0]			block8
block8_7_ac (Activation) _7[0][0]	(None, 5, 5, 2080)	0	block8
conv2d_192 (Conv2D) _7_ac[0][0]	(None, 5, 5, 192)	399360	block8
batch_normalization_192 (BatchN _192[0][0]	(None, 5, 5, 192)	576	conv2d
activation_192 (Activation) normalization_192[0][0]	(None, 5, 5, 192)	0	batch_
conv2d_193 (Conv2D) tion_192[0][0]	(None, 5, 5, 224)	129024	activa
batch_normalization_193 (BatchN _193[0][0]	(None, 5, 5, 224)	672	conv2d

activation_193 (Activation)	(None, 5, 5, 224)	0	batch_
normalization_193[0][0]			

conv2d_191 (Conv2D)	(None, 5, 5, 192)	399360	block8
_7_ac[0][0]			

conv2d_194 (Conv2D)	(None, 5, 5, 256)	172032	activa
tion_193[0][0]			

batch_normalization_191 (BatchN	(None, 5, 5, 192)	576	conv2d
_191[0][0]			

batch_normalization_194 (BatchN	(None, 5, 5, 256)	768	conv2d
_194[0][0]			

activation_191 (Activation)	(None, 5, 5, 192)	0	batch_
normalization_191[0][0]			

activation_194 (Activation)	(None, 5, 5, 256)	0	batch_
normalization_194[0][0]			

block8_8_mixed (Concatenate)	(None, 5, 5, 448)	0	activa
tion_191[0][0]			
			activa
tion_194[0][0]			

block8_8_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8
_8_mixed[0][0]			

block8_8 (Lambda)	(None, 5, 5, 2080)	0	block8
_7_ac[0][0]			
			block8
_8_conv[0][0]			

block8_8_ac (Activation)	(None, 5, 5, 2080)	0	block8
_8[0][0]			

conv2d_196 (Conv2D)	(None, 5, 5, 192)	399360	block8
_8_ac[0][0]			

batch_normalization_196 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_196[0][0]
activation_196 (Activation)	(None, 5, 5, 192)	0	batch_normalization_196[0][0]
conv2d_197 (Conv2D)	(None, 5, 5, 224)	129024	activation_196[0][0]
batch_normalization_197 (Batch Normalization)	(None, 5, 5, 224)	672	conv2d_197[0][0]
activation_197 (Activation)	(None, 5, 5, 224)	0	batch_normalization_197[0][0]
conv2d_195 (Conv2D)	(None, 5, 5, 192)	399360	block8_8_ac[0][0]
conv2d_198 (Conv2D)	(None, 5, 5, 256)	172032	activation_197[0][0]
batch_normalization_195 (Batch Normalization)	(None, 5, 5, 192)	576	conv2d_195[0][0]
batch_normalization_198 (Batch Normalization)	(None, 5, 5, 256)	768	conv2d_198[0][0]
activation_195 (Activation)	(None, 5, 5, 192)	0	batch_normalization_195[0][0]
activation_198 (Activation)	(None, 5, 5, 256)	0	batch_normalization_198[0][0]
block8_9_mixed (Concatenate)	(None, 5, 5, 448)	0	activation_195[0][0] activation_198[0][0]

block8_9_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8
_9_mixed[0][0]			
block8_9 (Lambda)	(None, 5, 5, 2080)	0	block8
_8_ac[0][0]			block8
_9_conv[0][0]			
block8_9_ac (Activation)	(None, 5, 5, 2080)	0	block8
_9[0][0]			
conv2d_200 (Conv2D)	(None, 5, 5, 192)	399360	block8
_9_ac[0][0]			
batch_normalization_200 (BatchN	(None, 5, 5, 192)	576	conv2d
_200[0][0]			
activation_200 (Activation)	(None, 5, 5, 192)	0	batch_
normalization_200[0][0]			
conv2d_201 (Conv2D)	(None, 5, 5, 224)	129024	activa
tion_200[0][0]			
batch_normalization_201 (BatchN	(None, 5, 5, 224)	672	conv2d
_201[0][0]			
activation_201 (Activation)	(None, 5, 5, 224)	0	batch_
normalization_201[0][0]			
conv2d_199 (Conv2D)	(None, 5, 5, 192)	399360	block8
_9_ac[0][0]			
conv2d_202 (Conv2D)	(None, 5, 5, 256)	172032	activa
tion_201[0][0]			
batch_normalization_199 (BatchN	(None, 5, 5, 192)	576	conv2d
_199[0][0]			

batch_normalization_202 (Batch Normalization)	(None, 5, 5, 256)	768	conv2d_202[0][0]
activation_199 (Activation)	(None, 5, 5, 192)	0	batch_normalization_199[0][0]
activation_202 (Activation)	(None, 5, 5, 256)	0	batch_normalization_202[0][0]
block8_10_mixed (Concatenate)	(None, 5, 5, 448)	0	activation_199[0][0] activation_202[0][0]
block8_10_conv (Conv2D)	(None, 5, 5, 2080)	933920	block8_10_mixed[0][0]
block8_10 (Lambda)	(None, 5, 5, 2080)	0	block8_9_ac[0][0] block8_10_conv[0][0]
conv_7b (Conv2D)	(None, 5, 5, 1536)	3194880	block8_10[0][0]
conv_7b_bn (Batch Normalization)	(None, 5, 5, 1536)	4608	conv_7b[0][0]
conv_7b_ac (Activation)	(None, 5, 5, 1536)	0	conv_7b_bn[0][0]
flatten (Flatten)	(None, 38400)	0	conv_7b_ac[0][0]
dense (Dense)	(None, 500)	19200500	flatten[0][0]

dense_1 (Dense)	(None, 250)	125250	dense[0][0]

dense_2 (Dense)	(None, 4)	1004	dense_1[0][0]
=====			
Total params: 73,663,490			
Trainable params: 73,602,946			
Non-trainable params: 60,544			

4.2 INCEPTION-RESNET-V2 TRAINING AND SAVE

In [12]:

```
tfb = TensorBoard('object_detection')
history = model.fit(x=x_train,y=y_train,batch_size=10,epochs=180,
                    validation_data=(x_test,y_test),callbacks=[tfb])
```

2022-06-17 11:07:09.212581: I tensorflow/core/profiler/lib/profiler_session.cc:131] Profiler session initializing.

2022-06-17 11:07:09.212630: I tensorflow/core/profiler/lib/profiler_session.cc:146] Profiler session started.

2022-06-17 11:07:09.214679: I tensorflow/core/profiler/internal/gpu/cupti_tracer.cc:1614] Profiler found 1 GPUs

2022-06-17 11:07:09.557052: I tensorflow/core/profiler/lib/profiler_session.cc:164] Profiler session tear down.

2022-06-17 11:07:09.557233: I tensorflow/core/profiler/internal/gpu/cupti_tracer.cc:1748] CUPTI activity buffer flushed

2022-06-17 11:07:10.429340: I tensorflow/compiler/mlir/mlir_graph_optimization_pass.cc:185] None of the MLIR Optimization Passes are enabled (registered 2)

Epoch 1/180

2022-06-17 11:07:26.532781: I tensorflow/stream_executor/cuda/cuda_dnn.cc:369] Loaded cuDNN version 8005

1/18 [>.....] - ETA: 6:24 - loss: 0.0252

2022-06-17 11:07:33.511392: I tensorflow/core/profiler/lib/profiler_session.cc:131] Profiler session initializing.

2022-06-17 11:07:33.511447: I tensorflow/core/profiler/lib/profiler_session.cc:146] Profiler session started.

2/18 [==>.....] - ETA: 17s - loss: 0.0498

2022-06-17 11:07:34.197508: I tensorflow/core/profiler/lib/profiler_session.cc:66] Profiler session collecting data.

2022-06-17 11:07:34.202716: I tensorflow/core/profiler/internal/gpu/cupti_tracer.cc:1748] CUPTI activity buffer flushed

2022-06-17 11:07:34.394793: I tensorflow/core/profiler/internal/gpu/cupti_collector.cc:673] GpuTracer has collected 5163 callback api events and 5161 activity events.

2022-06-17 11:07:34.528871: I tensorflow/core/profiler/lib/profiler_session.cc:164] Profiler session tear down.
2022-06-17 11:07:34.670969: I tensorflow/core/profiler/rpc/client/save_profile.cc:136] Creating directory: object_detection/train/plugins/profile/2022_06_17_11_07_34

2022-06-17 11:07:34.765101: I tensorflow/core/profiler/rpc/client/save_profile.cc:142] Dumped gzipped tool data for trace.json.gz to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.trace.json.gz

2022-06-17 11:07:34.941146: I tensorflow/core/profiler/rpc/client/save_profile.cc:136] Creating directory: object_detection/train/plugins/profile/2022_06_17_11_07_34

2022-06-17 11:07:34.950258: I tensorflow/core/profiler/rpc/client/save_profile.cc:142] Dumped gzipped tool data for memory_profile.json.gz to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.memory_profile.json.gz

2022-06-17 11:07:34.957428: I tensorflow/core/profiler/rpc/client/capture_profile.cc:251] Creating directory: object_detection/train/plugins/profile/2022_06_17_11_07_34

Dumped tool data for xplane.pb to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.xplane.pb

Dumped tool data for overview_page.pb to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.overview_page.pb

Dumped tool data for input_pipeline.pb to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.input_pipeline.pb

Dumped tool data for tensorflow_stats.pb to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.tensorflow_stats.pb

Dumped tool data for kernel_stats.pb to object_detection/train/plugins/profile/2022_06_17_11_07_34/65511e7e50cb.kernel_stats.pb

18/18 [=====] - 30s 462ms/step - loss: 0.0632
- val_loss: 0.0493

Epoch 2/180

18/18 [=====] - 3s 181ms/step - loss: 0.0241 -
val_loss: 0.0181

Epoch 3/180

18/18 [=====] - 3s 180ms/step - loss: 0.0089 -
val_loss: 0.0171

Epoch 4/180

18/18 [=====] - 3s 177ms/step - loss: 0.0056 -
val_loss: 0.0168

Epoch 5/180

18/18 [=====] - 3s 177ms/step - loss: 0.0041 -
val_loss: 0.0180

Epoch 6/180

18/18 [=====] - 3s 185ms/step - loss: 0.0034 -
val_loss: 0.0109

Epoch 7/180

```
18/18 [=====] - 3s 184ms/step - loss: 0.0029 -  
val_loss: 0.0113  
Epoch 8/180  
18/18 [=====] - 3s 180ms/step - loss: 0.0073 -  
val_loss: 0.0090  
Epoch 9/180  
18/18 [=====] - 3s 183ms/step - loss: 0.0044 -  
val_loss: 0.0097  
Epoch 10/180  
18/18 [=====] - 3s 180ms/step - loss: 0.0023 -  
val_loss: 0.0084  
Epoch 11/180  
18/18 [=====] - 3s 181ms/step - loss: 0.0018 -  
val_loss: 0.0094  
Epoch 12/180  
18/18 [=====] - 3s 180ms/step - loss: 0.0016 -  
val_loss: 0.0133  
Epoch 13/180  
18/18 [=====] - 3s 181ms/step - loss: 0.0012 -  
val_loss: 0.0141  
Epoch 14/180  
18/18 [=====] - 3s 177ms/step - loss: 0.0011 -  
val_loss: 0.0121  
Epoch 15/180  
18/18 [=====] - 3s 176ms/step - loss: 8.5752e-  
04 - val_loss: 0.0129  
Epoch 16/180  
18/18 [=====] - 4s 203ms/step - loss: 9.0463e-  
04 - val_loss: 0.0133  
Epoch 17/180  
18/18 [=====] - 3s 177ms/step - loss: 0.0014 -  
val_loss: 0.0341  
Epoch 18/180  
18/18 [=====] - 3s 180ms/step - loss: 0.0011 -  
val_loss: 0.0295  
Epoch 19/180  
18/18 [=====] - 3s 180ms/step - loss: 7.2101e-  
04 - val_loss: 0.0244  
Epoch 20/180  
18/18 [=====] - 3s 181ms/step - loss: 8.5819e-  
04 - val_loss: 0.0215  
Epoch 21/180  
18/18 [=====] - 3s 176ms/step - loss: 6.2261e-  
04 - val_loss: 0.0183  
Epoch 22/180  
18/18 [=====] - 3s 177ms/step - loss: 6.2524e-  
04 - val_loss: 0.0150  
Epoch 23/180  
18/18 [=====] - 3s 181ms/step - loss: 6.0729e-  
04 - val_loss: 0.0135
```

Epoch 24/180
18/18 [=====] - 3s 177ms/step - loss: 6.4947e-04 - val_loss: 0.0134
Epoch 25/180
18/18 [=====] - 3s 177ms/step - loss: 5.6002e-04 - val_loss: 0.0136
Epoch 26/180
18/18 [=====] - 3s 187ms/step - loss: 5.0687e-04 - val_loss: 0.0124
Epoch 27/180
18/18 [=====] - 3s 178ms/step - loss: 6.3071e-04 - val_loss: 0.0125
Epoch 28/180
18/18 [=====] - 3s 180ms/step - loss: 4.5222e-04 - val_loss: 0.0117
Epoch 29/180
18/18 [=====] - 3s 182ms/step - loss: 3.6329e-04 - val_loss: 0.0118
Epoch 30/180
18/18 [=====] - 3s 179ms/step - loss: 5.1769e-04 - val_loss: 0.0108
Epoch 31/180
18/18 [=====] - 3s 177ms/step - loss: 4.5084e-04 - val_loss: 0.0104
Epoch 32/180
18/18 [=====] - 3s 177ms/step - loss: 3.3693e-04 - val_loss: 0.0097
Epoch 33/180
18/18 [=====] - 3s 180ms/step - loss: 3.9427e-04 - val_loss: 0.0089
Epoch 34/180
18/18 [=====] - 3s 178ms/step - loss: 3.3008e-04 - val_loss: 0.0085
Epoch 35/180
18/18 [=====] - 3s 177ms/step - loss: 3.3242e-04 - val_loss: 0.0083
Epoch 36/180
18/18 [=====] - 4s 221ms/step - loss: 3.5103e-04 - val_loss: 0.0089
Epoch 37/180
18/18 [=====] - 3s 177ms/step - loss: 3.0543e-04 - val_loss: 0.0079
Epoch 38/180
18/18 [=====] - 3s 177ms/step - loss: 2.8094e-04 - val_loss: 0.0080
Epoch 39/180
18/18 [=====] - 3s 182ms/step - loss: 3.0287e-04 - val_loss: 0.0079
Epoch 40/180

18/18 [=====] - 3s 179ms/step - loss: 3.3246e-04 - val_loss: 0.0074
Epoch 41/180
18/18 [=====] - 3s 176ms/step - loss: 3.1612e-04 - val_loss: 0.0079
Epoch 42/180
18/18 [=====] - 3s 177ms/step - loss: 3.6023e-04 - val_loss: 0.0072
Epoch 43/180
18/18 [=====] - 3s 181ms/step - loss: 2.8871e-04 - val_loss: 0.0079
Epoch 44/180
18/18 [=====] - 3s 177ms/step - loss: 2.5653e-04 - val_loss: 0.0083
Epoch 45/180
18/18 [=====] - 3s 176ms/step - loss: 2.7350e-04 - val_loss: 0.0074
Epoch 46/180
18/18 [=====] - 4s 203ms/step - loss: 3.2693e-04 - val_loss: 0.0082
Epoch 47/180
18/18 [=====] - 3s 177ms/step - loss: 3.0728e-04 - val_loss: 0.0086
Epoch 48/180
18/18 [=====] - 3s 179ms/step - loss: 2.3513e-04 - val_loss: 0.0082
Epoch 49/180
18/18 [=====] - 3s 176ms/step - loss: 2.6588e-04 - val_loss: 0.0082
Epoch 50/180
18/18 [=====] - 3s 184ms/step - loss: 2.5553e-04 - val_loss: 0.0077
Epoch 51/180
18/18 [=====] - 3s 177ms/step - loss: 2.4150e-04 - val_loss: 0.0078
Epoch 52/180
18/18 [=====] - 3s 177ms/step - loss: 2.7731e-04 - val_loss: 0.0085
Epoch 53/180
18/18 [=====] - 3s 182ms/step - loss: 2.7688e-04 - val_loss: 0.0080
Epoch 54/180
18/18 [=====] - 3s 177ms/step - loss: 2.3691e-04 - val_loss: 0.0073
Epoch 55/180
18/18 [=====] - 3s 177ms/step - loss: 2.7263e-04 - val_loss: 0.0082
Epoch 56/180
18/18 [=====] - 3s 177ms/step - loss: 2.0742e-04 - val_loss: 0.0075

Epoch 57/180
18/18 [=====] - 4s 204ms/step - loss: 1.9083e-04 - val_loss: 0.0076
Epoch 58/180
18/18 [=====] - 3s 179ms/step - loss: 2.0246e-04 - val_loss: 0.0080
Epoch 59/180
18/18 [=====] - 3s 176ms/step - loss: 1.7790e-04 - val_loss: 0.0078
Epoch 60/180
18/18 [=====] - 3s 182ms/step - loss: 1.9031e-04 - val_loss: 0.0077
Epoch 61/180
18/18 [=====] - 3s 180ms/step - loss: 2.0385e-04 - val_loss: 0.0079
Epoch 62/180
18/18 [=====] - 3s 177ms/step - loss: 2.1916e-04 - val_loss: 0.0078
Epoch 63/180
18/18 [=====] - 3s 180ms/step - loss: 1.9533e-04 - val_loss: 0.0077
Epoch 64/180
18/18 [=====] - 3s 176ms/step - loss: 1.7168e-04 - val_loss: 0.0078
Epoch 65/180
18/18 [=====] - 3s 177ms/step - loss: 1.6540e-04 - val_loss: 0.0078
Epoch 66/180
18/18 [=====] - 3s 180ms/step - loss: 1.5014e-04 - val_loss: 0.0073
Epoch 67/180
18/18 [=====] - 4s 210ms/step - loss: 1.2901e-04 - val_loss: 0.0078
Epoch 68/180
18/18 [=====] - 3s 178ms/step - loss: 1.5214e-04 - val_loss: 0.0083
Epoch 69/180
18/18 [=====] - 3s 179ms/step - loss: 1.2878e-04 - val_loss: 0.0085
Epoch 70/180
18/18 [=====] - 3s 179ms/step - loss: 1.3750e-04 - val_loss: 0.0074
Epoch 71/180
18/18 [=====] - 3s 178ms/step - loss: 1.6853e-04 - val_loss: 0.0071
Epoch 72/180
18/18 [=====] - 3s 179ms/step - loss: 1.6168e-04 - val_loss: 0.0072
Epoch 73/180

18/18 [=====] - 3s 181ms/step - loss: 1.3989e-04 - val_loss: 0.0067
Epoch 74/180
18/18 [=====] - 3s 177ms/step - loss: 1.3245e-04 - val_loss: 0.0068
Epoch 75/180
18/18 [=====] - 3s 177ms/step - loss: 1.3745e-04 - val_loss: 0.0067
Epoch 76/180
18/18 [=====] - 3s 180ms/step - loss: 1.5385e-04 - val_loss: 0.0073
Epoch 77/180
18/18 [=====] - 4s 204ms/step - loss: 1.8950e-04 - val_loss: 0.0077
Epoch 78/180
18/18 [=====] - 3s 177ms/step - loss: 1.6494e-04 - val_loss: 0.0070
Epoch 79/180
18/18 [=====] - 3s 176ms/step - loss: 1.7439e-04 - val_loss: 0.0071
Epoch 80/180
18/18 [=====] - 3s 184ms/step - loss: 1.4345e-04 - val_loss: 0.0073
Epoch 81/180
18/18 [=====] - 3s 179ms/step - loss: 1.5315e-04 - val_loss: 0.0071
Epoch 82/180
18/18 [=====] - 3s 176ms/step - loss: 1.4757e-04 - val_loss: 0.0072
Epoch 83/180
18/18 [=====] - 3s 180ms/step - loss: 1.5262e-04 - val_loss: 0.0069
Epoch 84/180
18/18 [=====] - 3s 178ms/step - loss: 1.4000e-04 - val_loss: 0.0075
Epoch 85/180
18/18 [=====] - 3s 177ms/step - loss: 1.2905e-04 - val_loss: 0.0073
Epoch 86/180
18/18 [=====] - 3s 180ms/step - loss: 1.1496e-04 - val_loss: 0.0072
Epoch 87/180
18/18 [=====] - 4s 202ms/step - loss: 1.2455e-04 - val_loss: 0.0069
Epoch 88/180
18/18 [=====] - 3s 180ms/step - loss: 1.3334e-04 - val_loss: 0.0086
Epoch 89/180
18/18 [=====] - 3s 177ms/step - loss: 1.1222e-04 - val_loss: 0.0076

Epoch 90/180
18/18 [=====] - 3s 180ms/step - loss: 1.1846e-04 - val_loss: 0.0065
Epoch 91/180
18/18 [=====] - 3s 177ms/step - loss: 1.3428e-04 - val_loss: 0.0066
Epoch 92/180
18/18 [=====] - 3s 177ms/step - loss: 1.2116e-04 - val_loss: 0.0069
Epoch 93/180
18/18 [=====] - 3s 177ms/step - loss: 1.2350e-04 - val_loss: 0.0067
Epoch 94/180
18/18 [=====] - 3s 180ms/step - loss: 1.1575e-04 - val_loss: 0.0071
Epoch 95/180
18/18 [=====] - 3s 182ms/step - loss: 1.3703e-04 - val_loss: 0.0070
Epoch 96/180
18/18 [=====] - 3s 177ms/step - loss: 1.6852e-04 - val_loss: 0.0068
Epoch 97/180
18/18 [=====] - 4s 205ms/step - loss: 1.3040e-04 - val_loss: 0.0067
Epoch 98/180
18/18 [=====] - 3s 180ms/step - loss: 1.1534e-04 - val_loss: 0.0068
Epoch 99/180
18/18 [=====] - 3s 177ms/step - loss: 1.2002e-04 - val_loss: 0.0064
Epoch 100/180
18/18 [=====] - 4s 201ms/step - loss: 1.0505e-04 - val_loss: 0.0068
Epoch 101/180
18/18 [=====] - 3s 177ms/step - loss: 1.1929e-04 - val_loss: 0.0067
Epoch 102/180
18/18 [=====] - 3s 177ms/step - loss: 1.7024e-04 - val_loss: 0.0070
Epoch 103/180
18/18 [=====] - 3s 178ms/step - loss: 1.2909e-04 - val_loss: 0.0067
Epoch 104/180
18/18 [=====] - 3s 182ms/step - loss: 1.5226e-04 - val_loss: 0.0067
Epoch 105/180
18/18 [=====] - 3s 181ms/step - loss: 1.4285e-04 - val_loss: 0.0066
Epoch 106/180

18/18 [=====] - 3s 177ms/step - loss: 1.3594e-04 - val_loss: 0.0072
Epoch 107/180
18/18 [=====] - 4s 204ms/step - loss: 1.3416e-04 - val_loss: 0.0068
Epoch 108/180
18/18 [=====] - 3s 178ms/step - loss: 1.2329e-04 - val_loss: 0.0067
Epoch 109/180
18/18 [=====] - 3s 177ms/step - loss: 1.0934e-04 - val_loss: 0.0065
Epoch 110/180
18/18 [=====] - 4s 199ms/step - loss: 9.8583e-05 - val_loss: 0.0066
Epoch 111/180
18/18 [=====] - 3s 181ms/step - loss: 1.1918e-04 - val_loss: 0.0066
Epoch 112/180
18/18 [=====] - 3s 181ms/step - loss: 1.7080e-04 - val_loss: 0.0065
Epoch 113/180
18/18 [=====] - 3s 178ms/step - loss: 1.7788e-04 - val_loss: 0.0070
Epoch 114/180
18/18 [=====] - 3s 195ms/step - loss: 1.6266e-04 - val_loss: 0.0063
Epoch 115/180
18/18 [=====] - 3s 180ms/step - loss: 1.9065e-04 - val_loss: 0.0067
Epoch 116/180
18/18 [=====] - 3s 178ms/step - loss: 1.4408e-04 - val_loss: 0.0070
Epoch 117/180
18/18 [=====] - 4s 202ms/step - loss: 1.1291e-04 - val_loss: 0.0069
Epoch 118/180
18/18 [=====] - 3s 177ms/step - loss: 9.5333e-05 - val_loss: 0.0066
Epoch 119/180
18/18 [=====] - 3s 180ms/step - loss: 8.1524e-05 - val_loss: 0.0068
Epoch 120/180
18/18 [=====] - 3s 181ms/step - loss: 1.2272e-04 - val_loss: 0.0065
Epoch 121/180
18/18 [=====] - 3s 177ms/step - loss: 9.4329e-05 - val_loss: 0.0065
Epoch 122/180
18/18 [=====] - 3s 177ms/step - loss: 7.3594e-05 - val_loss: 0.0068

Epoch 123/180
18/18 [=====] - 3s 183ms/step - loss: 7.5413e-05 - val_loss: 0.0068
Epoch 124/180
18/18 [=====] - 3s 181ms/step - loss: 1.7459e-04 - val_loss: 0.0071
Epoch 125/180
18/18 [=====] - 3s 177ms/step - loss: 1.1409e-04 - val_loss: 0.0067
Epoch 126/180
18/18 [=====] - 3s 178ms/step - loss: 1.3270e-04 - val_loss: 0.0066
Epoch 127/180
18/18 [=====] - 4s 204ms/step - loss: 1.3608e-04 - val_loss: 0.0062
Epoch 128/180
18/18 [=====] - 3s 177ms/step - loss: 1.1815e-04 - val_loss: 0.0062
Epoch 129/180
18/18 [=====] - 3s 180ms/step - loss: 1.0859e-04 - val_loss: 0.0061
Epoch 130/180
18/18 [=====] - 3s 180ms/step - loss: 1.1539e-04 - val_loss: 0.0070
Epoch 131/180
18/18 [=====] - 3s 181ms/step - loss: 1.7011e-04 - val_loss: 0.0064
Epoch 132/180
18/18 [=====] - 3s 177ms/step - loss: 1.4650e-04 - val_loss: 0.0074
Epoch 133/180
18/18 [=====] - 3s 182ms/step - loss: 1.2307e-04 - val_loss: 0.0067
Epoch 134/180
18/18 [=====] - 3s 184ms/step - loss: 1.2227e-04 - val_loss: 0.0065
Epoch 135/180
18/18 [=====] - 3s 180ms/step - loss: 1.4197e-04 - val_loss: 0.0065
Epoch 136/180
18/18 [=====] - 3s 181ms/step - loss: 1.1630e-04 - val_loss: 0.0066
Epoch 137/180
18/18 [=====] - 4s 205ms/step - loss: 1.2912e-04 - val_loss: 0.0066
Epoch 138/180
18/18 [=====] - 3s 182ms/step - loss: 1.1751e-04 - val_loss: 0.0068
Epoch 139/180

18/18 [=====] - 3s 178ms/step - loss: 1.1681e-04 - val_loss: 0.0068
Epoch 140/180
18/18 [=====] - 3s 179ms/step - loss: 9.3450e-05 - val_loss: 0.0064
Epoch 141/180
18/18 [=====] - 3s 183ms/step - loss: 8.4480e-05 - val_loss: 0.0062
Epoch 142/180
18/18 [=====] - 3s 183ms/step - loss: 9.2226e-05 - val_loss: 0.0063
Epoch 143/180
18/18 [=====] - 3s 179ms/step - loss: 1.1219e-04 - val_loss: 0.0068
Epoch 144/180
18/18 [=====] - 3s 180ms/step - loss: 1.2974e-04 - val_loss: 0.0077
Epoch 145/180
18/18 [=====] - 3s 177ms/step - loss: 1.1895e-04 - val_loss: 0.0075
Epoch 146/180
18/18 [=====] - 3s 177ms/step - loss: 1.2025e-04 - val_loss: 0.0069
Epoch 147/180
18/18 [=====] - 4s 197ms/step - loss: 1.3721e-04 - val_loss: 0.0070
Epoch 148/180
18/18 [=====] - 3s 179ms/step - loss: 1.2100e-04 - val_loss: 0.0063
Epoch 149/180
18/18 [=====] - 3s 176ms/step - loss: 1.1397e-04 - val_loss: 0.0066
Epoch 150/180
18/18 [=====] - 3s 177ms/step - loss: 9.8503e-05 - val_loss: 0.0071
Epoch 151/180
18/18 [=====] - 3s 188ms/step - loss: 8.5244e-05 - val_loss: 0.0065
Epoch 152/180
18/18 [=====] - 3s 180ms/step - loss: 7.9321e-05 - val_loss: 0.0065
Epoch 153/180
18/18 [=====] - 3s 182ms/step - loss: 8.7068e-05 - val_loss: 0.0069
Epoch 154/180
18/18 [=====] - 3s 185ms/step - loss: 7.1885e-05 - val_loss: 0.0064
Epoch 155/180
18/18 [=====] - 3s 180ms/step - loss: 8.1814e-05 - val_loss: 0.0064

Epoch 156/180
18/18 [=====] - 3s 181ms/step - loss: 1.0209e-04 - val_loss: 0.0069
Epoch 157/180
18/18 [=====] - 4s 199ms/step - loss: 9.1655e-05 - val_loss: 0.0085
Epoch 158/180
18/18 [=====] - 3s 179ms/step - loss: 1.3150e-04 - val_loss: 0.0080
Epoch 159/180
18/18 [=====] - 3s 180ms/step - loss: 1.3249e-04 - val_loss: 0.0072
Epoch 160/180
18/18 [=====] - 3s 177ms/step - loss: 1.4734e-04 - val_loss: 0.0076
Epoch 161/180
18/18 [=====] - 3s 184ms/step - loss: 1.4219e-04 - val_loss: 0.0076
Epoch 162/180
18/18 [=====] - 3s 180ms/step - loss: 1.1701e-04 - val_loss: 0.0072
Epoch 163/180
18/18 [=====] - 3s 177ms/step - loss: 1.2523e-04 - val_loss: 0.0079
Epoch 164/180
18/18 [=====] - 3s 181ms/step - loss: 1.0676e-04 - val_loss: 0.0077
Epoch 165/180
18/18 [=====] - 3s 178ms/step - loss: 7.3352e-05 - val_loss: 0.0083
Epoch 166/180
18/18 [=====] - 3s 177ms/step - loss: 7.7341e-05 - val_loss: 0.0069
Epoch 167/180
18/18 [=====] - 3s 181ms/step - loss: 7.4907e-05 - val_loss: 0.0070
Epoch 168/180
18/18 [=====] - 4s 202ms/step - loss: 6.1625e-05 - val_loss: 0.0073
Epoch 169/180
18/18 [=====] - 3s 177ms/step - loss: 8.5513e-05 - val_loss: 0.0069
Epoch 170/180
18/18 [=====] - 3s 180ms/step - loss: 7.8864e-05 - val_loss: 0.0070
Epoch 171/180
18/18 [=====] - 3s 180ms/step - loss: 7.7951e-05 - val_loss: 0.0070
Epoch 172/180

```

18/18 [=====] - 3s 177ms/step - loss: 7.2048e-
05 - val_loss: 0.0072
Epoch 173/180
18/18 [=====] - 3s 178ms/step - loss: 5.7776e-
05 - val_loss: 0.0068
Epoch 174/180
18/18 [=====] - 3s 181ms/step - loss: 8.3889e-
05 - val_loss: 0.0078
Epoch 175/180
18/18 [=====] - 3s 181ms/step - loss: 8.6761e-
05 - val_loss: 0.0089
Epoch 176/180
18/18 [=====] - 3s 177ms/step - loss: 9.1527e-
05 - val_loss: 0.0080
Epoch 177/180
18/18 [=====] - 3s 178ms/step - loss: 6.3443e-
05 - val_loss: 0.0066
Epoch 178/180
18/18 [=====] - 4s 202ms/step - loss: 6.3016e-
05 - val_loss: 0.0064
Epoch 179/180
18/18 [=====] - 3s 181ms/step - loss: 6.6962e-
05 - val_loss: 0.0068
Epoch 180/180
18/18 [=====] - 3s 180ms/step - loss: 6.2468e-
05 - val_loss: 0.0066

```

In [13]:

```
model.save('./object_detection.h5')
```

4.3 TENSORBOARD

Lest have a look at on scalars on TensorBoard. In order to do it we will need to run simple command with right path for our "object detection". After we will see output with hosted link open it with Chrome. I was using VSCode for this project and for me it was way easy to run TensorBoard overview results, but in Kaggle it a bit more complicated and could be discussed in other topic. For now i will show one screenshot of result which we have. We can see on scalars *Figure 12* how is our model preform. Our train and validation set don't have over fitting behavior and our loss with respect of epochs is less.

You can simple type `!tensorboard --logdir="./object_detection"` it will generate link with text, click on link and here we go. Serving TensorBoard on localhost; to expose to the network, use a proxy or pass `--bind_all` TensorBoard 2.6.0 at `http://localhost:6006/` (Press CTRL+C to quit)

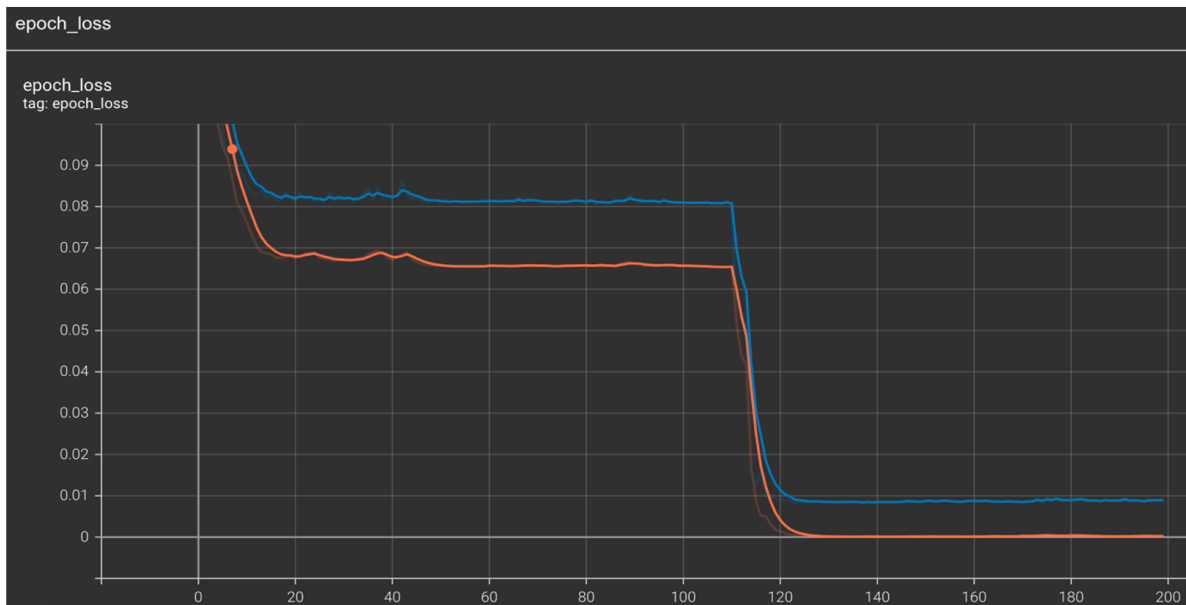


Figure 12 TensorBoard

5. PIPELINE OBJECT DETECTION MODEL

5.1 MAKE PREDICTIONS

This is the final step in object detection. In this step, we will put it all together and get the prediction for a given image. First, I would like to try with one of my test pictures of car. Let load our model.

In [14]:

```
# Load model
model = tf.keras.models.load_model('./object_detection.h5')
print('Model loaded Sucessfully')
```

Model loaded Sucessfully

Next is loading our TEST picture with right path to it. I loaded some more images for this purpose only - folder **TEST**.

In [15]:

```
path = '../input/number-plate-detection/TEST/TEST.jpeg'
image = load_img(path) # PIL object
image = np.array(image,dtype=np.uint8) # 8 bit array (0,255)
image1 = load_img(path,target_size=(224,224))
image_arr_224 = img_to_array(image1)/255.0 # Convert into array and get the normalized output
```

```
# Size of the original image
h,w,d = image.shape
print('Height of the image =',h)
print('Width of the image =',w)
```

Height of the image = 729

Width of the image = 901

Now we can have a look at our image *Figure 13*

```
In [16]:
fig = px.imshow(image)
fig.update_layout(width=700, height=500, margin=dict(l=10, r=10, b=10, t=10), xaxis_title='Figure 13 - TEST Image')
02004006008007006005004003002001000
Figure 13 - TEST Image
So, let's look into the shape of my image.
```

```
In [17]:
image_arr_224.shape
```

```
Out[17]:
(224, 224, 3)
But in order to pass this image of a model, we need to provide the data in the dynamic fourth dimension. And what one indicates is a number of images. So here we are just passing only one image.
```

```
In [18]:
test_arr = image_arr_224.reshape(1,224,224,3)
test_arr.shape
```

```
Out[18]:
(1, 224, 224, 3)
```

5.2 DE-NORMALIZE THE OUTPUT

```
In [19]:
# Make predictions
coords = model.predict(test_arr)
coords
```

```
Out[19]:
array([[0.35251912, 0.63733816, 0.65963215, 0.7349673 ]], dtype=float32)
We have got the output from the model and output what we got is the normalized output. So, what we need to do is to convert back into our original form values, which actually we did in during the training process, in the training process, we have the original form values and convert that normalized one. So basically, we will de-normalize the values back.
```

```
In [20]:
# Denormalize the values
denorm = np.array([w,w,h,h])
coords = coords * denorm
coords
```

```
Out[20]:
array([[317.6197314 , 574.24168348, 480.8718347 , 535.7911554 ]])
```

5.3 BOUNDING BOX

Now we will draw bounding box on top of the image. I just want to provide the two diagonal points. Let's make use of these points and let's draw the rectangle box.

```
In [21]:
```

```
coords = coords.astype(np.int32)
coords
```

Out[21]:

```
array([[317, 574, 480, 535]], dtype=int32)
```

In [22]:

```
# Draw bounding on top the image
xmin, xmax,ymin,ymax = coords[0]
pt1 =(xmin,ymin)
pt2 =(xmax,ymax)
print(pt1, pt2)
(317, 480) (574, 535)
```

In [23]:

```
cv2.rectangle(image,pt1,pt2,(0,255,0),3)
fig = px.imshow(image)
fig.update_layout(width=700, height=500, margin=dict(l=10, r=10, b=10, t=10
))
02004006008007006005004003002001000
```

5.4 CREATE PIPELINE

Now what we'll do, let's put it all together in one place and create function. And in the end visualize it. Our output will return image and coordinates of bounding box.

In [24]:

```
# Create pipeline
path = '../input/number-plate-detection/TEST/TEST.jpeg'
def object_detection(path):

    # Read image
    image = load_img(path) # PIL object
    image = np.array(image,dtype=np.uint8) # 8 bit array (0,255)
    image1 = load_img(path,target_size=(224,224))

    # Data preprocessing
    image_arr_224 = img_to_array(image1)/255.0 # Convert to array & normali
zed
    h,w,d = image.shape
    test_arr = image_arr_224.reshape(1,224,224,3)

    # Make predictions
    coords = model.predict(test_arr)

    # Denormalize the values
    denorm = np.array([w,w,h,h])
    coords = coords * denorm
    coords = coords.astype(np.int32)

    # Draw bounding on top the image
    xmin, xmax,ymin,ymax = coords[0]
    pt1 =(xmin,ymin)
```

```

    pt2 =(xmax,ymax)
    print(pt1, pt2)
    cv2.rectangle(image,pt1,pt2,(0,255,0),3)
    return image, coords

image, coords = object_detection(path)

fig = px.imshow(image)
fig.update_layout(width=700, height=500, margin=dict(l=10, r=10, b=10, t=10
),xaxis_title='Figure 14')
(317, 480) (574, 535)
02004006008007006005004003002001000
Figure 14

```

6. OPTICAL CHARACTER RECOGNITION - OCR

6.1 TESSERACT OCR

Optical character recognition (OCR) software that is used to extract text from the image. Tesseract OCR have a python API and it is open source. Firstly, we will do installation of it. It pretty simple and depend on you OS. You can find manual and files to download for installation [here](#).

6.2 LIMITATIONS OF PYTESSERACT

Tesseract works best when there is a clean segmentation of the foreground text from the background. In practice, it can be extremely challenging to guarantee these types of setups. There are a variety of reasons you might not get good quality output from Tesseract like if the image has noise on the background. The better the image quality (size, contrast, lightning) the better the recognition result. It requires a bit of preprocessing to improve the OCR results, images need to be scaled appropriately, have as much image contrast as possible, and the text must be horizontally aligned. Tesseract OCR is quite powerful but does have the following limitations.

Tesseract limitations summed in the list.

- The OCR is not as accurate as some commercial solutions available to us.
- Doesn't do well with images affected by artifacts including partial occlusion, distorted perspective, and complex background.
- It is not capable of recognizing handwriting.
- It may find gibberish and report this as OCR output.
- If a document contains languages outside of those given in the -l LANG arguments, results may be poor.
- It is not always good at analyzing the natural reading order of documents. For example, it may fail to recognize that a document contains two columns, and may try to join text across columns.
- Poor quality scans may produce poor quality OCR.
- It does not expose information about what font family text belongs to.

6.3 EXTRACT NUMBER PLATE TEXT FROM IMAGE

Firstly, we will load our image and convert to array. Crop our bounding box with coordinates of it. We will identify region of interest (ROI) and have look at our cropped image *Figure 15*.

In [25]:

```
img = np.array(load_img(path))
xmin ,xmax,ymin,ymax = cods[0]
roi = img[ymin:ymax,xmin:xmax]
fig = px.imshow(roi)
fig.update_layout(width=350, height=250, margin=dict(l=10, r=10, b=10, t=10),
),xaxis_title='Figure 15 Cropped image')
```

010020040200

Figure 15 Cropped image

With use of tesseract, we will extract the text from the mage.

In [26]:

```
# extract text from image
text = pt.image_to_string(roi)
print(text)
```

FKL Si K 4999

Obviously, we didn't get the proper text, but at least you can able to get 90 percent of the information. It just an example and again will need to say that as more data than better prediction. We will come to that point in future. What I realize here: First of all we don't have much data, to resolve this problem and I added to this topic more sets almost the same data sets from other kagglers posted recently. Secondly i don't see that this model perform well be honest with you, but full process which has been done gave as a chance to understand concept, we will build another model with help of YOLO and where we can see how it perform to compare our results. Second of of Tesseract, I already explain some limitation of it, but image preprocessing can be another topic and it could even require to build AI on it. So, now I want you to show how to build simple web and next step after it we will start with new model.

7. NUMBER PLATE WEB APP

This part I'm briefly will explain all about installation and integration it to web. I do understand that some steps are self-explained but I doing it as manual and if it helps at list for one person it is meaning something for me. In other case you can skip few steps and have a couple of coffee.

7.1 REQUIRED TOOLS

In order to run our project and we need to install [VSCode](#) and few extensions important one you can find in list below. Extensions which will help us in future.

- [All autocomplete](#)
- [Python extension pack](#)
- [Yaml](#)
- [Bootstrap 4, font awesome 4, font awesome 5 free & pro snippets](#)
- [Django](#)
- [Html boilerplate](#)

- [Html snippets](#)
- [Flask-snippets](#)

7.2 FLASK APP

Firstly, make sure install flask. !pip install flask. And crate new file **App.py**.

```
flask import Flask

#Webserver gateway interface

app = Flask (__name__)

@app.route('/')

def index():

    return "Hello World"

if __name__ == "__main__":

    app.run()>
```

Run in as on terminal (make sure you are in same directory where is you project.) You will see output like this Running on http://127.0.0.1:5000 (Press CTRL+C to quit) Open URL in Chrome to check it *Figure 16*

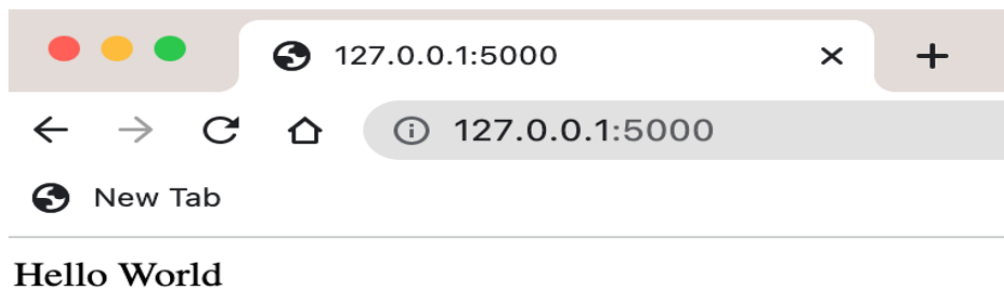


Figure 16 First page

Next create folder **templates** in our project in folder **templates** create new file as layout.html. Since we installed extensions just type html and choose boilerplate you will see something as on *Figure 17* and !DOCTYPE html

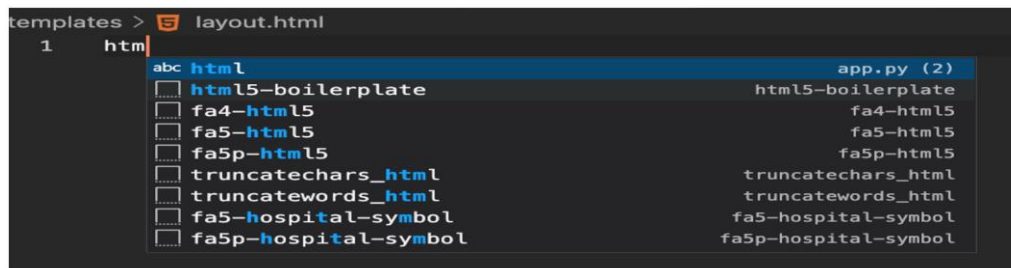


Figure 17

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="utf-8">

    <meta http-equiv="X-UA-Compatible" content="IE=edge">

    <title></title>

    <meta name="description" content="">

    <meta name="viewport" content="width=device-width, initial-scale=1">

    <link rel="stylesheet" href="">

  </head>

  <body>

    <!--[if lt IE 7]>

      <p class="browsehappy">You are using an <strong>outdated</strong> browser. Please <a href="#">upgrade your browser</a> to improve your experience.</p>

    <![endif]-->

    <script src="" async defer></script>

  </body>

</html>
```

And now we can start to build our page. We can change Title and add body type etc. We will need to modify our app.py, layout (and Bootstrap to layout you can find as well [link](#) here we are interesting) and adding footer. At this stage it will look as below.

app.py

```
from flask import Flask, render_template

#Webserver gateway interface

app = Flask(__name__)

@app.route('/')

def index():

    return render_template('layout.html')

if __name__ == "__main__":

    app.run(debug=True)
```

layout.html

```
<!DOCTYPE html>

<html>

  <head>

    <meta charset="utf-8" />

    <meta http-equiv="X-UA-Compatible" content="IE=edge" />

    <title>AUTOMATIC NUMBER-PLATE RECOGNITION</title>

    <link

      href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0-beta1/dist/css/bootstrap.min.css"

      rel="stylesheet"

      integrity="sha384-0evHe/X+R7YkIZDRvuzKMRqM+OrBnVFBL6DOitfPri4tjfhxaWutUpFmBp4vmVor"

      crossorigin="anonymous"

    />
```

```
<script
    src="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0-beta1/dist/js/b
ootstrap.bundle.min.js"

    integrity="sha384-pprn3073KE6tl6bjs2QrFaJGz5/SUsLqktiwsUTF55Jfv3q
YSDhgCecCxMW52nD2"

    crossorigin="anonymous"
></script>

<meta name="description" content="" />

<meta name="viewport" content="width=device-width, initial-scale=1"
/>

<link rel="stylesheet" href="" />
</head>
<body>

<!-- Navigation bar -->

<nav class="navbar navbar-light" style="background-color: #c1dce0">
    <div class="container">
        <a class="navbar-brand" href="/">
            <h1 class="display-6">NUMBER PLATE OCR</h1>
        </a>
    </div>
</nav>

<!-- Footer -->

<footer>
    <hr />
    <a href="http://aslanahmedov.com"> CONTACT ME</a>
</footer>

<script src="" async defer></script>
</body>
```

```
</html>
```

7.3 TEMPLATE INHERITANC

We will create new file as html and name it as index.html and we are planning to write all functionality to out index.html. First go to our layout.html under </nav> block our body - see below.

```
</nav>

{% block body %}


{% endblock %}
```

In index.html we need to extend layout.html and block body of it as below.

```
{% extends 'layout.html' %}

{% block body %}

{% endblock %}
```

In app.py change layout.html to index.html and save it. CTRL+S always-everywhere! We will add some forms as well to index.html see below.

```
{% extends 'layout.html' %}

{% block body %}

<div class="container">

    <br /><br /><br />

    <form action="#" method="POST" enctype="multipart/form-data">

        <div class="input-group">

            <input type="file" class="form-control" name="image_name" required />

            <input type="submit" value="UPLOAD" class="btn btn-outline-secondary" />

        </div>

    </form>

</div>
```

7.4 HTTP METHOD UPLOAD FILE IN FLASK

So now when we are upload the file, we need to receive it in flask and save it in some folder. That is called Static. For that create folder **static** and, in this folder, create folder **upload**. In app.py (our flask) import few necessary libraries as well add methods and create base path/upload path. See below

```
from flask import Flask, render_template, request

import os

# Webserver gateway interface

app = Flask(__name__)

BASE_PATH = os.getcwd()

UPLOAD_PATH = os.path.join(BASE_PATH, 'static/upload/')

@app.route('/', methods=['POST'])
def index():

    if request.method == 'POST':

        upload_file = request.files['image_name']

        filename = upload_file.filename

        path_save = os.path.join(UPLOAD_PATH, filename)

        upload_file.save(path_save)

        return render_template('index.html')

    return render_template('index.html')

if __name__ == "__main__":

    app.run(debug=True)
```

7.5 INTEGRATE NPD AND OCR TO FLASK APP

Now will make use of this upload image and we will make the predictions using deep learning model. Firstly, we will create one more file and name it as deeplearning.py in this file we are simply add our pipeline basically all necessary libraries and function with correct path to model. Create as well two more folder as roi and predict in static folder in the end your code will look like [this](#). On flask as well "add text" - you can find it [here](#). So, we are almost done. Now when we are uploading picture of car with plate number our model predicts bounding box, crop it at same time, save outputs to folder roi and folder predict. In order to display it on web we will modify index.html and add text future to save text as well. You can find ready code [here](#). And on *Figure 19* you can see result. All files and folder for Web please feel free to check [here](#).

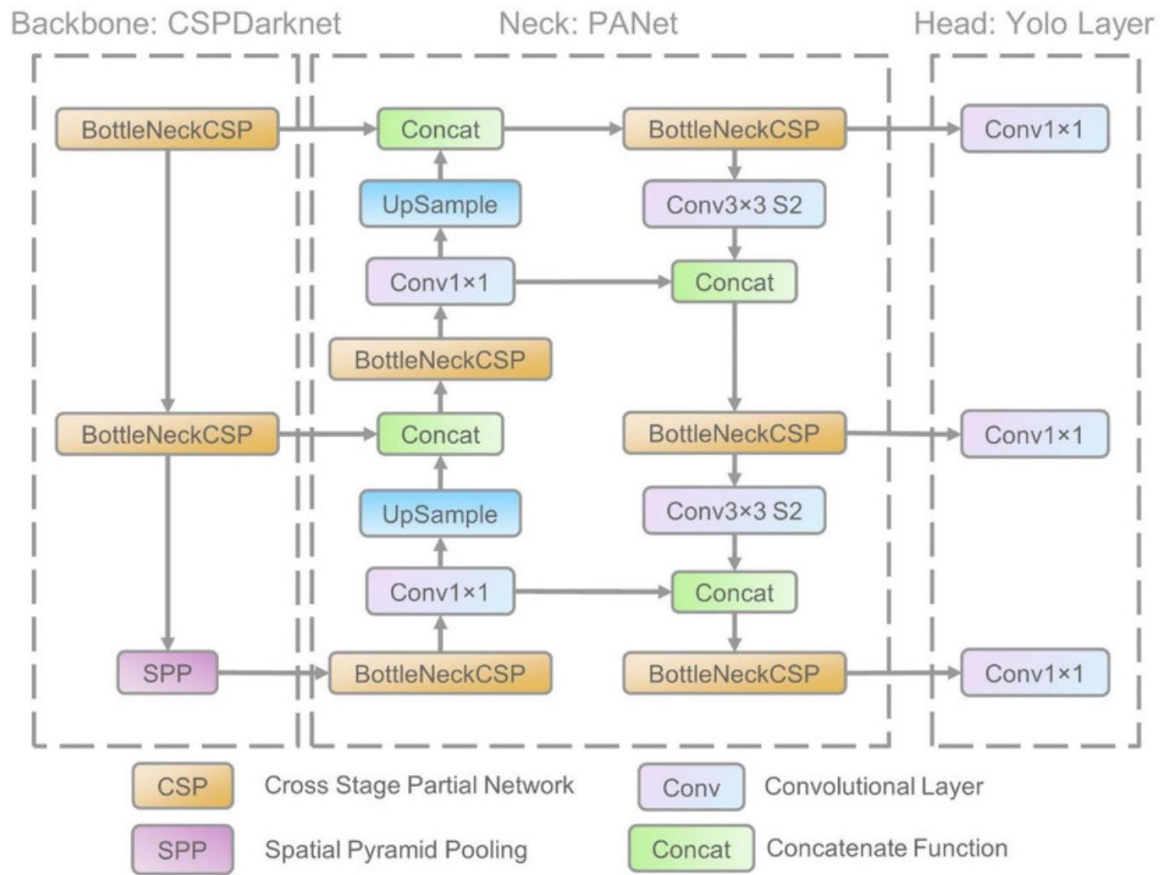


Figure 19

8. RAEAL TIME NUMBER PLATE RECOGNITIONT WITH YOLO

8.1 EXPLANATION OF REQUIRED DATA

YOLO or You only look once, is one of the most widely used, deep learning-based object detection algorithm out there. YOLO divides an image into a grid system, and each grid detects objects within itself. They can be used for real-time object detection based on the data streams. They require very few computational resources. The network architecture of YOLOv5. It consists of three parts: (1) Backbone: CSPDarknet, (2) Neck: PANet, and (3) Head: Yolo Layer. The data are first input to CSPDarknet for feature extraction, and then fed to PANet for feature fusion. Finally, Yolo Layer outputs detection results (class, score, location, size).



Our previous process is great, but there is a problem with the model. The major problem with the model is model has very low precision in detecting the license plate and also it is very, very slow processing. To overcome the above limitations, particularly the precision, we will use the most powerful of the reduction model available till now is YOLO. Now, let us see how we can use our existing data preparation to our YOLO model. The labeling and the data preprocessing work, what we have did for the existing work can be applied to the same Yola model. But there is one major difference we need to do in YOLO is our label. Previously that you have seen, we used the xmin, xmax, ymin, ymax. and VI Max is our actual output. But for YOLO is a center X-Y position of the bounding box and width and height of the bounding box, and particularly the X-Y position is referred to the center of the bounding box. Let me give you an example *Figure 20*. For this image, the labeling for the YOLO should be the center X and Y, and which should be the normalized between the width and height of the image. And W and H are the respective width and height of the bounding box normalized to width and height of the image.

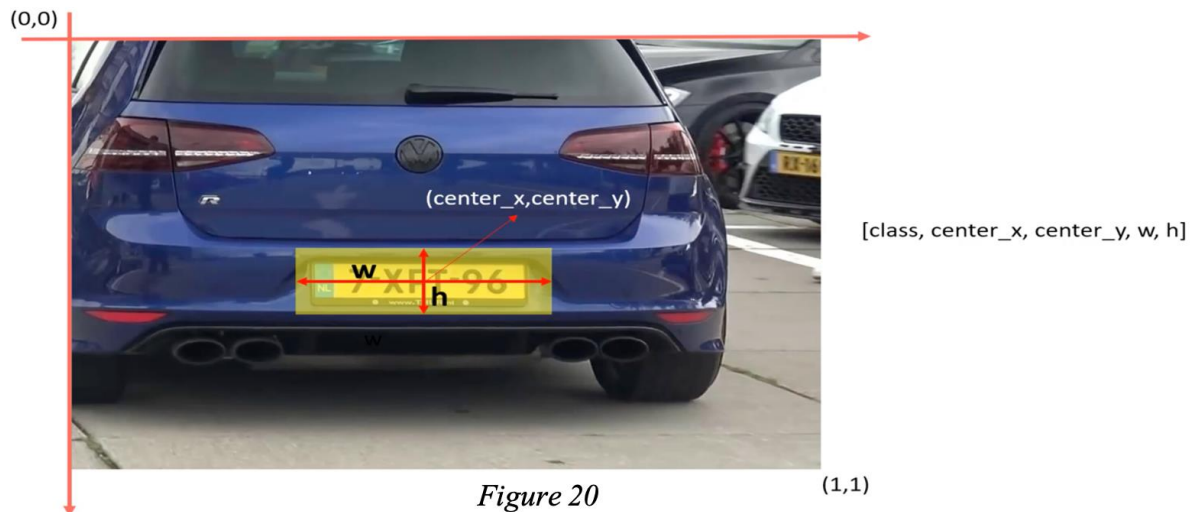


Figure 20

So, in the labeling process, basically this is what the format we now have to prepare our data plus center position of X center position of Y this refers to the bounding box and width and height of the bounding box. [class, center_x, center_y, w, h]. That is what one of the changes we need to make. And the next thing is that the folder structure which we need to prepare, particularly for the YOLO, will look something like Figure 21.

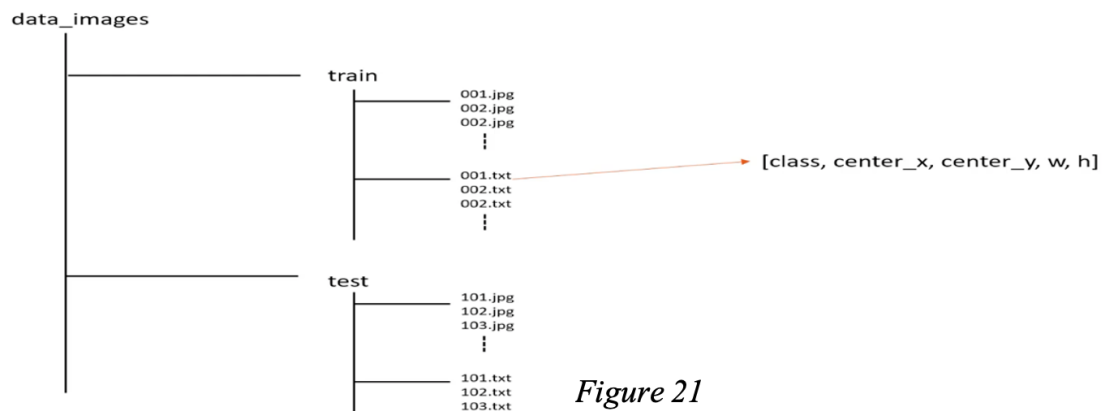


Figure 21

Let consider the data images. Inside the data images we should have train and test. In the training test folder, we have to maintain all our images related to the training and the corresponding labels should be there in .txt format. And moreover, the label information data should consist same name as our image. Just like our training data, we will also prepare our test data.

I will write function and using that function, I can try to extract weight of the image, height of the image and also the file name from the xml files. Creating function parsing and return will be our filename, width and height and apply all to our data frame as below.

In [27]:

```
# parsing
def parsing(path):
    parser = xet.parse(path).getroot()
    name = parser.find('filename').text
    filename = f'../input/number-plate-detection/images/{name}'

    # width and height
```

```

parser_size = parser.find('size')
width = int(parser_size.find('width').text)
height = int(parser_size.find('height').text)

return filename, width, height
df[['filename', 'width', 'height']] = df['filepath'].apply(parsing).apply(pd.
Series)
df.head()

```

Out[27]:

	filepath	xmin	xmax	ymin	ymax	filename	width	height
0	../input/number-plate-detection/images/N148.xml	244	369	240	293	../input/number-plate-detection/images/N148.jpeg	600	430
1	../input/number-plate-detection/images/N177.xml	331	538	263	317	../input/number-plate-detection/images/N177.jpeg	850	681
2	../input/number-plate-detection/images/N173.xml	80	335	150	243	../input/number-plate-detection/images/N173.jpeg	445	349
3	../input/number-plate-detection/images/N213.xml	131	209	129	153	../input/number-plate-detection/images/N213.jpeg	350	196
4	../input/number-plate-detection/images/N119.xml	180	559	216	314	../input/number-plate-detection/images/N119.jpeg	678	475

Now the next step is let me calculate the center_x, center_y, width and height, which is normalized to width and height. And as well normalize width and height of bounding box.

In [28]:

```

# center_x, center_y, width , height
df['center_x'] = (df['xmax'] + df['xmin'])/(2*df['width'])
df['center_y'] = (df['ymax'] + df['ymin'])/(2*df['height'])

df['bb_width'] = (df['xmax'] - df['xmin'])/df['width']
df['bb_height'] = (df['ymax'] - df['ymin'])/df['height']
df.head()

```

Out[28]:

	filepath	xmin	xmax	ymin	ymax	filename	width	height	center_x	center_y	bb_width	bb_height
0	../input/number-plate-detection/images/N148.xml	244	369	240	293	../input/number-plate-detection/images/N148.jpeg	600	430	0.510833	0.619767	0.208333	0.123256
1	../input/number-plate-detection/images/N177.xml	331	538	263	317	../input/number-plate-detection/images/N177.jpeg	850	681	0.511176	0.425844	0.243529	0.079295
2	../input/number-plate-detection/images/N173.xml	80	335	150	243	../input/number-plate-detection/images/N173.jpeg	445	349	0.466292	0.563037	0.573034	0.266476
3	../input/number-plate-detection/images/N213.xml	131	209	129	153	../input/number-plate-detection/images/N213.jpeg	350	196	0.485714	0.719388	0.222857	0.122449
4	../input/number-plate-detection/images/N119.xml	180	559	216	314	../input/number-plate-detection/images/N119.jpeg	678	475	0.544985	0.557895	0.558997	0.206316

8.2 DATA PREPARATION

One of the major challenges in the training of YOLO is hardware itself. YOLO requires a fast-processing chip used to get trained. It is extremely difficult to train the YOLO in normal CPUs. Firstly, we have to clone YOLOV5 to our work space. You can find it [here](#) and install all requirements. Make sure you in right folder type as below, let's do it.

In [29]:

```
!git clone https://github.com/ultralytics/yolov5
Cloning into 'yolov5'...
remote: Enumerating objects: 12264, done.
remote: Counting objects: 100% (35/35), done.
remote: Compressing objects: 100% (22/22), done.
remote: Total 12264 (delta 18), reused 22 (delta 13), pack-reused 12229
Receiving objects: 100% (12264/12264), 11.95 MiB | 21.74 MiB/s, done.
Resolving deltas: 100% (8488/8488), done.
```

In [30]:

```
!pip install -r ./yolov5/requirements.txt
```

Requirement already satisfied: matplotlib>=3.2.2 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 5)) (3.5.2)

Requirement already satisfied: numpy>=1.18.5 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 6)) (1.21.6)

Requirement already satisfied: opencv-python>=4.1.1 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 7)) (4.5.4.60)

Requirement already satisfied: Pillow>=7.1.2 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 8)) (9.1.0)

Requirement already satisfied: PyYAML>=5.3.1 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 9)) (6.0)

Requirement already satisfied: requests>=2.23.0 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 10)) (2.27.1)

Requirement already satisfied: scipy>=1.4.1 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 11)) (1.7.3)

Requirement already satisfied: torch>=1.7.0 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 12)) (1.11.0)

Requirement already satisfied: torchvision>=0.8.1 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 13)) (0.12.0)

Requirement already satisfied: tqdm>=4.41.0 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 14)) (4.64.0)

Requirement already satisfied: protobuf<=3.20.1 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 15)) (3.20.1)

Requirement already satisfied: tensorboard>=2.4.1 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 18)) (2.6.0)

Requirement already satisfied: pandas>=1.1.4 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 22)) (1.3.5)

Requirement already satisfied: seaborn>=0.11.0 in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 23)) (0.11.2)

Requirement already satisfied: ipython in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 35)) (7.33.0)

Requirement already satisfied: psutil in /opt/conda/lib/python3.7/site-packages (from -r ./yolov5/requirements.txt (line 36)) (5.9.0)

Collecting thop

Downloading thop-0.1.0.post2206102148-py3-none-any.whl (15 kB)

Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.7/site-packages (from matplotlib>=3.2.2->-r ./yolov5/requirements.txt (line 5)) (21.3)

Requirement already satisfied: python-dateutil>=2.7 in /opt/conda/lib/python3.7/site-packages (from matplotlib>=3.2.2->-r ./yolov5/requirements.txt (line 5)) (2.8.2)

Requirement already satisfied: fonttools>=4.22.0 in /opt/conda/lib/python3.7/site-packages (from matplotlib>=3.2.2->-r ./yolov5/requirements.txt (line 5)) (4.33.3)

Requirement already satisfied: kiwisolver>=1.0.1 in /opt/conda/lib/python3.7/site-packages (from matplotlib>=3.2.2->-r ./yolov5/requirements.txt (line 5)) (1.4.2)

Requirement already satisfied: cyclor>=0.10 in /opt/conda/lib/python3.7/site-packages (from matplotlib>=3.2.2->-r ./yolov5/requirements.txt (line 5)) (0.11.0)

Requirement already satisfied: pyparsing>=2.2.1 in /opt/conda/lib/python3.7/site-packages (from matplotlib>=3.2.2->-r ./yolov5/requirements.txt (line 5)) (3.0.9)

Requirement already satisfied: charset-normalizer~2.0.0 in /opt/conda/lib/python3.7/site-packages (from requests>=2.23.0->-r ./yolov5/requirements.txt (line 10)) (2.0.12)

Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.7/site-packages (from requests>=2.23.0->-r ./yolov5/requirements.txt (line 10)) (3.3)

Requirement already satisfied: urllib3<1.27,>=1.21.1 in /opt/conda/lib/python3.7/site-packages (from requests>=2.23.0->-r ./yolov5/requirements.txt (line 10)) (1.26.9)

Requirement already satisfied: certifi>=2017.4.17 in /opt/conda/lib/python3.7/site-packages (from requests>=2.23.0->-r ./yolov5/requirements.txt (line 10)) (2022.5.18.1)

Requirement already satisfied: typing-extensions in /opt/conda/lib/python3.7/site-packages (from torch>=1.7.0->-r ./yolov5/requirements.txt (line 12)) (4.2.0)

Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (0.4.6)

Requirement already satisfied: tensorboard-plugin-wit>=1.6.0 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (1.8.1)

Requirement already satisfied: werkzeug>=0.11.15 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (2.1.2)

Requirement already satisfied: wheel>=0.26 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (0.37.1)

Requirement already satisfied: setuptools>=41.0.0 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (59.8.0)

Requirement already satisfied: grpcio>=1.24.3 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (1.43.0)

Requirement already satisfied: google-auth<2,>=1.6.3 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (1.35.0)

Requirement already satisfied: tensorboard-data-server<0.7.0,>=0.6.0 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (0.6.1)

Requirement already satisfied: absl-py>=0.4 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (1.0.0)

Requirement already satisfied: markdown>=2.6.8 in /opt/conda/lib/python3.7/site-packages (from tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (3.3.7)

Requirement already satisfied: pytz>=2017.3 in /opt/conda/lib/python3.7/site-packages (from pandas>=1.1.4->-r ./yolov5/requirements.txt (line 22)) (2022.1)

Requirement already satisfied: jedi>=0.16 in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (0.18.1)

Requirement already satisfied: decorator in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (5.1.1)

Requirement already satisfied: pygments in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (2.12.0)

Requirement already satisfied: traitlets>=4.2 in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (5.2.0)

Requirement already satisfied: matplotlib-inline in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (0.1.3)

Requirement already satisfied: pickleshare in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (0.7.5)

Requirement already satisfied: pexpect>4.3 in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (4.8.0)

Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (3.0.29)

Requirement already satisfied: backcall in /opt/conda/lib/python3.7/site-packages (from ipython->-r ./yolov5/requirements.txt (line 35)) (0.2.0)

Requirement already satisfied: six in /opt/conda/lib/python3.7/site-packages (from absl-py>=0.4->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (1.16.0)

Requirement already satisfied: rsa<5,>=3.1.4 in /opt/conda/lib/python3.7/site-packages (from google-auth<2,>=1.6.3->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (4.8)

Requirement already satisfied: cachetools<5.0,>=2.0.0 in /opt/conda/lib/python3.7/site-packages (from google-auth<2,>=1.6.3->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (4.2.4)

Requirement already satisfied: pyasn1-modules<0.2.1 in /opt/conda/lib/python3.7/site-packages (from google-auth<2,>=1.6.3->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (0.2.7)

Requirement already satisfied: requests-oauthlib<0.7.0 in /opt/conda/lib/python3.7/site-packages (from google-auth-oauthlib<0.5,>=0.4.1->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (1.3.1)

Requirement already satisfied: parso<0.9.0,>=0.8.0 in /opt/conda/lib/python3.7/site-packages (from jedi>=0.16->ipython->-r ./yolov5/requirements.txt (line 35)) (0.8.3)

Requirement already satisfied: importlib-metadata>=4.4 in /opt/conda/lib/python3.7/site-packages (from markdown>=2.6.8->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (4.11.4)

Requirement already satisfied: ptyprocess>=0.5 in /opt/conda/lib/python3.7/site-packages (from pexpect>4.3->ipython->-r ./yolov5/requirements.txt (line 35)) (0.7.0)

Requirement already satisfied: wcwidth in /opt/conda/lib/python3.7/site-packages (from prompt-toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->-r ./yolov5/requirements.txt (line 35)) (0.2.5)

Requirement already satisfied: zipp>=0.5 in /opt/conda/lib/python3.7/site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (3.8.0)

Requirement already satisfied: pyasn1<0.5.0,>=0.4.6 in /opt/conda/lib/python3.7/site-packages (from pyasn1-modules>=0.2.1->google-auth<2,>=1.6.3->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (0.4.8)

Requirement already satisfied: oauthlib>=3.0.0 in /opt/conda/lib/python3.7/site-packages (from requests-oauthlib>=0.7.0->google-auth-oauthlib<0.5,>=0.4.1->tensorboard>=2.4.1->-r ./yolov5/requirements.txt (line 18)) (3.2.0)

Installing collected packages: thop

Successfully installed thop-0.1.0.post2206102148

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

We need to make a one more folder and make into it two more - name it as **data_images** and **train** and **test**. We will split data to train and split.

```
In [31]: mkdir /kaggle/working/yolov5/data_images/
```

```
In [32]: mkdir /kaggle/working/yolov5/data_images/test/
```

```
In [33]: mkdir /kaggle/working/yolov5/data_images/train/
```

```
In [34]: ### split the data into train and test
df_train = df.iloc[:200]
df_test = df.iloc[200:]
```

We will copy each image into the folder train and test and generate .txt which has label info.

In [35]:

```
train_folder = './yolov5/data_images/train'

values = df_train[['filename', 'center_x', 'center_y', 'bb_width', 'bb_height']]
values = values.values
for fname, x, y, w, h in values:
    image_name = os.path.splitext(fname)[-1]
    txt_name = os.path.splitext(image_name)[0]

    dst_image_path = os.path.join(train_folder, image_name)
    dst_label_file = os.path.join(train_folder, txt_name+'.txt')

    # copy each image into the folder
    copy(fname, dst_image_path)

    # generate .txt which has label info
    label_txt = f'0 {x} {y} {w} {h}'
    with open(dst_label_file, mode='w') as f:
        f.write(label_txt)

    f.close()

test_folder = './yolov5/data_images/test'

values = df_test[['filename', 'center_x', 'center_y', 'bb_width', 'bb_height']]
values = values.values
for fname, x, y, w, h in values:
    image_name = os.path.splitext(fname)[-1]
    txt_name = os.path.splitext(image_name)[0]

    dst_image_path = os.path.join(test_folder, image_name)
    dst_label_file = os.path.join(test_folder, txt_name+'.txt')

    # copy each image into the folder
    copy(fname, dst_image_path)

    # generate .txt which has label info
    label_txt = f'0 {x} {y} {w} {h}'
    with open(dst_label_file, mode='w') as f:
        f.write(label_txt)

    f.close()
```

As well I will create **data.yaml** file with path to train and validation, as well number of classes as 1, name it in list as license plate. You will need to save this in your dataset because we will need to provide the path on it later.

```
train: data_images/train

val: data_images/test

nc: 1

names: [
```

```
'license_plate'
]
```

8.3 TRAINING YOLO

Next step is basically train model. It could take time be ready for it. You can use Kaggle or GoogleColab for it. We already set out data.yaml file, lets us give path to it and train model. Below you will find code to cleare you GPU(It helped me a lot and will save you time.)

In [36]:

```
!pip install GPUtil

import torch
from GPUtil import showUtilization as gpu_usage
from numba import cuda

def free_gpu_cache():
    print("Initial GPU Usage")
    gpu_usage()

    torch.cuda.empty_cache()

    cuda.select_device(0)
    cuda.close()
    cuda.select_device(0)

    print("GPU Usage after emptying the cache")
    gpu_usage()

free_gpu_cache()
Collecting GPUtil
  Downloading GPUtil-1.4.0.tar.gz (5.5 kB)
  Preparing metadata (setup.py) ... - done
Building wheels for collected packages: GPUtil
  Building wheel for GPUtil (setup.py) ... - \ done
  Created wheel for GPUtil: filename=GPUtil-1.4.0-py3-none-any.whl size
=7411 sha256=cc480cf92970055e92e6f363d9212080b9e8864225da31712be62e02e5
f746b7
  Stored in directory: /root/.cache/pip/wheels/6e/f8/83/534c52482d6da64
622ddb72cd93c35d2ef2881b78fd08ff0c
Successfully built GPUtil
Installing collected packages: GPUtil
Successfully installed GPUtil-1.4.0
WARNING: Running pip as the 'root' user can result in broken permission
s and conflicting behaviour with the system package manager. It is reco
mmended to use a virtual environment instead: https://pip.pypa.io/warni
ngs/venv
Initial GPU Usage
| ID | GPU | MEM |
-----
```

```

| 0 | 0% | 98% |
GPU Usage after emptying the cache
| ID | GPU | MEM |
-----
| 0 | 100% | 2% |

```

In [37]:

```
!python ./yolov5/train.py --data ../input/number-plate-detection/data.yaml
--cfg ./yolov5/models/yolov5s.yaml --batch-size 8 --name Model --epochs 100
```

wandb: (1) Create a W&B account

wandb: (2) Use an existing W&B account

wandb: (3) Don't visualize my results

wandb: Enter your choice: (30 second timeout)

wandb: W&B disabled due to login timeout.

train: weights=yolov5/yolov5s.pt, cfg=./yolov5/models/yolov5s.yaml, dat
a=../input/number-plate-detection/data.yaml, hyp=yolov5/data/hyps/hyp.s
cratch-low.yaml, epochs=100, batch_size=8, imgsz=640, rect=False, resum
e=False, nosave=False, noval=False, noautoanchor=False, noplots=False,
evolve=None, bucket=, cache=None, image_weights=False, device=, multi_s
cale=False, single_cls=False, optimizer=SGD, sync_bn=False, workers=8,
project=yolov5/runs/train, name=Model, exist_ok=False, quad=False, cos_
lr=False, label_smoothing=0.0, patience=100, freeze=[0], save_period=-1
, local_rank=-1, entity=None, upload_dataset=False, bbox_interval=-1, a
rtifact_alias=latest

github: up to date with <https://github.com/ultralytics/yolov5> ☒

YOLOv5  v6.1-253-g75bbaa8 Python-3.7.12 torch-1.11.0 CUDA:0 (Tesla P1
00-PCIE-16GB, 16281MiB)

hyperparameters: lr0=0.01, lrf=0.01, momentum=0.937, weight_decay=0.000
5, warmup_epochs=3.0, warmup_momentum=0.8, warmup_bias_lr=0.1, box=0.05
, cls=0.5, cls_pw=1.0, obj=1.0, obj_pw=1.0, iou_t=0.2, anchor_t=4.0, fl
_gamma=0.0, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, degrees=0.0, translate=0
.1, scale=0.5, shear=0.0, perspective=0.0, flipud=0.0, fliplr=0.5, mosa
ic=1.0, mixup=0.0, copy_paste=0.0

Weights & Biases: run 'pip install wandb' to automatically track and vi
sualize YOLOv5  runs (RECOMMENDED)

TensorBoard: Start with 'tensorboard --logdir yolov5/runs/train', view
at <http://localhost:6006/>

Downloading <https://ultralytics.com/assets/Arial.ttf> to /root/.config/U
ltralytics/Arial.ttf...

100%|██| 755k/755k [00:00<
00:00, 26.3MB/s]

Downloading [https://github.com/ultralytics/yolov5/releases/download/v6.
1/yolov5s.pt](https://github.com/ultralytics/yolov5/releases/download/v6.1/yolov5s.pt) to yolov5/yolov5s.pt...

100%|██| 14.1M/14.1M [00:00
<00:00, 132MB/s]

Overriding model.yaml nc=80 with nc=1

	arguments	from	n	params	module
	0	-1	1	3520	models.common.Conv
	[3, 32, 6, 2, 2]				
	1	-1	1	18560	models.common.Conv
	[32, 64, 3, 2]				
	2	-1	1	18816	models.common.C3
	[64, 64, 1]				
	3	-1	1	73984	models.common.Conv
	[64, 128, 3, 2]				
	4	-1	2	115712	models.common.C3
	[128, 128, 2]				
	5	-1	1	295424	models.common.Conv
	[128, 256, 3, 2]				
	6	-1	3	625152	models.common.C3
	[256, 256, 3]				
	7	-1	1	1180672	models.common.Conv
	[256, 512, 3, 2]				
	8	-1	1	1182720	models.common.C3
	[512, 512, 1]				
	9	-1	1	656896	models.common.SPPF
	[512, 512, 5]				
	10	-1	1	131584	models.common.Conv
	[512, 256, 1, 1]				
	11	-1	1	0	torch.nn.modules.upsampling.Upsamp1
e	[None, 2, 'nearest']				
	12	[-1, 6]	1	0	models.common.Concat
	[1]				
	13	-1	1	361984	models.common.C3
	[512, 256, 1, False]				
	14	-1	1	33024	models.common.Conv
	[256, 128, 1, 1]				
	15	-1	1	0	torch.nn.modules.upsampling.Upsamp1
e	[None, 2, 'nearest']				
	16	[-1, 4]	1	0	models.common.Concat
	[1]				
	17	-1	1	90880	models.common.C3
	[256, 128, 1, False]				
	18	-1	1	147712	models.common.Conv
	[128, 128, 3, 2]				
	19	[-1, 14]	1	0	models.common.Concat
	[1]				
	20	-1	1	296448	models.common.C3
	[256, 256, 1, False]				
	21	-1	1	590336	models.common.Conv
	[256, 256, 3, 2]				
	22	[-1, 10]	1	0	models.common.Concat
	[1]				
	23	-1	1	1182720	models.common.C3
	[512, 512, 1, False]				

```
24      [17, 20, 23] 1      16182 models.yolo.Detect
[1, [[10, 13, 16, 30, 33, 23], [30, 61, 62, 45, 59, 119], [116, 90, 156
, 198, 373, 326]], [128, 256, 512]]
YOLOv5s summary: 270 layers, 7022326 parameters, 7022326 gradients
```

Transferred 342/349 items from yolov5/yolov5s.pt

AMP: checks passed ☒

Scaled weight_decay = 0.0005

optimizer: SGD with parameter groups 57 weight (no decay), 60 weight, 60 bias

augmentations: Blur(always_apply=False, p=0.01, blur_limit=(3, 7)), MedianBlur(always_apply=False, p=0.01, blur_limit=(3, 7)), ToGray(always_apply=False, p=0.01), CLAHE(always_apply=False, p=0.01, clip_limit=(1, 4.0), tile_grid_size=(8, 8))

train: Scanning '/kaggle/working/yolov5/data_images/train' images and labels...2

train: New cache created: /kaggle/working/yolov5/data_images/train.cache

val: Scanning '/kaggle/working/yolov5/data_images/test' images and labels...25 f

val: New cache created: /kaggle/working/yolov5/data_images/test.cache
Plotting labels to yolov5/runs/train/Model/labels.jpg...

AutoAnchor: 3.46 anchors/target, 1.000 Best Possible Recall (BPR). Current anchors are a good fit to dataset ☒

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to **yolov5/runs/train/Model**

Starting training for 100 epochs...

Epoch	gpu_mem	box	obj	cls	labels	img_size
0/99	1.94G	0.1113	0.02923	0	9	640:
100%						
	Class	Images	Labels	P	R	mA
P@.5 mAP@	WARNING: NMS time limit 0.570s exceeded					
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	23	0.00626	0.0435	0.0
0477	0.000584					

Epoch	gpu_mem	box	obj	cls	labels	img_size
1/99	2.02G	0.0895	0.02403	0	10	640:
100%						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.00267	0.8	0.
0287	0.00577					

Epoch	gpu_mem	box	obj	cls	labels	img_size
-------	---------	-----	-----	-----	--------	----------

2/99 2.02G 0.08548 0.02064 0 15 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.00253 0.76 0.
0304 0.00464

Epoch gpu_mem box obj cls labels img_size
3/99 2.02G 0.08297 0.01755 0 13 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.0028 0.84 0.
0558 0.00967

Epoch gpu_mem box obj cls labels img_size
4/99 2.02G 0.07817 0.01615 0 16 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.00307 0.92 0
.156 0.036

Epoch gpu_mem box obj cls labels img_size
5/99 2.02G 0.06983 0.01898 0 11 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.0766 0.72 0
.382 0.151

Epoch gpu_mem box obj cls labels img_size
6/99 2.02G 0.06894 0.01614 0 18 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.544 0.717 0
.641 0.314

Epoch gpu_mem box obj cls labels img_size
7/99 2.02G 0.0568 0.01708 0 15 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.55 0.48
0.44 0.0787

Epoch gpu_mem box obj cls labels img_size
8/99 2.02G 0.05308 0.01872 0 26 640:
100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.756	0.621	0
.671	0.177					

Epoch	gpu_mem	box	obj	cls	labels	img_size
9/99	2.02G	0.05659	0.01573	0	14	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.95	0.753	0
.835	0.313					

Epoch	gpu_mem	box	obj	cls	labels	img_size
10/99	2.02G	0.05794	0.01519	0	15	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.952	0.79	0
.883	0.516					

Epoch	gpu_mem	box	obj	cls	labels	img_size
11/99	2.02G	0.05122	0.01467	0	8	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.835	0.813	0
.846	0.367					

Epoch	gpu_mem	box	obj	cls	labels	img_size
12/99	2.02G	0.04583	0.01385	0	15	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.963	0.8	0
.901	0.426					

Epoch	gpu_mem	box	obj	cls	labels	img_size
13/99	2.02G	0.04247	0.01479	0	18	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.984	0.8	0
.897	0.594					

Epoch	gpu_mem	box	obj	cls	labels	img_size
14/99	2.02G	0.03978	0.01352	0	9	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@						

		all	25	25	0.966	0.8	0
.902	0.262						

Epoch	gpu_mem	box	obj	cls	labels	img_size	
15/99	2.02G	0.04239	0.01264	0	12	640:	
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	0.957	0.89	0
.946	0.557						

Epoch	gpu_mem	box	obj	cls	labels	img_size	
16/99	2.02G	0.0394	0.01234	0	15	640:	
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	0.956	0.864	0
.943	0.446						

Epoch	gpu_mem	box	obj	cls	labels	img_size	
17/99	2.02G	0.04438	0.01108	0	15	640:	
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	0.988	0.88	0
.944	0.421						

Epoch	gpu_mem	box	obj	cls	labels	img_size	
18/99	2.02G	0.03982	0.01153	0	15	640:	
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	0.898	0.8	
0.91	0.338						

Epoch	gpu_mem	box	obj	cls	labels	img_size	
19/99	2.02G	0.04133	0.01102	0	13	640:	
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	0.994	0.88	0
.966	0.416						

Epoch	gpu_mem	box	obj	cls	labels	img_size	
20/99	2.02G	0.03976	0.01114	0	18	640:	
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.91	0
.973	0.515						

Epoch	gpu_mem	box	obj	cls	labels	img_size
21/99	2.02G	0.03741	0.01067	0	19	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.885	0.92	0
.962	0.524					

Epoch	gpu_mem	box	obj	cls	labels	img_size
22/99	2.02G	0.03543	0.00989	0	12	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.919	0.913	0
.957	0.467					

Epoch	gpu_mem	box	obj	cls	labels	img_size
23/99	2.02G	0.03809	0.01015	0	17	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.922	0.96	
0.98	0.632					

Epoch	gpu_mem	box	obj	cls	labels	img_size
24/99	2.02G	0.03496	0.0105	0	11	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.98	0.96	0
.984	0.535					

Epoch	gpu_mem	box	obj	cls	labels	img_size
25/99	2.02G	0.0363	0.009415	0	11	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	1	0.946	0
.987	0.667					

Epoch	gpu_mem	box	obj	cls	labels	img_size
26/99	2.02G	0.03264	0.01044	0	20	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.96	0.952	0
.986	0.56					

Epoch	gpu_mem	box	obj	cls	labels	img_size
-------	---------	-----	-----	-----	--------	----------

27/99 2.02G 0.03544 0.008677 0 20 640:
 100%|██████████
 Class Images Labels P R mA
 P@.5 mAP@
 all 25 25 0.991 0.92 0
 .982 0.604

Epoch gpu_mem box obj cls labels img_size
 28/99 2.02G 0.03457 0.00969 0 15 640:
 100%|██████████
 Class Images Labels P R mA
 P@.5 mAP@
 all 25 25 0.996 0.92 0
 .977 0.541

Epoch gpu_mem box obj cls labels img_size
 29/99 2.02G 0.03273 0.009109 0 15 640:
 100%|██████████
 Class Images Labels P R mA
 P@.5 mAP@
 all 25 25 0.864 0.96 0
 .977 0.656

Epoch gpu_mem box obj cls labels img_size
 30/99 2.02G 0.02857 0.008326 0 12 640:
 100%|██████████
 Class Images Labels P R mA
 P@.5 mAP@
 all 25 25 0.961 0.994 0
 .989 0.665

Epoch gpu_mem box obj cls labels img_size
 31/99 2.02G 0.02974 0.008331 0 12 640:
 100%|██████████
 Class Images Labels P R mA
 P@.5 mAP@
 all 25 25 0.961 0.997
 0.99 0.671

Epoch gpu_mem box obj cls labels img_size
 32/99 2.02G 0.03072 0.008841 0 16 640:
 100%|██████████
 Class Images Labels P R mA
 P@.5 mAP@
 all 25 25 0.992 0.96 0
 .993 0.61

Epoch gpu_mem box obj cls labels img_size
 33/99 2.02G 0.03573 0.00819 0 17 640:
 100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	1	0.956	0
.993	0.681					

Epoch	gpu_mem	box	obj	cls	labels	img_size
34/99	2.02G	0.03354	0.008475	0	12	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.958	0.96	0
.976	0.639					

Epoch	gpu_mem	box	obj	cls	labels	img_size
35/99	2.02G	0.0312	0.008276	0	17	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	1	0.953	0
.993	0.655					

Epoch	gpu_mem	box	obj	cls	labels	img_size
36/99	2.02G	0.03132	0.00869	0	14	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.997	1	0
.995	0.683					

Epoch	gpu_mem	box	obj	cls	labels	img_size
37/99	2.02G	0.02638	0.00728	0	19	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.962	0.999	0
.992	0.636					

Epoch	gpu_mem	box	obj	cls	labels	img_size
38/99	2.02G	0.02823	0.008421	0	18	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.993	0.96	0
.992	0.716					

Epoch	gpu_mem	box	obj	cls	labels	img_size
39/99	2.02G	0.02686	0.00762	0	16	640:

100%|██████████

	Class	Images	Labels	P	R	mA
P@.5 mAP@						

		all	25	25	0.924	1	
0.99	0.585						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	40/99	2.02G	0.02894	0.007577	0	16	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	1	0.954	0
.992	0.662						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	41/99	2.02G	0.02987	0.007458	0	15	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.961	1	0
.993	0.626						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	42/99	2.02G	0.02882	0.008058	0	13	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.993	0.96	0
.992	0.699						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	43/99	2.02G	0.02675	0.008497	0	9	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.996	0.96	0
.992	0.52						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	44/99	2.02G	0.02897	0.007826	0	13	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.998	0.96	0
.992	0.676						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	45/99	2.02G	0.02612	0.00776	0	15	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.998	0.96	0
.993	0.654						

Epoch	gpu_mem	box	obj	cls	labels	img_size
46/99	2.02G	0.02492	0.007731	0	12	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.958	1	0
.993	0.676					

Epoch	gpu_mem	box	obj	cls	labels	img_size
47/99	2.02G	0.0236	0.006941	0	17	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.961	1	0
.993	0.695					

Epoch	gpu_mem	box	obj	cls	labels	img_size
48/99	2.02G	0.02759	0.007434	0	13	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.957	1	0
.993	0.673					

Epoch	gpu_mem	box	obj	cls	labels	img_size
49/99	2.02G	0.02769	0.007337	0	18	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.959	1	0
.993	0.703					

Epoch	gpu_mem	box	obj	cls	labels	img_size
50/99	2.02G	0.02562	0.007327	0	16	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.961	0.997	0
.992	0.704					

Epoch	gpu_mem	box	obj	cls	labels	img_size
51/99	2.02G	0.02445	0.006713	0	12	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5 mAP@						
	all	25	25	0.961	0.996	0
.992	0.635					

Epoch	gpu_mem	box	obj	cls	labels	img_size
-------	---------	-----	-----	-----	--------	----------

52/99 2.02G 0.02551 0.007237 0 14 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.961 0.996 0
.993 0.679

Epoch gpu_mem box obj cls labels img_size
53/99 2.02G 0.02391 0.007192 0 17 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.961 0.999
0.99 0.672

Epoch gpu_mem box obj cls labels img_size
54/99 2.02G 0.02185 0.006582 0 15 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.962 1 0
.993 0.7

Epoch gpu_mem box obj cls labels img_size
55/99 2.02G 0.02463 0.00661 0 17 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.961 0.994 0
.992 0.713

Epoch gpu_mem box obj cls labels img_size
56/99 2.02G 0.02295 0.007 0 12 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.96 0.952
0.99 0.699

Epoch gpu_mem box obj cls labels img_size
57/99 2.02G 0.02324 0.00638 0 19 640:
100%|██████████
Class Images Labels P R mA
P@.5 mAP@
all 25 25 0.947 1 0
.992 0.723

Epoch gpu_mem box obj cls labels img_size
58/99 2.02G 0.02355 0.006503 0 16 640:
100%|██████████

	Class	Images	Labels	P	R	mA
P@.5	all	25	25	0.958	0.96	0
mAP@	0.69					

Epoch	gpu_mem	box	obj	cls	labels	img_size
60/99	2.02G	0.0226	0.006495	0	17	640:
100% ██████████						
P@.5	Class	Images	Labels	P	R	mA
mAP@						
	all	25	25	0.961	0.994	0
.992	0.731					

Epoch	gpu_mem	box	obj	cls	labels	img_size
62/99	2.02G	0.02364	0.006801	0	13	640:
100% ██████████						
P@.5	Class	Images	Labels	P	R	mA
mAP@						
	all	25	25	0.961	0.995	0
.993	0.689					

Epoch	gpu_mem	box	obj	cls	labels	img_size
64/99	2.02G	0.0226	0.006393	0	16	640:
100% ██████████						
	Class	Images	Labels		P	R
P@.5	mAP@					mA

		all	25	25	0.961	0.992	0
.993	0.669						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	65/99	2.02G	0.02393	0.006678	0	20	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.961	0.997	0
.993	0.711						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	66/99	2.02G	0.02248	0.006522	0	17	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.955	1	0
.992	0.694						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	67/99	2.02G	0.02297	0.006624	0	24	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.961	0.998	0
.993	0.648						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	68/99	2.02G	0.02134	0.006086	0	13	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.961	0.996	0
.993	0.665						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	69/99	2.02G	0.02152	0.006414	0	13	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.959	1	0
.993	0.641						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	70/99	2.02G	0.02107	0.006244	0	17	640:
100%							
		Class	Images	Labels	P	R	mA
P@.5	mAP@						
		all	25	25	0.96	1	0
.993	0.7						

Epoch	gpu_mem	box	obj	cls	labels	img_size
71/99	2.02G	0.02067	0.006157	0	17	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.962	0.999	0
.993	0.691					

Epoch	gpu_mem	box	obj	cls	labels	img_size
72/99	2.02G	0.01909	0.006087	0	15	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.96	1	0
.993	0.696					

Epoch	gpu_mem	box	obj	cls	labels	img_size
73/99	2.02G	0.0201	0.005953	0	17	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.959	1	0
.993	0.696					

Epoch	gpu_mem	box	obj	cls	labels	img_size
74/99	2.02G	0.01886	0.006252	0	15	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.961	0.996	0
.993	0.7					

Epoch	gpu_mem	box	obj	cls	labels	img_size
75/99	2.02G	0.0199	0.006393	0	14	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	0.998	0.96	0
.992	0.71					

Epoch	gpu_mem	box	obj	cls	labels	img_size
76/99	2.02G	0.01893	0.006115	0	14	640:
100% ██████████						
	Class	Images	Labels	P	R	mA
P@.5	mAP@					
	all	25	25	1	0.958	0
.992	0.7					

Epoch	gpu_mem	box	obj	cls	labels	img_size
-------	---------	-----	-----	-----	--------	----------

```

    77/99      2.02G   0.01818  0.005814      0      16      640:
100%|█████
          Class      Images      Labels      P      R      mA
P@.5 mAP@
          all      25      25      0.963      0.96      0
.992      0.707

```

```

    Epoch      gpu_mem      box      obj      cls      labels      img_size
    78/99      2.02G   0.01897  0.006313      0      17      640:
100%|█████
          Class      Images      Labels      P      R      mA
P@.5 mAP@
          all      25      25      0.961      0.993      0
.993      0.697

```

```

    Epoch      gpu_mem      box      obj      cls      labels      img_size
    79/99      2.02G   0.02006  0.006036      0      20      640:
100%|█████
          Class      Images      Labels      P      R      mA
P@.5 mAP@
          all      25      25      0.961      0.992      0
.993      0.706

```

```

    Epoch      gpu_mem      box      obj      cls      labels      img_size
    80/99      2.02G   0.0198   0.006134      0      14      640:
100%|█████
          Class      Images      Labels      P      R      mA
P@.5 mAP@
          all      25      25      0.961      0.996      0
.993      0.685

```

```

    Epoch      gpu_mem      box      obj      cls      labels      img_size
    81/99      2.02G   0.01773  0.005746      0      16      640:
100%|█████
          Class      Images      Labels      P      R      mA
P@.5 mAP@
          all      25      25      0.962      1      0
.993      0.693

```

```

    Epoch      gpu_mem      box      obj      cls      labels      img_size
    82/99      2.02G   0.01937  0.0062      0      22      640:
100%|█████
          Class      Images      Labels      P      R      mA
P@.5 mAP@
          all      25      25      0.96      1      0
.993      0.692

```

```

    Epoch      gpu_mem      box      obj      cls      labels      img_size
    83/99      2.02G   0.01794  0.005616      0      13      640:
100%|█████

```

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.959	1	0
.993	0.689					

Epoch	gpu_mem	box	obj	cls	labels	img_size
84/99	2.02G	0.01816	0.005511	0	17	640:
100%						

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.957	1	0
.993	0.694					

Epoch	gpu_mem	box	obj	cls	labels	img_size
85/99	2.02G	0.01736	0.005212	0	14	640:
100%						

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.957	1	0
.993	0.698					

Epoch	gpu_mem	box	obj	cls	labels	img_size
86/99	2.02G	0.01833	0.00545	0	16	640:
100%						

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.957	1	0
.993	0.697					

Epoch	gpu_mem	box	obj	cls	labels	img_size
87/99	2.02G	0.01692	0.005607	0	15	640:
100%						

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.957	1	0
.993	0.682					

Epoch	gpu_mem	box	obj	cls	labels	img_size
88/99	2.02G	0.01832	0.005639	0	15	640:
100%						

	Class	Images	Labels	P	R	mA
P@.5 mAP@	all	25	25	0.956	1	0
.993	0.695					

Epoch	gpu_mem	box	obj	cls	labels	img_size
89/99	2.02G	0.01717	0.005699	0	11	640:
100%						

	Class	Images	Labels	P	R	mA
P@.5 mAP@						

		all	25	25	0.955	1	0
.993	0.701						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	90/99	2.02G	0.01719	0.00589	0	16	640:
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.993	0
.995	0.706						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	91/99	2.02G	0.01795	0.00593	0	14	640:
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.993	0
.995	0.694						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	92/99	2.02G	0.01761	0.005412	0	18	640:
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.994	0
.995	0.681						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	93/99	2.02G	0.01683	0.004922	0	19	640:
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.994	0
.995	0.682						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	94/99	2.02G	0.0169	0.00542	0	14	640:
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.994	0
.995	0.701						
	Epoch	gpu_mem	box	obj	cls	labels	img_size
	95/99	2.02G	0.01639	0.005673	0	13	640:
100%							
	Class	Images	Labels		P	R	mA
P@.5	mAP@						
		all	25	25	1	0.996	0
.995	0.708						

Epoch	gpu_mem	box	obj	cls	labels	img_size
96/99	2.02G	0.01692	0.005484	0	19	640:

100%|██████████|

Class	Images	Labels	P	R	mA
all	25	25	1	0.996	0

P@.5 mAP@ .995 0.705

Epoch	gpu_mem	box	obj	cls	labels	img_size
97/99	2.02G	0.01678	0.005144	0	11	640:

100%|██████████|

Class	Images	Labels	P	R	mA
all	25	25	1	0.997	0

P@.5 mAP@ .995 0.705

Epoch	gpu_mem	box	obj	cls	labels	img_size
98/99	2.02G	0.01691	0.006163	0	23	640:

100%|██████████|

Class	Images	Labels	P	R	mA
all	25	25	1	0.995	0

P@.5 mAP@ .995 0.701

Epoch	gpu_mem	box	obj	cls	labels	img_size
99/99	2.02G	0.01649	0.005696	0	17	640:

100%|██████████|

Class	Images	Labels	P	R	mA
all	25	25	1	0.995	0

P@.5 mAP@ .995 0.706

100 epochs completed in 0.289 hours.

Optimizer stripped from yolov5/runs/train/Model/weights/last.pt, 14.5MB

Optimizer stripped from yolov5/runs/train/Model/weights/best.pt, 14.5MB

Validating yolov5/runs/train/Model/weights/best.pt...

Fusing layers...

YOLOv5s summary: 213 layers, 7012822 parameters, 0 gradients

Class	Images	Labels	P	R	mA
all	25	25	0.961	0.994	0

P@.5 mAP@ .992 0.731

Results saved to **yolov5/runs/train/Model**

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7fda7e0aab00>

Traceback (most recent call last):

File "/opt/conda/lib/python3.7/site-packages/torch/utils/data/dataloader.py", line 1358, in __del__

```
File "/opt/conda/lib/python3.7/site-packages/torch/utils/data/dataloader.py", line 1283, in _shutdown_workers
AttributeError: 'NoneType' object has no attribute 'python_exit_status'
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__
at 0x7fda7e0aab00>
```

Traceback (most recent call last):

```
File "/opt/conda/lib/python3.7/site-packages/torch/utils/data/dataloader.py", line 1358, in __del__
```

```
File "/opt/conda/lib/python3.7/site-packages/torch/utils/data/dataloader.py", line 1283, in _shutdown_workers
```

```
AttributeError: 'NoneType' object has no attribute 'python_exit_status'
When model trained, we need to save our it in order to use it at OCR in onnx format as below:
```

In [38]:

```
!python ./yolov5/export.py --weight ./yolov5/runs/train/Model/weights/best.pt --include torchscript onnx
```

```
export: data=yolov5/data/coco128.yaml, weights=['./yolov5/runs/train/Model/weights/best.pt'], imgsz=[640, 640], batch_size=1, device=cpu, half=False, inplace=False, train=False, keras=False, optimize=False, int8=False, dynamic=False, simplify=False, opset=12, verbose=False, workspace=4, nms=False, agnostic_nms=False, topk_per_class=100, topk_all=100, iou_thres=0.45, conf_thres=0.25, include=['torchscript', 'onnx']
YOLOv5 🚀 v6.1-253-g75bbaa8 Python-3.7.12 torch-1.11.0 CPU
```

Fusing layers...

YOLOv5s summary: 213 layers, 7012822 parameters, 0 gradients

PyTorch: starting from yolov5/runs/train/Model/weights/best.pt with output shape (1, 25200, 6) (13.8 MB)

TorchScript: starting export with torch 1.11.0...

TorchScript: export success, saved as yolov5/runs/train/Model/weights/best.torchscript (27.2 MB)

requirements: onnx not found and is required by YOLOv5, attempting auto-update...

WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: <https://pip.pypa.io/warnings/venv>

Collecting onnx

Downloading onnx-1.11.0-cp37-cp37m-manylinux_2_12_x86_64.manylinux2010_x86_64.whl (12.8 MB)

— 12.8/12.8 MB 36.7 MB/s eta 0:00:00

Requirement already satisfied: typing-extensions>=3.6.2.1 in /opt/conda/lib/python3.7/site-packages (from onnx) (4.2.0)

Requirement already satisfied: protobuf>=3.12.2 in /opt/conda/lib/python3.7/site-packages (from onnx) (3.20.1)

Requirement already satisfied: numpy>=1.16.6 in /opt/conda/lib/python3.7/site-packages (from onnx) (1.21.6)

Installing collected packages: onnx
Successfully installed onnx-1.11.0

requirements: 1 package updated per ['onnx']

requirements: ⚠ Restart runtime or rerun command for updates to take effect

ONNX: starting export with onnx 1.11.0...

ONNX: export success, saved as yolov5/runs/train/Model/weights/best.onnx (27.2 MB)

Export complete (18.15s)

Results saved to **/kaggle/working/yolov5/runs/train/Model/weights**

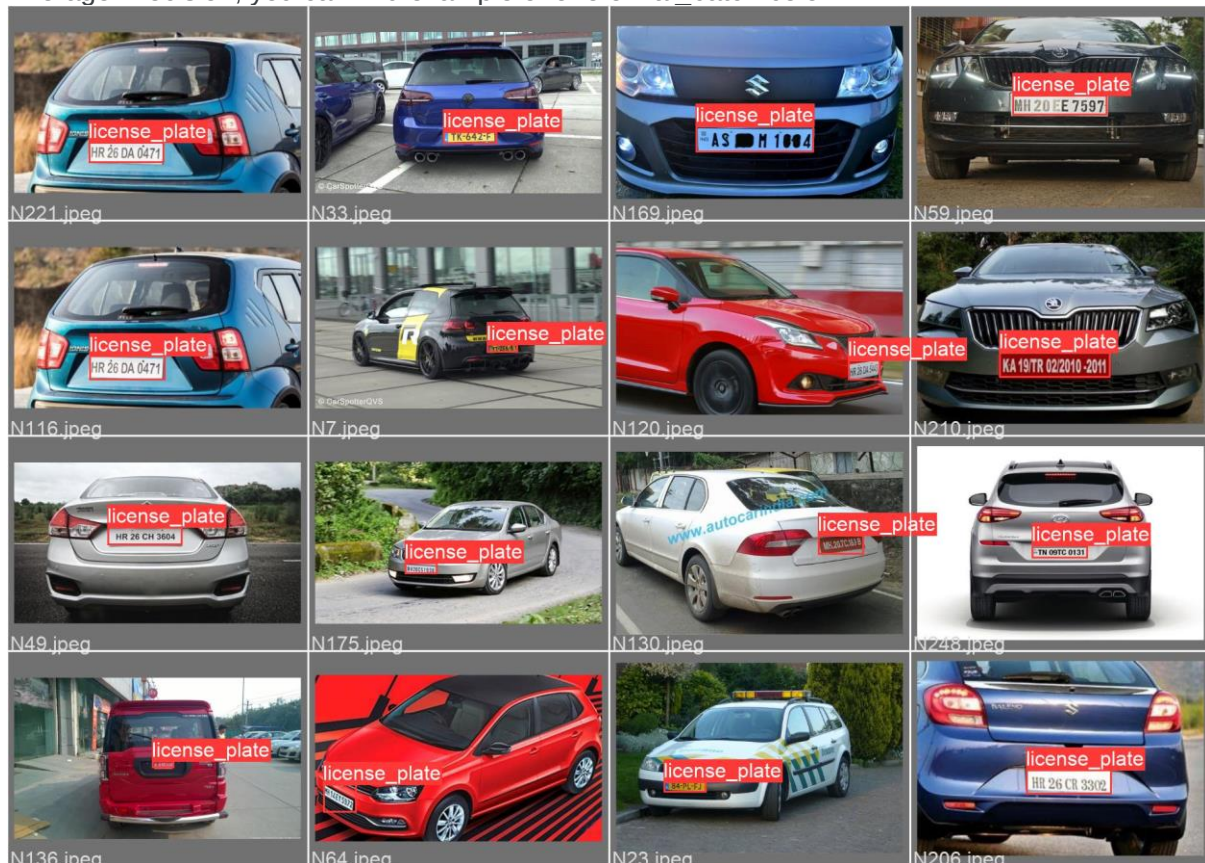
Detect: `python detect.py --weights yolov5/runs/train/Model/weights/best.onnx`

PyTorch Hub: `model = torch.hub.load('ultralytics/yolov5', 'custom', 'yolov5/runs/train/Model/weights/best.onnx')`

Validate: `python val.py --weights yolov5/runs/train/Model/weights/best.onnx`

Visualize: `https://netron.app`

All right, we have successfully saved our model after unzip it. We can notice that there is a model folder. We can find predicted pictures and we are able to detect the license plate very accurately. There might be the few wrong results, but still our model detects the license plate accurately. PR cow is eventually very important for me that will tell you the detection rate. We can see that our precision recall tell us that we can able to detect our model with 0.92 Precision and 0.5 mean Average Precision, you can find example of one of val_batch below



8.4 PREDICTING RESULTS FROM YOLO

Let's do some images settings which is important to provide in model construction. Input settings we need to pass as 640 by 640.

In [39]:

```
# settings
INPUT_WIDTH = 640
INPUT_HEIGHT = 640
```

Let me first load TEST image, it will have a plate number on it.

In [40]:

```
# LOAD THE IMAGE
img = io.imread('../input/number-plate-detection/TEST/TEST.jpeg')

fig = px.imshow(img)
fig.update_layout(width=700, height=400, margin=dict(l=10, r=10, b=10, t=10))
fig.update_xaxes(showticklabels=False).update_yaxes(showticklabels=False)
fig.show()
```

Now, we will load our YOLO model.

In [41]:

```
# LOAD YOLO MODEL
net = cv2.dnn.readNetFromONNX('../yolov5/runs/train/Model/weights/best.onnx')
net.setPreferableBackend(cv2.dnn.DNN_BACKEND_OPENCV)
net.setPreferableTarget(cv2.dnn.DNN_TARGET_CPU)
```

The next step is I need to parse this image to my YOLO model and get the prediction. Important - size we need to set as input width a 640 and input height is 640.

We have the 25200 rows and 6 columns. The first four columns will give you the information about the bounding box, which is center X and, Center Y, W and H. And these values are normalized to 640 by 640. And the next two values are the confidence and the probabilities. Confidence tells us what is the confidence for the detecting the bounding box and the next one is the probability score of the class, since we have only one class that is number plate.

Let's make a list of boxes and confidence and place it order. As well we need to identify our X_factor and Y_factor of image.

We have to filter detections based on the confidence score and the probability score. Let me take the confidence >0.4 and filter from class >0.25.

There is will be one of major issue with YOLO modeling it will give you the repeated boxes, but the same object it might give you that multiple boxes. For that we need to perform the non-maximum suppression. To do it we need to provide all our clean boxes and confidences.

In the end of that part of code I will take bounding boxes and draw the rectangle box to our image. And will write it all in functions.

In [42]:

```
def get_detections(img, net):
    # 1.CONVERT IMAGE TO YOLO FORMAT
    image = img.copy()
```

```

row, col, d = image.shape

max_rc = max(row,col)
input_image = np.zeros((max_rc,max_rc,3),dtype=np.uint8)
input_image[0:row,0:col] = image

# 2. GET PREDICTION FROM YOLO MODEL
blob = cv2.dnn.blobFromImage(input_image,1/255,(INPUT_WIDTH,INPUT_HEIGHT),swapRB=True,crop=False)
net.setInput(blob)
preds = net.forward()
detections = preds[0]

return input_image, detections

def non_maximum_supression(input_image,detections):

# 3. FILTER DETECTIONS BASED ON CONFIDENCE AND PROBABILITY SCORE

# center x, center y, w , h, conf, proba
boxes = []
confidences = []

image_w, image_h = input_image.shape[:2]
x_factor = image_w/INPUT_WIDTH
y_factor = image_h/INPUT_HEIGHT

for i in range(len(detections)):
    row = detections[i]
    confidence = row[4] # confidence of detecting license plate
    if confidence > 0.4:
        class_score = row[5] # probability score of license plate
        if class_score > 0.25:
            cx, cy , w, h = row[0:4]

            left = int((cx - 0.5*w)*x_factor)
            top = int((cy-0.5*h)*y_factor)
            width = int(w*x_factor)
            height = int(h*y_factor)
            box = np.array([left,top,width,height])

            confidences.append(confidence)
            boxes.append(box)

# 4.1 CLEAN
boxes_np = np.array(boxes).tolist()
confidences_np = np.array(confidences).tolist()

# 4.2 NMS
index = cv2.dnn.NMSBoxes(boxes_np,confidences_np,0.25,0.45)

return boxes_np, confidences_np, index

```

```

def drawings(image, boxes_np, confidences_np, index):
    # 5. Drawings
    for ind in index:
        x,y,w,h = boxes_np[ind]
        bb_conf = confidences_np[ind]
        conf_text = 'plate: {:.0f}%'.format(bb_conf*100)
        license_text = extract_text(image, boxes_np[ind])

        cv2.rectangle(image, (x,y), (x+w,y+h), (255,0,255), 2)
        cv2.rectangle(image, (x,y-30), (x+w,y), (255,0,255), -1)
        cv2.rectangle(image, (x,y+h), (x+w,y+h+25), (0,0,0), -1)

        cv2.putText(image, conf_text, (x,y-10), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (
255,255,255), 1)
        cv2.putText(image, license_text, (x,y+h+27), cv2.FONT_HERSHEY_SIMPLEX,
0.7, (0,255,0), 1)

    return image

```

In [43]:

```

# predictions flow with return result
def yolo_predictions(img, net):
    # step-1: detections
    input_image, detections = get_detections(img, net)
    # step-2: NMS
    boxes_np, confidences_np, index = non_maximum_supression(input_image, d
etections)
    # step-3: Drawings
    result_img = drawings(img, boxes_np, confidences_np, index)
    return result_img

```

Here we will not only detect license plate but else extract test from it. We already done it before let's repeat it again. We will write function. First from bounding box we need to take our x,y,w,h and extract ROI. I place it at top of our functions in drawing part.

In [44]:

```

# extrating text
def extract_text(image, bbox):
    x,y,w,h = bbox
    roi = image[y:y+h, x:x+w]

    if 0 in roi.shape:
        return 'no number'

    else:
        text = pt.image_to_string(roi)
        text = text.strip()

    return text

```

In [45]:

```
# test
img = io.imread('../input/number-plate-detection/TEST/TEST.jpeg')
results = yolo_predictions(img,net)
```

In [46]:

```
fig = px.imshow(img)
fig.update_layout(width=700, height=400, margin=dict(l=10, r=10, b=10, t=10))
fig.update_xaxes(showticklabels=False).update_yaxes(showticklabels=False)
fig.show()
```

The accuracy of the model is really pretty accurate. However, there are few misclassification, but still our model is performing really well. If you want to get more and more good accuracy, what you have to do is feed more and more data and then you can able to predict very good results.

8.5 REAL TIME OBJECT DETECTION

In real time prediction my point is that the detection is very perfect and in order to get very, very good predictions, we need to get very high 4K definition camera so that we can able to extract the text properly.

```
cap = cv2.VideoCapture('../input/number-plate-detection/TEST/TEST.mp4')

while True:

    ret, frame = cap.read()

    if ret == False:

        print('Unable to read video')

        break

    results = yolo_predictions(frame,net)

    cv2.namedWindow('YOLO',cv2.WINDOW_KEEPRATIO)

    cv2.imshow('YOLO',results)

    if cv2.waitKey(30) == 27 :

        break

cv2.destroyAllWindows()

cap.release()
```



9. CONCLUSION

With the increase in the number of vehicles, vehicle tracking has become an important research area for efficient traffic control, surveillance, and finding stolen cars. For this purpose, efficient real-time license plate detection and recognition are of great importance. Due to the variation in the background and font color, font style, size of the license plate, and non-standard characters, license plate recognition is a great challenge in developing countries. To overcome such issues, this study applies a deep-learning strategy to improve license plate recognition efficiency. The collected images have been captured under various lighting/contrast conditions, distance from the camera, varying angle of rotation, and validated to produce a high recognition rate. The approach can be effectively used by law enforcement agencies and private organizations to improve homeland security. Future work may include training and validation of the existing algorithm using the hybrid classifier method and improvement of the robustness of the license plate recognition system in varying weather conditions.

*All files you can find as well on my [GITHUB](#) If you find this post useful please **UPVOTE**, and if you have any question feel free to [CONTACT ME](#).*